PEARSON EDEXCEL INTERNATIONAL A LEVEL CHEMISTRY · WCH13

Unit 3: Practical Skills in Chemistry I

Revision notes: practical skills P1–P5 and the eight AS core practicals

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

The WCH13 paper is 1 hour 20 minutes, 50 marks, all questions compulsory, with at least 6 marks of maths. A typical paper has one question on tests for ions or organic groups, one on titration, one on an enthalpy or gas-volume experiment, and one on preparing and purifying an organic liquid.

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Core practicals

CP1 Measuring the molar volume of a gas

CP2 Enthalpy change of a reaction using Hess's law

CP3 Finding the concentration of a solution of hydrochloric acid

CP4 Making a standard solution and standardising sodium hydroxide

CP5 Rates of hydrolysis of halogenoalkanes

CP6 Chlorination of 2-methylpropan-2-ol with concentrated hydrochloric acid

CP7 Oxidation of propan-1-ol to propanal and propanoic acid

CP8 Analysis of some inorganic and organic unknowns

Practical skills

SPEC P1 Planning an investigation

WHAT THIS SPEC POINT ASKS

Plan an experiment: pick suitable apparatus, decide which variables to change, measure and control, write a method someone else could follow, and spot the hazards with a matching precaution. Expect to do this for the core practicals and for set-ups you have never seen.

P1 Variables

- **Independent variable (IV)**: the one you change on purpose (e.g. mass of solid added).
- **Dependent variable (DV)**: the one you measure (e.g. volume of gas, temperature change, time for a precipitate).
- **Control variables**: everything else that could change the DV. Keep them the same, and say **how**:
  - temperature → water bath (or all at room temperature);
  - volume of a solution → same pipette or burette;
  - concentration → same stock solution;
  - mass and particle size of a solid → same balance, same form (powder or ribbon);
  - stirring, timing, apparatus.
- An experiment is **valid** if it really tests what it claims to, which needs the control variables to be kept constant.

P1 Choosing apparatus

You need to…	Best choice	Why
deliver an exact fixed volume (e.g. 25.0 cm^3)	volumetric **pipette** + filler	most precise for one fixed volume
add a variable volume and know it exactly	**burette**	reads to 0.05 cm^3
make exactly 250.0 cm^3 of a solution	**volumetric flask**	one calibrated line
measure a rough volume (reagent in excess)	**measuring cylinder**	quick; precision not needed
measure a gas volume	**gas syringe**, or inverted measuring cylinder or burette over water	see CP1
weigh a solid	balance (2 or 3 d.p.), **weighing bottle or boat**	weigh **by difference**
measure temperature	thermometer (0.1–1 °C divisions) or temperature probe	finer divisions → smaller uncertainty
break up lumps of solid	**pestle and mortar**	bigger surface area
keep heat in during a reaction	**polystyrene cup with a lid**, standing in a beaker	insulates; the beaker stops it tipping
heat a flammable liquid	**water bath** or **electric heater** (heating mantle)	no naked flame
time a reaction	stopwatch / stop clock	

P1 Writing a method that scores

- Number the steps. Each step names a **quantity, the apparatus and the action**: "Use a pipette to transfer 25.0 cm^3 of the sodium carbonate solution into a conical flask", not "add some solution".
- Say what you **measure, and when**: mass of spirit burner **before and after**, temperature **before and after**, volume of water used.
- Say how you get reliable data: repeat until **concordant** (titration) or repeat and take a **mean**.
- Add safety only where it matters, and match it to the hazard.

P1 Methods you should be able to write from memory

These are the planning questions that come up again and again. The full method for each is in the section shown.

- Make a **standard solution** (4–6 marks): see CP4.
- Carry out an **accurate titration** after a rough one: see P5 (titration).
- Carry out a **flame test**: see P5 (cations).
- **Collect a gas** over water and draw the apparatus: see CP1.
- Draw **reflux**, **distillation** and **separating funnel** set-ups: see P5 (organic techniques).
- List the **measurements** for a combustion experiment: see CP2.
- Plan a **sequence of tests** to identify unknowns from a given set of reagents: see CP8.

P1 Hazard, risk and precautions

- A **hazard** is something that could cause harm (a property of a chemical or a procedure). The **risk** is the chance of harm actually happening. A **precaution** lowers the risk.
- The precaution must fit the hazard. Goggles and a lab coat are taken as given, so questions usually say "other than eye protection and a lab coat".

Pictogram	Meaning	Examples in this unit	Precaution
flame	**flammable**	ethanol, propan-1-ol, propanal, cyclohexane, halogenoalkanes	no naked flames; heat with a water bath or electric heater
test tubes dripping onto a hand and a surface	**corrosive**	concentrated HCl, H_2SO_4, concentrated NaOH	gloves; add slowly; fume cupboard for fumes
skull and crossbones	**toxic**	barium compounds, chlorine	gloves; fume cupboard or mask for dusts and gases
exclamation mark	**harmful / irritant**	dilute NaOH, dilute $AgNO_3$	gloves; keep below eye level
flame over a circle	**oxidising**	potassium dichromate(VI), nitrates, concentrated H_2O_2	keep away from flammable substances
person with a damaged chest	**health hazard** (e.g. carcinogen)	potassium dichromate(VI)	gloves; avoid skin contact
dead tree and fish	**hazardous to the environment**	silver nitrate, halogenoalkanes	don't pour down the sink

- Toxic or corrosive **gases and vapours** (NO_2, Cl_2, SO_2, HCl fumes, H_2S) → **fume cupboard**.
- **Exothermic** reactions or ones that make gas → add the reagent **slowly / in portions**, cool in an **ice bath**, use a tray to catch overflow.

MARK-SCHEME LANGUAGE

- Standard solution: "dissolve in distilled / deionised water in a beaker" → "transfer to a volumetric flask" → "add the washings" → "make up to the mark" → "invert / shake to mix". "Pure water" and "test tube" are not accepted (WCH13 Oct 2019).
- Flame test: "nichrome / platinum wire", "dip in concentrated hydrochloric acid", "hold in a hot / blue / non-luminous / roaring Bunsen flame".
- Measurement lists: name each measurement precisely ("mass of burner before and after burning", "initial and final temperature of the water").
- Hazard of about 0.25 mol dm^{-3} NaOH: "irritant / harmful", with a precaution such as "keep the burette below eye level". Corrosive, caustic or toxic was not accepted for this dilute solution (WCH13 Jan 2023).

Often asked as: describe how to make a standard solution; describe a flame test; list the measurements you would take; draw the apparatus; name the hazard shown by this pictogram and give a precaution; suggest an outline method for an unfamiliar task.

COMMON MISTAKE

"Measure the amount of fuel used" scores nothing. Say *what* you weigh, and *when*.
Using a wooden splint or a copper wire for a flame test, or naming the wrong acid to clean the wire.
Giving a precaution that doesn't match the hazard (e.g. "use a fume cupboard" for a flammable liquid).
Writing "pure water" instead of distilled or deionised water.

BEYOND THE SPEC, BUT EXAMINED

- Comparing the **thermal stability of Group 2 carbonates**: heat the same number of moles of each, at the same height above the same Bunsen flame, with the delivery tube in the same volume of limewater; time how long it takes to turn cloudy. Limitation: a Bunsen may not get hot enough to decompose $SrCO_3$ or $BaCO_3$ (WCH13 Oct 2024).
- **Mr of a gas from a canister**: weigh the canister, release gas into an inverted measuring cylinder over water, make the water levels inside and outside the same (so the gas is at atmospheric pressure), then reweigh (WCH13 Jan 2026).
- **Combustion analysis**: burn the compound in pure oxygen (not air, which contains water vapour and CO_2), and weigh the tubes that absorb H_2O and CO_2 before and after (WCH13 Jan 2022).

SPEC P2 Carrying out practical work: apparatus, techniques and measurement

WHAT THIS SPEC POINT ASKS

Use standard laboratory apparatus and techniques properly, read scales correctly, know the range and resolution of the apparatus, and explain why each step in a standard technique is done.

P2 Range, resolution and uncertainty

- **Range**: the smallest to the largest value an instrument can measure (a 50 cm^3 burette, a 100 cm^3 gas syringe, a −10 to 110 °C thermometer). If the experiment could exceed it, change the quantities or the apparatus.
- **Resolution**: the smallest change the instrument can show (one division, or the last digit of a digital display). Finer resolution → smaller uncertainty.

- **Uncertainty**: the range either side of a reading within which the true value is expected to lie. The question normally tells you the value to use. Typical values:

Apparatus	Record to	Typical uncertainty
50 cm^3 burette	2 d.p., last digit 0 or 5 (e.g. 23.45 cm^3)	±0.05 cm^3 per reading
25.0 cm^3 pipette	the volume on the label	±0.04 to ±0.06 cm^3
250.0 cm^3 volumetric flask	the volume on the label	±0.15 to ±0.2 cm^3
100 cm^3 measuring cylinder	nearest 1 cm^3 (or 0.5)	±0.5 to ±1 cm^3
100 cm^3 gas syringe	nearest 1 cm^3 (or 0.5)	±0.5 cm^3
2 d.p. balance	0.01 g	±0.005 g (or ±0.01 g)
3 d.p. balance	0.001 g	±0.0005 g
thermometer, 1 °C divisions	nearest 0.5 °C	±0.5 °C
thermometer, 0.2 °C divisions or digital	0.1 °C	±0.1 °C

P2 Reading scales

- Read the **bottom of the meniscus**, with your **eye level** with it to avoid **parallax**.
- A burette scale reads **downwards** (0.00 at the top). Record both readings to **2 d.p.**: 23.40, not 23.4.
- A burette turned upside down to collect gas reads the other way round: work out which reading is the start (WCH13 Jan 2025).
- Read a thermometer to **half a division** (e.g. 27.5 °C on a scale marked every 1 °C).
- Record every digit a digital instrument shows.

P2 Why each step is done

Step	Reason
Rinse the burette with the solution it will hold	water left inside would **dilute** it → titre too large
Rinse the pipette with the solution it will deliver	water would dilute it → fewer moles delivered
Rinse the conical flask with **water** only (it can be wet)	water doesn't change the **number of moles** put in the flask
Remove the funnel before titrating; fill the jet (no air bubble)	drops from the funnel or a bubble leaving the jet make the reading wrong
Let the pipette drain and touch the tip on the surface; **don't blow out** the last drop	the pipette is calibrated to deliver with that drop left in
White tile under the flask; swirl	see the colour change clearly; mix the solution
Rough titration first, then dropwise near the end-point	find the approximate end-point, then don't overshoot
Rinse the beaker, rod and funnel into the volumetric flask	**all** the solute ends up in the flask
Invert the stoppered volumetric flask several times	solution has the **same concentration throughout**
Weigh by difference (bottle before and after tipping)	allows for solid left in the bottle
Heat to **constant mass**	reaction (decomposition, water loss) is **complete**
Leave the crucible lid slightly open	gas escapes, solid stays in
Take the delivery tube out of the water **before** you stop heating	prevents **suck-back** of cold water into the hot tube, which could crack it
Add **anti-bumping granules**	smooth boiling: small bubbles form on them instead of large violent ones
Condenser water **in at the bottom**, out at the top	the condenser stays **full** of water, so cooling is efficient
Never heat a **sealed** apparatus	pressure builds up and it can burst
Polystyrene cup with a lid; stir	**reduces** heat loss; even temperature throughout
Draught shield round a spirit burner	reduces heat loss to moving air
Ice bath while adding a reagent; add slowly	reaction is **exothermic**; stops the mixture boiling over and volatile substances escaping
Grind a solid	larger surface area → faster reaction or dissolving
Fume cupboard	toxic or corrosive gases or vapours

P2 Heating methods

- **Bunsen burner**: flame tests and thermal decomposition of solids. Never for flammable organic liquids.
- **Water bath**: gentle, even heating up to 100 °C, no flame.
- **Electric heating mantle / hot plate**: heating flammable liquids, including above 100 °C.
- **Oil bath** (or sand bath): even heating above 100 °C, e.g. distilling propanoic acid, boiling temperature 141 °C (WCH13 Jan 2026).

MARK-SCHEME LANGUAGE

- Rinse the burette with the acid "to prevent dilution of the acid". The conical flask needn't be rinsed with the solution because "dilution does not affect the number of moles in the flask". "Does not change the concentration" was not credited (WCH13 Oct 2025).
- Meniscus: "bottom of the meniscus on the line" **and** "eye level with the mark".
- Inverting the volumetric flask: "so the solution has the same concentration throughout / is homogeneous".
- Suck-back: "to stop water being sucked back into the hot tube" (or "the tube cracking"). "Gases sucking back" is not credited.
- Anti-bumping granules: "smooth / even boiling", "sites for small bubbles to form". "Prevent bumping" on its own is ignored.
- Condenser water in at the bottom: "condenser is full of water / no air pockets" and "more efficient cooling".
- Electric heater instead of a Bunsen: "alcohols and products are flammable". "Even heating" is ignored (WCH13 Oct 2022).

Often asked as: name the best apparatus to measure 25.0 cm^3; read this burette or thermometer; give a reason for this step; what happens to the titre if the pipette is rinsed with water; name the hazard and give a precaution.

COMMON MISTAKE

Blowing out the last drop from a pipette. That delivers *more* than 25.0 cm^3.
Saying sodium hydroxide "evaporates". It is the **water** that evaporates, so the solution becomes more concentrated.
"Prevents heat loss". Insulation and lids **reduce** heat loss; nothing stops it completely.
Burette readings to 1 d.p., or a last digit that isn't 0 or 5.

BEYOND THE SPEC, BUT EXAMINED

- Why the polystyrene cup stands in a beaker: **stability**, because the light cup tips over easily. Heat loss was not the answer wanted (WCH13 Jun 2019).
- **Gas syringe** compared with collecting over water: no gas dissolves in water (CO_2 is fairly soluble) and the syringe has finer graduations (WCH13 Jun 2024).
- Collecting H_2 in an **inverted burette**: acid is layered under water, and the magnesium doesn't react at once because the acid has to diffuse down to it (WCH13 Jan 2025).

SPEC P3

Analysing data: tables, graphs and calculations

WHAT THIS SPEC POINT ASKS

Record data in clear tables, plot and interpret graphs, process data with the right equations, and give answers to a sensible number of significant figures with units.

P3 Tables

- Column headings carry the **quantity and unit**: "Temperature / °C", "Volume of gas / cm^3". Don't write units in the body of the table.
- Every value in a column has the **same number of decimal places**, set by the instrument's resolution.

- Titration tables: rows for final reading, initial reading and titre; columns for the rough and accurate runs. Tick or circle the concordant titres.

P3 Significant figures and rounding

- Give the final answer to the same number of significant figures as the **least precise** piece of data, usually **3 s.f.** (2 or 3 is usually accepted).
- **Don't round intermediate values.** Keep the full calculator value and round only at the end.
- **Round, don't truncate**: 0.1035… → 0.104, not 0.103.
- Some mark schemes penalise wrong rounding once in the paper; others ignore it. Always round correctly.

P3 Graphs

- **Axes**: independent variable on x, dependent on y. Label each with quantity / unit.
- **Scales**: plotted points should cover **at least half the grid** in both directions. Use even steps (1, 2, 5 or 10 per large square). The axes needn't start at zero unless you need the origin.
- **Plot** to within half a small square, with small neat crosses.
- **Line of best fit**: a single straight line or smooth curve with points scattered evenly either side. Ignore (circle) anomalies. **Never join the dots.**
- **Gradient**: use a large triangle (at least half the line). Gradient = $\Delta y \div \Delta x$, with units.
- **Tangent**: for the rate at one moment on a curve; its gradient is the rate.
- **Intersection of two lines**: the end-point in a thermometric titration, or the point where one reagent stops being in excess in a series of mixtures.
- **Extrapolation**: extend the line beyond the data (dashed) to a value you couldn't measure, e.g. the temperature at the moment of mixing (CP2).
- **Line through the origin**: when theory says zero in gives zero out (gas volume against mass of carbonate).
- **Calibration graph**: measure standards of known value, plot, then read off the unknown.

EXAM TIP

When a question says "show your working on the graph", the construction lines (extrapolation, intersection, tangent, gradient triangle) must be visible. Without them the reading scores nothing.

P3 Calculation toolkit

Quantity	Equation	Watch out
moles from mass	n = m ÷ M	M of the whole formula, including water of crystallisation
moles in solution	n = c × V	V in **dm³** (cm^3 ÷ 1000)
mass concentration	g dm^{-3} = mol dm^{-3} × M	
moles of gas	n = V ÷ 24.0 dm^3 mol^{-1} (or the value given)	
energy transferred	Q = m × c × ΔT	m = mass of **solution** or water, not of the solid
enthalpy change	ΔH = −Q ÷ n	n of the **limiting** reagent; kJ; sign
percentage yield	actual ÷ theoretical × 100	theoretical from the limiting reagent
percentage purity	mass of pure substance ÷ mass of sample × 100	
rate	1 ÷ time, or gradient	units s^{-1} or cm^3 s^{-1}

WORKED EXAMPLE: PERCENTAGE PURITY BY TITRATION

2.00 g of impure sodium carbonate is dissolved and made up to 250.0 cm^3. 25.0 cm^3 portions need a mean titre of 23.50 cm^3 of 0.150 mol dm^{-3} HCl.

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

1. n(HCl) = 0.150 × 23.50 ÷ 1000 = 3.525×10^{-3} mol
2. n(Na_2CO_3) in 25.0 cm^3 = $3.525 \times 10^{-3} \div 2 = 1.7625 \times 10^{-3}$ mol
3. n(Na_2CO_3) in 250.0 cm^3 = $1.7625 \times 10^{-3} \times 10 = 1.7625 \times 10^{-2}$ mol
4. mass of Na_2CO_3 = $1.7625 \times 10^{-2} \times 106.0 = 1.868$ g
5. purity = 1.868 ÷ 2.00 × 100 = **93.4%**

WORKED EXAMPLE: IDENTIFY A METAL FROM MR

An experiment gives Mr = 138 for the Group 1 carbonate M_2CO_3.

2 × Ar(M) = 138 − 60.0 (for CO_3) = 78, so Ar(M) = 39 → **potassium**.

Always take off the known part of the formula, and pick the nearest element **from the group the question names**.

MARK-SCHEME LANGUAGE

- Plotting (often 3 marks): axes the right way round with labels and units; scale covering at least half the grid; points within half a small square. Uneven scales and joined-up dots lose marks.
- Calculations use transferred error (TE): a wrong earlier value carried through correctly still earns the later marks.
- Final answers to 2 or 3 s.f.; enthalpy changes need a **sign and unit** (e.g. −201 kJ mol^{-1}). Molar volume is in dm^3 mol^{-1} or cm^3 mol^{-1}, not mol dm^{-3} (WCH13 Oct 2023).
- Percentage yield above 100% is never credited.

Often asked as: complete the table; plot the graph and draw lines of best fit; use your graph to find…; calculate the concentration / purity / Mr / value of x; give your answer to an appropriate number of significant figures.

COMMON MISTAKE

Using the mass of the solid in Q = mcΔT instead of the mass of the solution.
Forgetting the ×10 scale-up from a 25.0 cm^3 portion to the 250.0 cm^3 flask, or ignoring the mole ratio in the equation.
Not converting cm^3 to dm^3.
Rounding early, so the final answer drifts out of the accepted range.

BEYOND THE SPEC, BUT EXAMINED

- **Back-titration** (antacid tablet): add a known excess of acid, titrate the acid left over with NaOH; moles that reacted = moles added − moles left (WCH13 Oct 2022).
- **Calibration graph** from density: density = mass ÷ volume for known ethanol–water mixtures, then read the % ethanol of an unknown (WCH13 Jan 2024).
- **Precipitation titration** with silver nitrate and a chromate(VI) indicator: red silver chromate forms only after all the chloride has precipitated, because AgCl is less soluble (WCH13 Jan 2022).

SPEC P4 Evaluating: errors, uncertainty and improvements

WHAT THIS SPEC POINT ASKS

Judge the quality of data and methods: tell accuracy from precision, calculate percentage uncertainty and percentage error, identify the main sources of error, and suggest improvements with a reason for each.

P4 Words you must use correctly

Term	Meaning
true value	the value you'd get with perfect measurement
accurate	close to the true (or data-book) value
precise	repeat readings are close to each other (small spread)
uncertainty	the range either side of a measurement within which the true value is expected to lie
random error	unpredictable scatter either side of the true value; reduced by repeating and taking a mean
systematic error	always pushes the result the same way (e.g. heat loss always makes ΔT too small); repeating does **not** remove it
repeatable	the same person, method and equipment give similar results
reproducible	a different person or method gives similar results
valid	the experiment measures what it is meant to, with other variables controlled
anomalous result	one that doesn't fit the pattern; leave it out of means and best-fit lines

P4 Percentage uncertainty

$$\%\ \text{uncertainty} = (\text{uncertainty} \div \text{measured value}) \times 100$$

- A value found from **two readings** (titre, temperature change, mass by difference) carries **two** uncertainties, so double it.
- To estimate the uncertainty in a final result, **add the percentage uncertainties** of the measurements that went into it.

WORKED EXAMPLE: PERCENTAGE UNCERTAINTIES

Titre: 24.15 cm^3 from a burette read to ±0.05 cm^3.
% uncertainty = (2 × 0.05) ÷ 24.15 × 100 = **0.41%**
Temperature change: ΔT = 6.4 °C, thermometer ±0.1 °C.
% uncertainty = (2 × 0.1) ÷ 6.4 × 100 = **3.1%**
Mass by difference: 1.57 g on a balance read to ±0.005 g.
% uncertainty = (2 × 0.005) ÷ 1.57 × 100 = **0.64%**
Overall for the NaOH concentration in CP4: balance 0.64% + volumetric flask (0.15 ÷ 250.0 × 100 = 0.06%) + pipette (0.04 ÷ 25.0 × 100 = 0.16%) + burette (2 × 0.05 ÷ 23.80 × 100 = 0.42%) = **1.3%**.

P4 Reducing percentage uncertainty

- Make the **measured quantity bigger**, so the same absolute uncertainty is a smaller fraction of it:
  - larger mass of solid;
  - larger titre (more dilute solution in the burette, or a larger portion in the flask);
  - larger temperature change (more solid, or a smaller volume of solution);
  - more gas (more solid), as long as it stays within the range of the syringe or cylinder.
- Use apparatus with **finer resolution** (if the question allows a change of apparatus).
- Repeating and averaging reduces the effect of **random** error. It doesn't change the uncertainty of each reading.

P4 Percentage error: comparing with the data-book value

$$\%\ \text{error} = (\text{data-book value} - \text{experimental value}) \div \text{data-book value} \times 100$$

Divide by the **data-book (accepted) value**, not by your own result. Ignore the sign of the difference.

WORKED EXAMPLE: IS THE METHOD TO BLAME?

A spirit-burner experiment gives ΔcH(ethanol) = −924 kJ mol^{-1}. The data-book value is −1367 kJ mol^{-1}.
% error = (1367 − 924) ÷ 1367 × 100 = **32.4%**
The total measurement uncertainty is only about 6% (see CP2). The error is far bigger than the uncertainty, so the measurements can't explain it. There must be **systematic errors** in the method: heat loss to the air and the can, incomplete combustion, ethanol evaporating from the wick.

WORKED EXAMPLE: IS A RESULT CONSISTENT WITH THE DATA BOOK?

A result of 37.5 has a total uncertainty of 4.5%.
4.5% of 37.5 = 1.7, so the true value lies between **35.8 and 39.2**.
If the data-book value (e.g. 39.1) lies inside this range, the result is consistent with it. Apply the percentage to **your experimental value**, not to the data-book value.

P4 Common systematic errors and how to reduce them

Experiment	Main error	Effect	Improvement (with reason)
Solution calorimetry	heat lost to the surroundings	ΔT too small → ΔH **less exothermic**	lid on the cup (reduces heat loss); extrapolate a cooling line back to the time of mixing
Solution calorimetry	assumptions: density 1.00 g cm^{-3}, c = 4.18 J g^{-1} K^{-1}, solid's mass ignored, cup absorbs no heat	small errors	state them; use a more precise thermometer
Combustion	heat lost to the air; heat used to warm the can; incomplete combustion; fuel evaporates from the wick	ΔcH far less exothermic	draught shield, lid on the burner when not burning, flame close to a copper can, stir the water
Gas collection	gas escapes before the bung goes on; CO_2 dissolves in water	volume too small	tip the solid in after sealing; use a gas syringe
Titration	overshooting; judging the colour; air bubble in the jet	titre wrong	rough titration first, dropwise near the end-point, white tile
Preparations	incomplete reaction, side reactions, losses on transfer and separation, volatile product evaporates	yield low	cool during addition; rinse apparatus; collect in an ice-cooled receiver

EXAM TIP

An improvement needs a **reason**. "Use a burette, because its smaller uncertainty (±0.05 cm^3) gives a more accurate volume" scores. "Use a burette, it is more accurate" usually doesn't.

MARK-SCHEME LANGUAGE

- Reduce percentage uncertainty without changing apparatus: "use a greater mass of solid / smaller volume of acid" **and** "so the temperature change is larger". Reducing the acid concentration, or anything to cut heat loss, is not credited for *uncertainty* (WCH13 Jan 2020).
- Glass beaker instead of polystyrene: "**less exothermic** / less negative" (not just "smaller") **and** "more heat lost / glass conducts heat / glass absorbs heat".
- Assumptions (any two): "density of the solution is 1 g cm^{-3}", "specific heat capacity of the solution is the same as water", "the mass of solid is ignored", "the heat capacity of the cup is ignored". Mentioning water without "solution" isn't enough.
- Uncertainty against accuracy: the uncertainty is the range within which the true value lies; accuracy is how close the result is to the data-book value. Burning ethanol for longer reduces the percentage uncertainty but not the inaccuracy, because more heat is lost (WCH13 Jan 2023).

Often asked as: calculate the percentage uncertainty in the titre / ΔT / mass; how could the percentage uncertainty be reduced without changing the apparatus; calculate the percentage error; suggest two improvements and justify each; explain why the value is less exothermic than the data-book value.

COMMON MISTAKE

Dividing by the experimental value in a percentage error.
Applying the percentage uncertainty to the data-book value instead of your own result.
Forgetting to double the uncertainty for a value found from two readings.
"Human error" or "measured wrongly". The stem often says to assume no measurement mistakes, and these never score.
Mixing up accuracy, precision, reliability and validity. Examiners report this every year.

BEYOND THE SPEC, BUT EXAMINED

- Comparing two pieces of apparatus by percentage uncertainty: volumetric flask (±0.15 in 250 cm^3 = 0.06%) against pipette (±0.04 in 25 cm^3 = 0.16%). The flask contributes less (WCH13 Jan 2026).
- Judging whether an impurity explains a discrepancy: an MgO impurity in Mg means *fewer* moles of Mg react, so it would make the molar volume too *low*. It can't explain a result that is too high, and 2% impurity can't explain a 20% error (WCH13 Jan 2025).
- Deciding whether a repeat is needed: one clear anomaly and an obvious trend means no repeat is needed (WCH13 Oct 2021).

SPEC P5 Chemistry-specific skills: tests, titration and organic techniques

WHAT THIS SPEC POINT ASKS

Recall and interpret tests for cations, anions, gases and organic functional groups from Units 1 and 2. Carry out titrations to a high standard. Use the organic techniques of reflux, distillation, solvent extraction with a separating funnel, washing, drying and boiling-temperature (or melting-temperature) checks.

P5 Tests for cations: flame tests

METHOD: FLAME TEST

1. Clean a **nichrome (or platinum) wire** by dipping it in **concentrated hydrochloric acid** and holding it in a flame until it gives no colour.

2. Dip the wire in the acid again, then into the **solid sample**, so some sticks to it.
3. Hold it in the edge of a **hot, blue (non-luminous) Bunsen flame** (air hole open).
4. Note the colour. Clean the wire between samples.

Ion	Flame colour
Li^+	red (crimson)
Na^+	yellow (orange-yellow)
K^+	**lilac**
Rb^+	red-violet
Cs^+	blue-violet
Mg^{2+}	**no colour**
Ca^{2+}	**brick-red** (orange-red / yellow-red)
Sr^{2+}	**red / crimson**
Ba^{2+}	**(apple) green**
Cu^{2+} (not Group 1 or 2)	blue-green

- **Why there are colours**: heat promotes electrons to **higher energy levels**. When they drop back down they release energy as light. Each energy gap gives a specific **frequency (wavelength)**, and for these ions some of it is in the **visible** region. Different ions have different energy gaps, so different colours.
- Mg^{2+} gives no colour because its emissions are not in the visible region.
- Why concentrated HCl: it cleans the wire and turns the sample into the **chloride**, which is more **volatile**, so it gets into the flame better.

P5 Tests for cations: precipitation and ammonium

- **Ammonium, NH_4^+**: add **sodium hydroxide solution and warm**. Hold **damp red litmus** at the mouth of the tube: it turns **blue** (ammonia).
  $NH_4^+(aq) + OH^-(aq) \rightarrow NH_3(g) + H_2O(l)$
- **Group 2 ions** with dilute NaOH(aq) and a soluble sulfate (hydroxide solubility rises down the group; sulfate solubility falls):

Ion	+ NaOH(aq)	+ sulfate solution	+ carbonate solution
Mg^{2+}	white precipitate ($Mg(OH)_2$)	no change ($MgSO_4$ soluble)	white precipitate
Ca^{2+}	slight white precipitate	slight white precipitate	white precipitate
Ba^{2+}	no change ($Ba(OH)_2$ soluble)	dense white precipitate ($BaSO_4$)	white precipitate

- **Li^+ and Sr^{2+} both give red flames.** Add a soluble sulfate (or a carbonate): a white precipitate means Sr^{2+}; no precipitate means Li^+.

P5 Tests for anions

Ion	Test	Positive result	Ionic equation
CO_3^{2-}	add a dilute acid; bubble the gas through limewater	effervescence; limewater turns **milky / cloudy**	$2H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2O(l) + CO_2(g)$
HCO_3^-	same as carbonate	same	$H^+(aq) + HCO_3^-(aq) \rightarrow H_2O(l) + CO_2(g)$
SO_4^{2-}	add **dilute HCl**, then **barium chloride** solution	**white precipitate**	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$
Cl^-	add **dilute nitric acid**, then **silver nitrate** solution	**white** precipitate; dissolves in **dilute** $NH_3(aq)$	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
Br^-	as above	**cream** precipitate; dissolves in **concentrated** $NH_3(aq)$ only	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$
I^-	as above	**pale yellow** precipitate; insoluble in both	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$

- **Why acidify first?** The acid removes carbonate (and sulfite, hydroxide) ions, which would also give white precipitates ($BaCO_3$, Ag_2CO_3).
  - In the sulfate test, don't use sulfuric acid: it adds sulfate ions.
  - In the halide test, use nitric acid, never HCl (adds Cl^-) or H_2SO_4.
- **Dissolve the solid in water first** before adding the reagents.
- **Carbonate or hydrogencarbonate?** Heat the solid. A Group 1 hydrogencarbonate gives off CO_2 and water (condensation at the top of the tube); a Group 1 carbonate (apart from Li_2CO_3) doesn't decompose. $2KHCO_3(s) \rightarrow K_2CO_3(s) + H_2O(g) + CO_2(g)$

Solid halides with concentrated sulfuric acid (fume cupboard):

Halide	Observations	Products
Cl^-	**steamy / misty fumes** only	HCl
Br^-	steamy fumes **and brown / orange fumes**	HBr, Br_2, SO_2
I^-	some steamy fumes, **purple vapour** / black solid, yellow solid, **bad-egg smell**	HI, I_2, S, H_2S

Halogen displacement (shows which halide is present, and the trend in oxidising power):

Halogen	Colour in water	Colour in hexane / cyclohexane (upper layer)
Cl_2	very pale green (almost colourless)	very pale green (almost colourless)
Br_2	yellow-orange	orange / red-brown
I_2	brown (yellow when dilute)	**purple / violet**

- Example: chlorine water added to an iodide solution turns brown; shake with hexane and the **upper organic layer turns purple**. $Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$

P5 Tests for gases

Gas	Test	Positive result
H_2	lighted splint	**squeaky pop**
O_2	**glowing** splint (just blown out)	**relights**
CO_2	bubble through limewater	turns **milky / cloudy** (white precipitate)
NH_3	**damp red** litmus	turns **blue**; also white smoke with HCl fumes (glass rod dipped in concentrated HCl)
HCl	damp blue litmus; ammonia	**steamy fumes** in moist air; litmus turns red; **white smoke** with NH_3
Cl_2	damp blue litmus	turns red then **bleached white**; green-yellow gas
NO_2	look; damp blue litmus	**brown** gas; acidic, turns litmus red
H_2O	anhydrous copper(II) sulfate; or cobalt(II) chloride paper	**white** → **blue**; or **blue** → **pink**

P5 Thermal decomposition of nitrates and carbonates

Heat the solid in a boiling tube. Test any gas: glowing splint (O_2), limewater (CO_2), colour (NO_2 is brown). Do nitrates in a fume cupboard.

Compound	Equation	What you see
Group 1 nitrate (not Li)	$2NaNO_3(s) \rightarrow 2NaNO_2(s) + O_2(g)$	glowing splint relights; no brown gas
Group 2 nitrate (and $LiNO_3$)	$2Mg(NO_3)_2(s) \rightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$	**brown gas**; glowing splint relights
Group 2 carbonate (and Li_2CO_3)	$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$	limewater turns milky
Group 1 carbonate (not Li)	no decomposition at Bunsen temperatures	no change

Thermal stability **increases down both groups**: larger cations with the same charge have a lower charge density, so they **polarise** (distort) the anion's electron cloud less.

MARK-SCHEME LANGUAGE

- Ions need their **charge**: Ba^{2+}, not Ba. "Iodide", not "iodine". If you give both name and formula, both must be right.
- Flame colours: K^+ "lilac" (purple was refused in some years), Ba^{2+} "green / apple green" (not blue-green), Sr^{2+} "red / crimson" (not brick-red), Ca^{2+} "brick-red / orange-red / yellow-red".
- Ammonium test needs **both** the reagent (NaOH) **and** warming, plus damp red litmus turning blue. Putting indicator paper *into* the solution isn't credited.
- O_2 test: the splint must be **glowing**. "Pops" belongs to H_2.
- NH_3 with HCl gives "white smoke", not "steamy fumes".
- Silver halide test: name **dilute nitric acid**, and say **dilute** or **concentrated** ammonia.
- Ionic equations: no spectator ions, with state symbols when asked.

Often asked as: describe a flame test and give the colour for…; identify the ions in A and B from these observations; give the ionic equation (with state symbols); explain why the acid is added first; how could you tell Li^+ from Sr^{2+}; deduce the identity of five solutions from a mixing grid.

COMMON MISTAKE

Brick-red for strontium (it's calcium); purple for iodine in *water* (purple only in hexane, or as vapour).
Adding an extra observation that is wrong, e.g. "effervescence" where there is none. A wrong extra observation can cost the mark.
Mixing up the ammonium-ion test (NaOH + warm) with the ammonia-gas test (damp red litmus).
"Halogens" when you mean halides; "clear" when you mean colourless.

BEYOND THE SPEC, BUT EXAMINED

- Why a **wooden splint** is a poor flame-test tool: it burns with a yellow flame that hides the colour, although it is cheap and needs no cleaning (WCH13 Jun 2025).
- Distinguishing **acid from hydrogencarbonate** in a mixture: add $NaHCO_3(aq)$ for effervescence, or use blue litmus or universal indicator paper, or magnesium ribbon. Phenolphthalein can't do it, because it is colourless in acid and in neutral solution (WCH13 Jun 2024).
- **Ammonium carbonate** on heating gives three gases: $(NH_4)_2CO_3(s) \rightarrow 2NH_3(g) + H_2O(g) + CO_2(g)$. Test each one (WCH13 Jun 2019).

P5 Titration technique

METHOD: AN ACID–BASE TITRATION

1. Rinse the burette with a little of the acid, then fill it (funnel removed, jet filled, no air bubble). Read the initial volume.
2. Rinse a pipette with the solution to be analysed, then use a pipette filler to transfer **25.0 cm^3** into a conical flask.
3. Add **2–3 drops** of indicator. Stand the flask on a **white tile**.
4. **Rough titration**: run in acid while swirling until the colour changes. Record the volume.
5. **Accurate titrations**: run in acid quickly to about **1–2 cm^3 short of the rough end-point**, then add it **drop by drop**, swirling, until the indicator **just** changes colour.
6. Read the burette to the nearest 0.05 cm^3 (bottom of the meniscus, eye level).
7. Repeat until you have **concordant** titres (within 0.20 cm^3 of each other; 0.10 cm^3 is better). Average the concordant titres only.

Indicator	In acid	In alkali	Acid added to alkali (end-point)	Alkali added to acid (end-point)
methyl orange	red	yellow	**yellow → orange**	red → orange
phenolphthalein	colourless	pink	**pink → colourless**	**colourless → (pale) pink**

- **Which indicator?**
  - Strong acid + strong base (HCl + NaOH): either.
  - Strong acid + weak base (HCl + Na_2CO_3 or $NaHCO_3$, or + ammonia): **methyl orange**.
  - Weak acid + strong base (ethanedioic, citric or ethanoic acid + NaOH): **phenolphthalein**.

WORKED EXAMPLE: CONCORDANT TITRES AND THE MEAN

	Rough	1	2	3
Final reading / cm^3	24.60	24.70	48.80	24.35
Initial reading / cm^3	0.00	0.50	24.70	0.20
Titre / cm^3	24.60	24.20	24.10	24.15

Titres 1, 2 and 3 are all within 0.20 cm^3 of each other, so they are concordant. The rough titre is left out.
Mean titre = (24.20 + 24.10 + 24.15) ÷ 3 = **24.15 cm^3**
This mean titre is used in the CP3 worked example.

What happens to the titre if… (acid in the burette, alkali pipetted into the flask)

Mistake	Effect on titre	Why
burette rinsed with water, not acid	**larger**	acid diluted, so more volume is needed
pipette rinsed with water, not alkali	**smaller**	alkali diluted, so fewer moles in the flask
conical flask wet with water	**no change**	same number of moles of alkali
conical flask rinsed with the alkali	**larger**	extra moles of alkali in the flask
last drop blown out of the pipette	**larger**	more than 25.0 cm^3 delivered
air bubble in the jet escapes during the titration	**larger** (recorded)	the reading change includes the bubble
end-point overshot	**larger**	acid added after the colour change
old NaOH solution (absorbed CO_2 or water evaporated)	changes	concentration is no longer the stated value

MARK-SCHEME LANGUAGE

- Concordant: "within 0.20 cm^3 of each other" (0.10 also allowed). "±0.2 of the mean" is not the definition. The rough titre is left out because it is "rough / a trial", not because of "uncertainty".
- Accurate titration: add acid "quickly to just short of the rough end-point" (1–4 cm^3 short), then "drop by drop, swirling, until the colour just changes". "Until the rough value is reached" is not credited (WCH13 Oct 2023).
- Methyl orange end-point: "yellow to orange" (not red or pink). Phenolphthalein: "pink to colourless" or "colourless to pink" depending on which solution is in the burette.
- Standard solution: "a solution of **accurately known** concentration". "Constant" or "fixed" concentration is not credited (WCH13 Jan 2025).
- Reason for a rough titration: "to find the approximate end-point" so the accurate runs can be done dropwise near it.

Often asked as: complete the titration table; which titres are concordant; calculate the mean titre; state the colour change at the end-point; why was the burette rinsed with acid; what is the effect on the titre of…; why was the solution diluted before titrating (otherwise the titre would be bigger than the burette).

COMMON MISTAKE

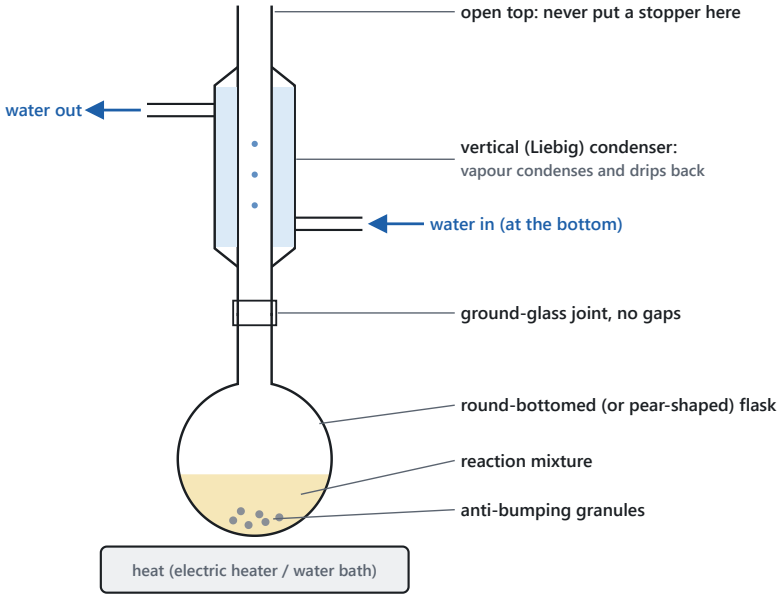
Including the rough titre, or a non-concordant one, in the mean.
Giving the indicator colours the wrong way round for the order of addition.
Writing "concordant" when you mean "anomalous".

BEYOND THE SPEC, BUT EXAMINED

- **Crystals from a titration**: repeat the titration without indicator, evaporate to the point of crystallisation (not to dryness), leave to cool and crystallise, then dry the crystals between filter papers (WCH13 Oct 2020).
- A **concentrated solution** is diluted by a factor of 10 (25.0 cm^3 made up to 250.0 cm^3) so that the titre fits in one burette (WCH13 Jan 2023).

P5 Organic techniques: preparing and purifying a liquid

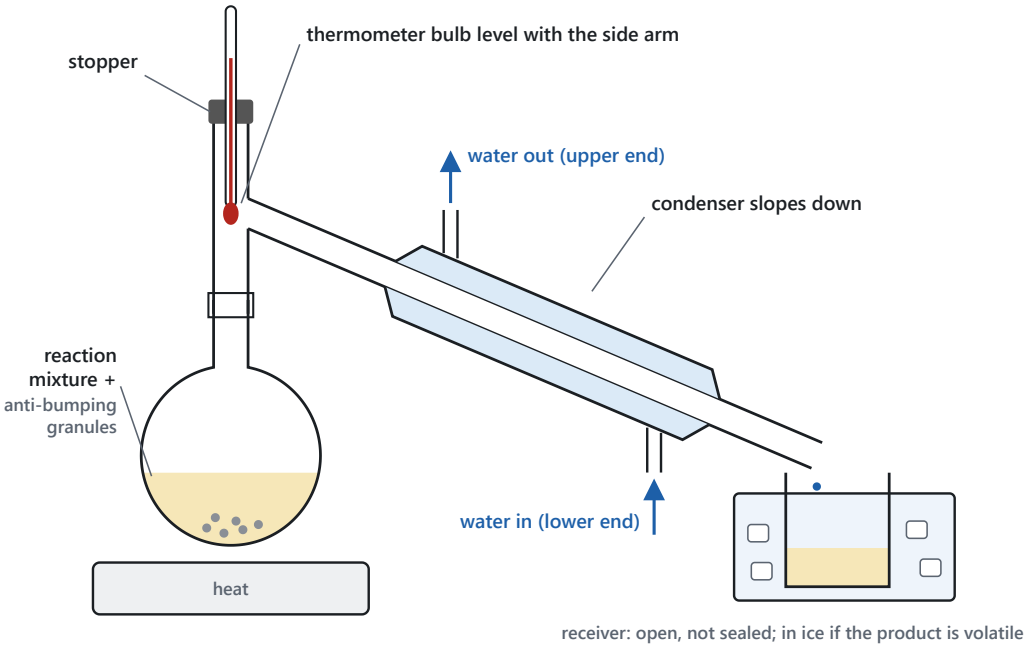
Heating under reflux



Heating under reflux. The condenser is vertical and open at the top. Water enters at the bottom.

- **What it does**: heats a reaction mixture for a long time without losing volatile reactants or products. Vapour rises, condenses in the vertical condenser and **drips back** into the flask.
- **When**: slow reactions, and whenever you want the reaction to go to **completion** (e.g. a primary alcohol to a carboxylic acid: the aldehyde formed first is returned to the oxidising agent).
- Never stopper the top.

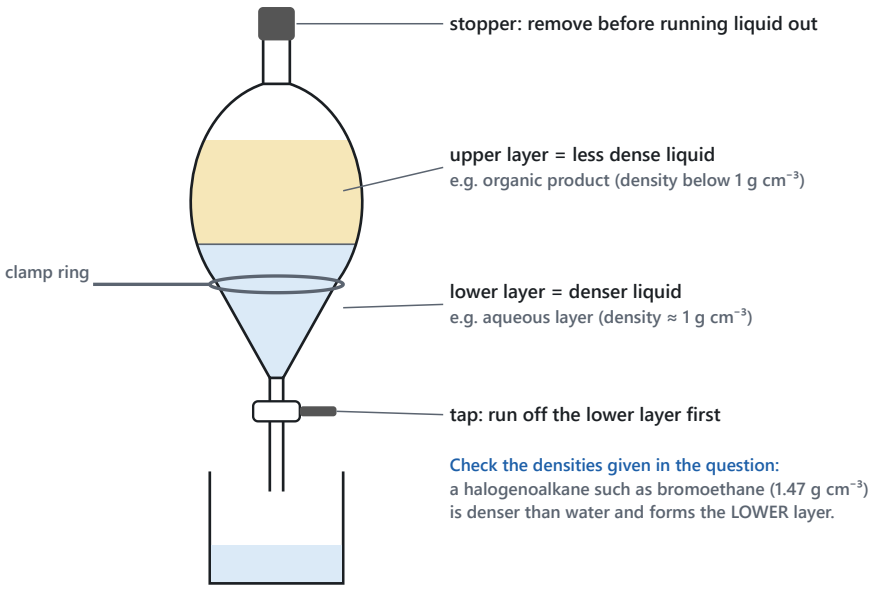
Simple distillation



Simple distillation. The thermometer bulb is level with the side arm, so it reads the temperature of the vapour leaving. Water enters the condenser at the lower end.

- **What it does**: separates a liquid from a mixture by boiling point. The vapour passes into a sloping condenser and is collected.
- Uses: to remove a product **as it forms** (the aldehyde in CP7); to **purify** a crude product, collecting only the fraction that boils within about **2 °C of the product's boiling temperature**.
- **Fractional distillation** adds a fractionating column. It separates liquids whose boiling temperatures are **close together** (e.g. cyclohexene from cyclohexanol and water).
- **Steam distillation**: steam carries over an oil that doesn't mix with water and boils above 100 °C (e.g. extracting oils from plants).

Separating funnel: extraction and washing



Two immiscible liquids in a separating funnel. The denser liquid is the lower layer.

1. Pour the mixture into the funnel (tap closed). Add the washing liquid, e.g. **sodium hydrogencarbonate solution** to remove acid.
2. Stopper, **invert**, shake gently, and **open the tap regularly to release pressure** (CO_2 forms when hydrogencarbonate meets acid). Point the tip away from people.
3. Stand the funnel in a clamp ring, **remove the stopper**, and let the layers **settle**.
4. **Run off the lower layer through the tap.** Keep whichever layer holds your product. Decide which layer is which from the **densities** given.
5. A final wash with water removes water-soluble (ionic) impurities.

Drying

- Run the organic layer into a dry conical flask and add a **drying agent**: an **anhydrous** salt such as anhydrous **calcium chloride, sodium sulfate or magnesium sulfate**.
- **Swirl** and leave until the liquid turns from **cloudy to clear** (add more if the solid all clumps together).
- **Decant** or filter off the liquid.
- Don't use anhydrous copper(II) sulfate, NaOH, KOH or concentrated H_2SO_4. Don't dry a volatile product in a warm oven, because it would evaporate.

Checking purity

- **Liquid**: redistil it, or measure its **boiling temperature**. A pure liquid boils at a fixed temperature close to the data-book value.

- **Solid**: measure its **melting temperature** (sample in a sealed capillary tube, heated slowly in a melting-point apparatus). A pure solid melts sharply, over a narrow range (about 1–2 °C), at the data-book value. Impurities **lower** the melting temperature and make it melt over a **wider range**.

Typical purification sequence for a liquid organic product (e.g. a halogenoalkane):
1. separate the organic layer from the aqueous layer (separating funnel);
2. wash with $NaHCO_3$(aq) to remove acid, releasing the pressure;
3. wash with water; separate again;
4. dry with an anhydrous salt; decant;
5. redistil, collecting the fraction at the product's boiling temperature.

P5 Tests for organic functional groups

Group	Test	Positive result
C=C (alkene)	shake with **bromine water**	**orange → colourless**
C=C (alkene)	acidified potassium manganate(VII)	purple → colourless
–OH (alcohol or acid)	add **phosphorus(V) chloride, PCl_5** (dry)	**steamy (misty) fumes** of HCl; turn damp blue litmus red, white smoke with NH_3
–OH	add sodium	effervescence (H_2)
–COOH (carboxylic acid)	add **sodium carbonate or sodium hydrogencarbonate**	**effervescence**; gas turns limewater milky
–CHO (aldehyde)	warm with **Fehling's or Benedict's** solution	**blue solution → red (brick-red) precipitate**
–CHO (aldehyde)	warm with Tollens' reagent	silver mirror
primary or secondary alcohol, or aldehyde	warm with **acidified potassium dichromate(VI)**	**orange → green**
tertiary alcohol	warm with acidified potassium dichromate(VI)	stays orange
halogenoalkane	warm with NaOH(aq), **acidify with dilute HNO_3**, add $AgNO_3$(aq)	white / cream / yellow precipitate (Cl / Br / I)

- **PCl_5 doesn't tell an alcohol from an acid**: both give fumes. Use $NaHCO_3$ to pick out the acid.
- **Ketones** give no change with Fehling's, Benedict's or Tollens'.

P5 Infrared and mass spectra as a final check

- **IR**: look up the absorption ranges in the data booklet and quote the bond **and** the range.
  - O–H in an alcohol: 3750–3200 cm^{-1} (broad).
  - O–H in a carboxylic acid: 3300–2500 cm^{-1} (very broad).
  - C=O: about 1740–1680 cm^{-1}.
  - C=C: 1669–1645 cm^{-1}.
  - C–H: about 2962–2853 cm^{-1}.
- An alcohol turning into a chloroalkane: the broad O–H absorption **disappears** (WCH13 Oct 2025).
- **Mass spectrum**: the **molecular ion** peak gives Mr. Fragment ions are **positive**, e.g. CH_2OH^+ at m/z = 31 points to a primary alcohol; CH_3CO^+ at m/z = 43. Isomers have the same molecular ion but different fragments.

MARK-SCHEME LANGUAGE

- Bromine water: "orange / yellow / brown to colourless". "Clear" and "red" are not accepted.
- Fehling's / Benedict's: give the **start** colour (blue) and say **precipitate** (red / brick-red). "Red solution" isn't enough.
- PCl_5: "steamy / misty fumes". "White smoke" and "PCl_3" are not accepted.
- Acidified dichromate: "orange to green". Use **sulfuric** acid. HCl, HNO_3, potassium chromate(VI) and manganate(VII) were rejected as the oxidant in these questions.
- Separating funnel: "invert and open the tap / release the pressure"; "allow the layers to separate"; "run off the lower layer through the tap". Pouring or pipetting off the top layer is not credited (WCH13 Oct 2019).
- Drying: name an **anhydrous** salt, "swirl", "until the liquid is clear", "decant / filter".
- Distillation diagram (3 marks): heated flask with the thermometer bulb **at the side arm**; condenser sloping down with water **in at the lower end**; no gaps, and a receiver that is **not sealed**.
- Reflux diagram: round-bottomed or pear-shaped flask with heat; **vertical** condenser with water in at the bottom; **open** at the top.

Often asked as: draw / complete / find the errors in a distillation or reflux diagram; explain why the mixture is heated under reflux; which layer is the organic layer; why is the funnel tap opened; why is sodium hydrogencarbonate used; name a drying agent and say how you know it's dry; give a test for the functional group and its result; which IR absorption shows…

COMMON MISTAKE

A stopper on top of the reflux condenser, or a sealed receiver: never heat a closed system.
Condenser water going in at the top.
Thermometer bulb dipping into the liquid during distillation (it must read the vapour at the side arm).
Assuming the organic layer is always on top. Halogenoalkanes such as bromoethane (1.47 g cm^{-3}) are **denser** than water.

BEYOND THE SPEC, BUT EXAMINED

- **Choosing a drying agent from a list**: $MgSO_4$ is right because it is anhydrous and doesn't react. The others are hydrated, flammable (ethanol), basic ($Ca(OH)_2$) or a reactive acid (H_2SO_4) (WCH13 Jun 2022).
- Using $NaHCO_3$ rather than NaOH to remove acid: NaOH could **hydrolyse the halogenoalkane** product (WCH13 Jan 2025).
- **Steam distillation** of geraniol (insoluble in water, boils above 100 °C): collect water + oil, separate in a separating funnel, then dry the oil (WCH13 Oct 2021).

Core practicals

SPEC CP1 Measuring the molar volume of a gas

WHAT THIS SPEC POINT ASKS

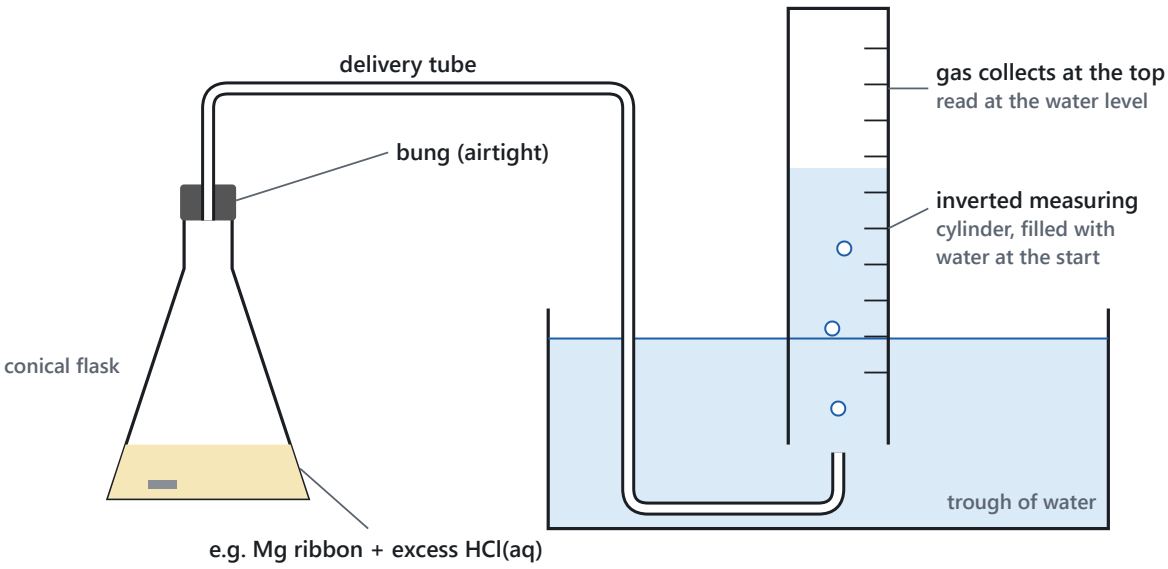
Unit 1 core practical (1.11): measure the volume of gas produced from a known amount of a reactant, and use it to find the volume occupied by one mole of gas.

CP1 The idea

- React a **known mass** of a solid that is the **limiting reagent** with an **excess** of acid (or water), collect all the gas, and measure its volume.
- Moles of gas come from the equation. Then:

$$\text{molar volume} = \text{volume of gas} \div \text{moles of gas}$$

- The expected value is about **24.0 dm^3 mol^{-1}** (24 000 cm^3 mol^{-1}) at room temperature and pressure.
- Common reactions:
  - $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
  - $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$
  - $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$



Collecting gas over water. A gas syringe connected to the delivery tube is the alternative.

CORE PRACTICAL 1: METHOD IN BRIEF

1. Clean a strip of magnesium ribbon with emery paper (removes the oxide coating). Weigh about 0.04–0.05 g of it on a 3 d.p. balance.
2. Measure 50 cm^3 of 1.0 mol dm^{-3} hydrochloric acid (an **excess**) into a conical flask.
3. Fill a 100 cm^3 measuring cylinder with water and turn it upside down in a trough of water (or connect a 100 cm^3 gas syringe). Put the delivery tube under its mouth.
4. Drop the magnesium into the acid and **immediately** put the bung in.
5. When no more bubbles form, read the gas volume.

6. Repeat with different masses. Plot volume against mass for a line through the origin.

CP1 Variables and safety

- **IV**: mass of solid (if you plot a graph). **DV**: volume of gas.
- **Control**: volume and concentration of acid (always in excess), temperature and pressure (room), same apparatus.
- **Safety**: hydrogen is flammable, so no flames nearby. Dilute HCl is an irritant. Calcium reacts with water to give an alkaline (irritant) solution.

CP1 Processing the results

WORKED EXAMPLE: MOLAR VOLUME FROM MAGNESIUM

0.0480 g of Mg gives 46.5 cm^3 of hydrogen.
1. $n(Mg) = 0.0480 \div 24.3 = 1.975 \times 10^{-3}$ mol
2. $Mg : H_2 = 1 : 1$, so $n(H_2) = 1.975 \times 10^{-3}$ mol
3. molar volume $= 46.5 \div 1.975 \times 10^{-3} = 23\,500\ cm^3\ mol^{-1} =$ **$23.5\ dm^3\ mol^{-1}$**

Check the acid is in excess: $n(HCl) = 1.0 \times 50 \div 1000 = 0.050$ mol. Only $2 \times 1.975 \times 10^{-3} = 0.00395$ mol is needed. ✓

Mass uncertainty: one reading on a 3 d.p. balance (±0.0005 g) gives $0.0005 \div 0.0480 \times 100 =$ **1.0%**. On a 2 d.p. balance (±0.005 g) it would be about 10%, which is far too big.

WORKED EXAMPLE: IDENTIFY A GROUP 2 CARBONATE FROM THE GAS VOLUME

0.250 g of MCO_3 gives 70.0 cm^3 of CO_2 with acid. Molar volume = 24.0 $dm^3\ mol^{-1}$.
1. $n(CO_2) = 70.0 \div 24\,000 = 2.917 \times 10^{-3}$ mol $= n(MCO_3)$
2. $Mr(MCO_3) = 0.250 \div 2.917 \times 10^{-3} = 85.7$
3. $Ar(M) = 85.7 - 60.0 = 25.7 \rightarrow$ nearest Group 2 metal is **magnesium** (24.3)

- **Graph method**: plot volume of CO_2 against mass of $CaCO_3$. Draw a best-fit straight line **through the origin**. Gradient = volume per gram; multiply by $M(CaCO_3) = 100.1$ to get the molar volume. (Use M of the **carbonate**, not of CO_2.)

CP1 Sources of error and improvements

Error	Effect on molar volume	Improvement
gas escapes before the bung goes in	too low	put the solid in a small tube inside the flask, seal, then tip it over; or use a weaker acid (e.g. ethanoic acid) so the reaction starts more slowly
CO_2 dissolves in the water	too low	use a gas syringe
oxide coating on Mg (fewer moles of Mg react)	too low	clean with emery paper
very small mass → large % uncertainty	random scatter	use a 3 d.p. balance; use more solid if the syringe or cylinder can hold the gas
room temperature and pressure aren't those of the 24.0 value	small difference	measure T and p; use pV = nRT
syringe plunger sticks	too low	twist the plunger gently

MARK-SCHEME LANGUAGE

- Weigh **by difference** ("weighing bottle before and after") to allow for solid left behind (WCH13 Oct 2023).
- Ethanoic acid instead of HCl: "ethanoic acid is weak, so the reaction is slower" and "less gas escapes before the bung is replaced".
- Gas syringe over collecting over water: no CO_2 dissolves; finer graduations. The acid can be tipped onto the solid after sealing.
- Too-low result: "gas escaped before the bung was replaced", "gas dissolved in the water", "the metal had an oxide coating".
- Molar volume units: $dm^3\ mol^{-1}$ or $cm^3\ mol^{-1}$.
- Using a larger mass: "would reduce the percentage uncertainty" but "would produce more gas than the measuring cylinder / burette can hold" (WCH13 Jan 2021).

Often asked as: draw the gas-collection apparatus (boiling tube or flask, bung, delivery tube into an inverted water-filled cylinder in a trough); calculate the molar volume; identify the metal; explain why the value is lower than 24.0 $dm^3\ mol^{-1}$; plot volume against mass and use the gradient.

COMMON MISTAKE

Using the Mr of CO_2 (44.0) instead of the Mr of the carbonate when working out moles from the mass of solid.
Giving molar volume in mol dm^{-3}.
Forgetting to subtract 60.0 for CO_3 when finding Ar of the metal, or picking a metal from the wrong group.

BEYOND THE SPEC, BUT EXAMINED

- **Calcium and water**: you see bubbles, the calcium disappears, and the solution goes cloudy (white $Ca(OH)_2$, only slightly soluble). Not "coloured products" (WCH13 Jun 2023).
- A result **above** 24.0 dm^3 mol^{-1} (e.g. 28.8 dm^3 mol^{-1} from Mg in an inverted burette) can't be blamed on an MgO impurity, which would make the result *lower* (WCH13 Jan 2025).
- Heating a Group 2 carbonate and collecting the CO_2: take the delivery tube out of the water before stopping heating, to prevent suck-back (WCH13 Jan 2021).

SPEC CP2

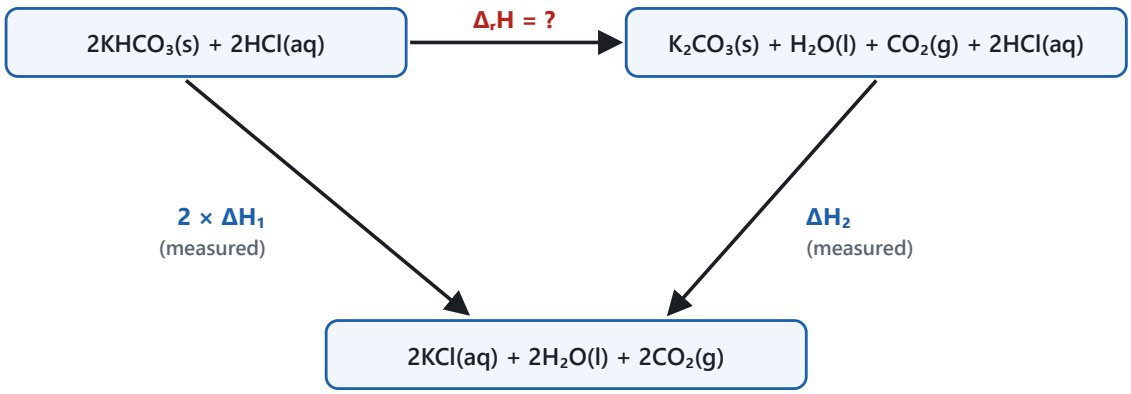
Enthalpy change of a reaction using Hess's law

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (6.7): measure the enthalpy changes of two reactions you can do in a polystyrene cup, then combine them in a Hess cycle to get the enthalpy change of a reaction you can't measure directly. Evaluate the results, including heat loss, assumptions and cooling corrections (6.8).

CP2 The idea

- Some enthalpy changes can't be measured directly. Thermal decomposition needs strong heating and you can't measure the temperature change; hydrating anhydrous $CuSO_4$ exactly can't be controlled.
- Instead, react **each** substance with an excess of acid in an insulated cup. These reactions are quick and happen in solution, so ΔT can be measured. Hess's law then links them.
- The usual school version: the decomposition of potassium hydrogencarbonate.
  $2KHCO_3(s) \rightarrow K_2CO_3(s) + H_2O(l) + CO_2(g)$



Route 1 = Route 2: $2\Delta H_1 = \Delta_r H + \Delta H_2$, so $\Delta_r H = 2\Delta H_1 - \Delta H_2$

Both solids react with hydrochloric acid to give the same products, so the two routes can be compared.

CORE PRACTICAL 2: METHOD IN BRIEF

1. Measure 30.0 cm^3 of 2.0 mol dm^{-3} HCl (an excess) into a **polystyrene cup** standing in a beaker.
2. Weigh about 3.0 g of $KHCO_3$ in a weighing bottle.
3. Record the temperature of the acid every 30 s for a few minutes to get a steady starting value.
4. Tip in the solid at a recorded time, **stir**, and put the **lid** on. Keep reading the temperature every 30 s for several more minutes.
5. Reweigh the bottle (mass added = by difference).
6. Repeat the whole thing with about 3.0 g of K_2CO_3.

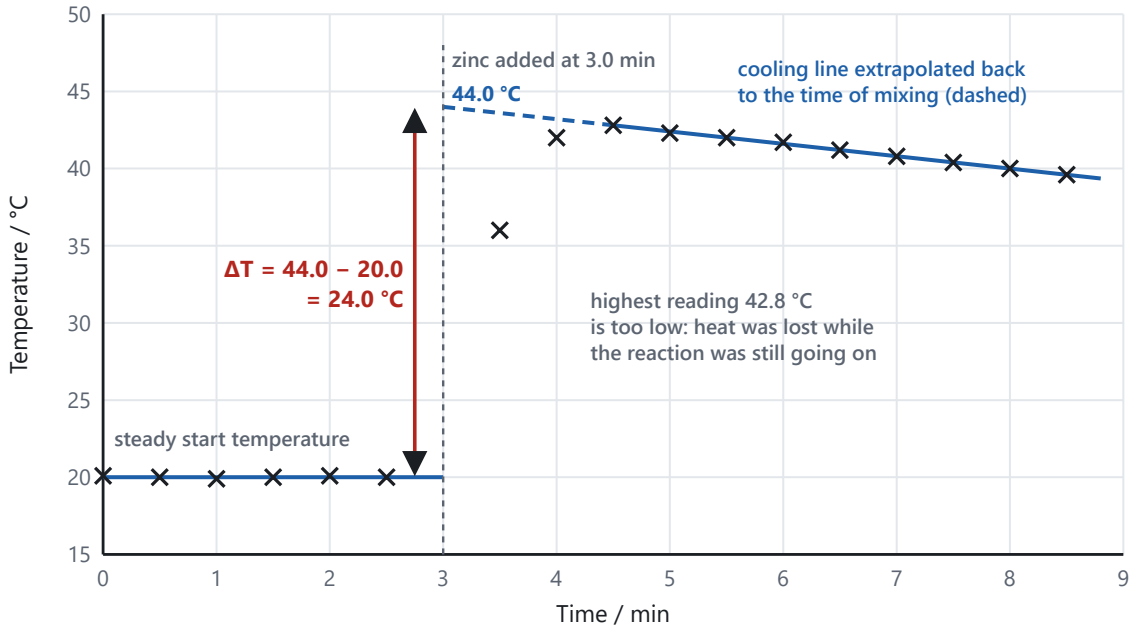
CP2 The calorimetry equations

$$Q = m \times c \times \Delta T$$

$$\Delta H = -Q \div n$$

- **Q** = energy transferred (J). **m** = mass of **solution** in g (volume in cm^3, taking the density as 1.00 g cm^{-3}; ignore the mass of the solid). **c** = 4.18 J g^{-1} K^{-1}. **ΔT** = temperature change (°C = K for a change).
- **n** = moles of the **limiting** reagent (the one not in excess).
- **Sign**: temperature **rises** → exothermic → ΔH **negative**. Temperature **falls** → endothermic → ΔH **positive**.
- Divide J by 1000 to get kJ.

CP2 Correcting for heat loss: the temperature–time graph



Excess zinc added to 50.0 cm^3 of 0.500 mol dm^{-3} $CuSO_4(aq)$ at 3.0 min. Extrapolating the cooling line back to the time of mixing gives the temperature the mixture would have reached with no heat loss.

- The reaction isn't instantaneous, and the mixture loses heat while it is still reacting. So the highest reading is **lower** than the true maximum.
- Draw a best-fit line through the steady readings before mixing, and another through the cooling readings after the peak. **Extrapolate the cooling line back to the time of mixing**. ΔT is the vertical gap between the two lines at that time.
- No reading is taken at the moment of mixing: the solid is being added, so the temperature is neither that of the starting solution nor that of the products.
- For an **endothermic** reaction the graph is upside down: extrapolate the warming line back to find the lowest temperature.

WORKED EXAMPLE: ΔH FROM THE GRAPH

50.0 cm^3 of 0.500 mol dm^{-3} $CuSO_4$ with excess zinc (3.00 g). From the graph ΔT = 24.0 °C.
$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$
1. Zinc is in excess: n(Zn) = 3.00 ÷ 65.4 = 0.0459 mol; $n(CuSO_4)$ = 0.500 × 50.0 ÷ 1000 = 0.0250 mol, the limiting reagent.
2. Q = 50.0 × 4.18 × 24.0 = 5016 J = 5.016 kJ (m = 50.0 g of solution, not the zinc)
3. ΔH = −5.016 ÷ 0.0250 = **−201 kJ mol^{-1}** (sign, unit, 3 s.f.)
Using the highest reading instead (42.8 − 20.0 = 22.8 °C) would give a value that is even less exothermic.

WORKED EXAMPLE: THE HESS'S LAW RESULT

Reaction 1: 3.00 g $KHCO_3$ (M = 100.1) in 30.0 cm^3 acid; temperature **falls** by 7.0 °C.
n = 3.00 ÷ 100.1 = 0.02997 mol; Q = 30.0 × 4.18 × 7.0 = 877.8 J
ΔH_1 = +0.8778 ÷ 0.02997 = **+29.3 kJ mol^{-1}** (endothermic)
Reaction 2: 3.00 g K_2CO_3 (M = 138.2) in 30.0 cm^3 acid; temperature **rises** by 5.5 °C.
n = 3.00 ÷ 138.2 = 0.02171 mol; Q = 30.0 × 4.18 × 5.5 = 689.7 J
ΔH_2 = −0.6897 ÷ 0.02171 = **−31.8 kJ mol^{-1}** (exothermic)
Cycle: $\Delta_r H = 2\Delta H_1 - \Delta H_2 = 2(+29.3) - (-31.8)$ = **+90.4 kJ mol^{-1}**
The value calculated from standard enthalpies of formation is about +96 kJ mol^{-1}, so this result is about 6% low.
Note the small ΔT values: (2 × 0.1) ÷ 5.5 × 100 = 3.6% uncertainty for reaction 2. More solid, or less acid, would give larger temperature changes.

CP2 Combustion of an alcohol (spirit burner)

MEASURING AN ENTHALPY CHANGE OF COMBUSTION

1. Measure 100 cm^3 of water into a **copper can** (a calorimeter) clamped above a spirit burner. Record the water's temperature.
2. Weigh the spirit burner **with its cap on** (stops the alcohol evaporating).
3. Light it, with a **draught shield** round it, and stir the water. Heat until the temperature has risen by about 20–30 °C.
4. Put out the flame, record the **highest temperature**, put the cap back on and **reweigh** the burner straight away.

WORKED EXAMPLE: ΔCH OF ETHANOL

100 g of water rises by 25.0 °C when 0.520 g of ethanol (M = 46.0) burns.
Q = 100 × 4.18 × 25.0 = 10 450 J = 10.45 kJ
n = 0.520 ÷ 46.0 = 0.01130 mol
ΔcH = −10.45 ÷ 0.01130 = **−924 kJ mol^{-1}**
Here m is the mass of **water**, not the mass of ethanol. The data-book value is −1367 kJ mol^{-1}. For the percentage error and what it shows, see P4.
Uncertainty: ΔT (2 × 0.5) ÷ 25.0 × 100 = 4.0%; mass (2 × 0.005) ÷ 0.520 × 100 = 1.9%; total about **6%**.

CP2 Errors, assumptions and improvements

- **Solution reactions** come out **less exothermic** than data-book values mainly because of **heat loss**. Fixes: lid, insulation, extrapolation. The **assumptions** also add small errors:
  - density of the solution = 1.00 g cm^{-3};

  - c of the solution = 4.18 J g^{-1} K^{-1};
  - the mass of solid is ignored;
  - the cup absorbs no heat;
  - the reaction is complete.
- **Combustion** comes out far less exothermic:
  - heat lost to the air and used to warm the can and the clamp;
  - **incomplete combustion** (soot on the can);
  - alcohol evaporates from the wick;
  - a delay before putting the flame out or reweighing.
- **Combustion fixes**: draught shield, flame close to a copper can, lid on the burner, stirring. A sealed apparatus with an oxygen supply and a copper coil gives complete combustion and better heat transfer.
- Results aren't **standard** values, because the conditions and concentrations aren't standard.

MARK-SCHEME LANGUAGE

- Excess reagent: "to make sure all the HCl reacts / is used up". "So the reaction goes to completion" was not credited (WCH13 Oct 2020).
- Stirring (dissolving): "so the solid dissolves" and "so the temperature is uniform".
- Readings before mixing: "to find a steady / average starting temperature".
- Extrapolation: the best-fit line "allows for cooling while the reaction was happening", because the reaction is not instantaneous.
- Answer: sign + value + kJ mol^{-1}, 2 or 3 s.f. (e.g. "−201 kJ mol^{-1}").
- Glass instead of polystyrene: "less exothermic" and "more heat lost / glass conducts heat".
- Combustion measurements: "mass of burner before and after", "volume / mass of water", "initial and final temperature".

Often asked as: plot the temperature–time graph and extrapolate to find ΔT; calculate ΔH with sign and units; show which reagent is in excess; construct the Hess cycle and calculate ΔH; give two assumptions; explain why the value is less exothermic than the data book; suggest improvements.

COMMON MISTAKE

Using the mass of the solid in Q = mcΔT.
Taking ΔT as highest minus lowest reading instead of extrapolating to the time of mixing.
Leaving out the sign, or writing "+" for an exothermic reaction.
Saying the value is "smaller" when you mean **less exothermic** (less negative).
Saying a lid **prevents** heat loss; it **reduces** it.

BEYOND THE SPEC, BUT EXAMINED

- **Thermometric titration**: add acid to alkali in portions and record the temperature. Draw two best-fit lines (rising, then falling); where they cross is the end-point volume. This gives the concentration and ΔneutH. The total volume of both solutions is the mass in Q = mcΔT (WCH13 Oct 2024).
- **Hydration of $CuSO_4$** by Hess's law: dissolve anhydrous and hydrated $CuSO_4$ separately. ΔhydH = ΔH(anhydrous) − ΔH(hydrated), e.g. −39.0 − (+5.6) = −44.6 kJ mol^{-1} (WCH13 Oct 2021).
- **Endothermic dissolving** (NH_4Cl): extrapolate the warming line back to the time of mixing to find the lowest temperature; ΔH is positive (WCH13 Jan 2022).
- **H_2SO_4 + NaOH**: divide by the moles of NaOH (or of water formed), because 2NaOH react with 1H_2SO_4 (WCH13 Jun 2024).

SPEC CP3 Finding the concentration of a solution of hydrochloric acid

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (8.21): titrate hydrochloric acid against a standard alkali or carbonate to find its concentration, and minimise the measurement uncertainty (8.22).

CP3 The idea

- Pipette a known volume of a **standard solution** (accurately known concentration), e.g. sodium carbonate or sodium hydroxide, into a flask. Titrate it with the HCl from a burette.
- Use the mole ratio in the equation to get moles of HCl, then c = n ÷ V.
- With **sodium carbonate** use **methyl orange** (weak base): end-point **yellow** → **orange**. With **NaOH**, either indicator works.

CORE PRACTICAL 3: METHOD IN BRIEF

1. Rinse and fill a burette with the hydrochloric acid.
2. Pipette 25.0 cm^3 of standard 0.0500 mol dm^{-3} Na_2CO_3 into a conical flask; add 2–3 drops of methyl orange.
3. Do a rough titration, then accurate titrations: add acid dropwise near the end-point, swirling, until the solution **just** turns orange.
4. Repeat until concordant (within 0.20 cm^3) and calculate the mean of the concordant titres.
   (Full technique and the reason for each step: P5, titration technique.)

CP3 Variables, safety and uncertainty

- Not a variables experiment. "Control" means technique: same pipette, same indicator volume, same end-point colour.
- **Safety**: dilute HCl and Na_2CO_3 are irritants; wear eye protection. Fill the burette below eye level, using a funnel, then remove the funnel.
- **Minimising uncertainty** (8.22):
  - Choose concentrations that give a titre of about 20–25 cm^3 (a bigger titre has a smaller % uncertainty).
  - Use a pipette and a volumetric flask rather than measuring cylinders.
  - Read to ±0.05 cm^3 at eye level.
  - Use concordant titres.
  - The overall uncertainty is about the sum of the pipette, burette (and flask, balance) percentage uncertainties.

WORKED EXAMPLE: CONCENTRATION OF HCL

25.0 cm^3 of 0.0500 mol dm^{-3} Na_2CO_3 needs a mean titre of 24.15 cm^3 of HCl (the titration table is in P5).

$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

1. $n(Na_2CO_3) = 0.0500 \times 25.0 \div 1000 = 1.25 \times 10^{-3}$ mol
2. $n(HCl) = 2 \times 1.25 \times 10^{-3} = 2.50 \times 10^{-3}$ mol
3. $c(HCl) = 2.50 \times 10^{-3} \div (24.15 \div 1000) =$ **0.104 mol dm^{-3}**
4. In g dm^{-3}: $0.1035 \times 36.5 =$ **3.78 g dm^{-3}**

Titre uncertainty: (2 × 0.05) ÷ 24.15 × 100 = 0.41%.

WORKED EXAMPLE: WATER OF CRYSTALLISATION

3.58 g of $Na_2CO_3 \cdot nH_2O$ is made up to 250.0 cm^3. 25.0 cm^3 portions need 25.00 cm^3 of 0.100 mol dm^{-3} HCl.

1. $n(HCl) = 0.100 \times 25.00 \div 1000 = 2.50 \times 10^{-3}$ mol
2. $n(Na_2CO_3)$ in 25.0 $cm^3 = 1.25 \times 10^{-3}$ mol; in 250.0 $cm^3 = 1.25 \times 10^{-2}$ mol
3. $Mr(Na_2CO_3 \cdot nH_2O) = 3.58 \div 1.25 \times 10^{-2} = 286.4$
4. 286.4 − 106.0 = 180.4 = mass of water per mole; n = 180.4 ÷ 18.0 = **10**

MARK-SCHEME LANGUAGE

- Procedure: "pipette 25.0 cm^3 of the standard solution into a conical flask", "add a few drops of methyl orange", "run in acid until just before the rough end-point", "add dropwise, swirling, until the colour just changes", "repeat until concordant".
- Methyl orange: "yellow to orange". Red or pink is not accepted.
- Ratio: Na_2CO_3 : HCl = 1 : 2; $Ba(OH)_2$ or $Ca(OH)_2$: HCl = 1 : 2; NaOH : HCl = 1 : 1.
- Purity and formula questions: scale up from 25.0 to 250.0 cm^3 (× 10) before finding the mass.

Often asked as: complete and process the titration table; state the colour change; calculate the concentration of the acid; calculate the purity of the base, the Ar of M in M_2CO_3, or the value of n; calculate the percentage uncertainty in the titre.

COMMON MISTAKE

Using 1 : 1 for Na_2CO_3 and HCl.
Missing the × 10 scale-up from 25.0 cm^3 to 250.0 cm^3.
Averaging all the titres, including the rough one.

BEYOND THE SPEC, BUT EXAMINED

- **Back-titration** of an antacid: excess H_2SO_4 is added, and the acid left over is titrated with NaOH. Use the NaOH concentration, not the acid's, in the step for moles of NaOH (WCH13 Oct 2022).
- **Saturated limewater or $Ba(OH)_2$** is already dilute, so it is titrated directly with dilute HCl and methyl orange (WCH13 Jun 2023).

SPEC CP4

Making a standard solution and standardising sodium hydroxide

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (8.23): make up a standard solution of a solid acid in a volumetric flask, then titrate it to find the concentration of a sodium hydroxide solution.

CP4 The idea

- NaOH can't be weighed accurately (it absorbs water and CO_2 from the air), so its solution has to be **standardised** against a **primary standard**: a pure, stable solid acid such as **ethanedioic acid dihydrate**, $H_2C_2O_4 \cdot 2H_2O$ (M = 126.0).
- A **standard solution** is a solution of **accurately known** concentration.
- Weak acid + strong base → **phenolphthalein**.

CORE PRACTICAL 4: METHOD IN BRIEF

Making 250.0 cm^3 of the standard solution

1. Calculate the mass needed. For 0.0500 mol dm^{-3}: n = 0.0500 × 0.2500 = 0.0125 mol; mass = 0.0125 × 126.0 = **1.58 g**.
2. Weigh the solid in a weighing bottle. Tip it into a beaker and **reweigh the bottle** (mass by difference).
3. Dissolve in a small volume (less than 250 cm^3) of **distilled / deionised water**, stirring.
4. Pour through a funnel into a **250.0 cm^3 volumetric flask**. **Rinse the beaker, rod and funnel** with distilled water and add the washings to the flask.
5. Add distilled water until the **bottom of the meniscus is on the mark** (dropwise near the end, eye level).
6. Stopper and **invert several times** to mix.

Standardising the NaOH

7. Pipette 25.0 cm^3 of the acid into a conical flask; add phenolphthalein.
8. Titrate with NaOH from the burette until the solution turns **(pale) pink** that stays. Repeat until concordant.

CP4 Safety

- Ethanedioic acid is **harmful / toxic** if swallowed and an irritant: gloves, no dust.
- NaOH solution is an **irritant** (corrosive when concentrated): eye protection, burette below eye level.

WORKED EXAMPLE: FROM THE WEIGHING TO THE NAOH CONCENTRATION

Weighing: bottle + acid 12.84 g; bottle after tipping 11.27 g → mass = **1.57 g**.
c(acid) = (1.57 ÷ 126.0) ÷ 0.2500 = **0.0498 mol dm^{-3}**
Titration: 25.0 cm^3 of the acid needs a mean titre of 23.80 cm^3 of NaOH.
$H_2C_2O_4(aq) + 2NaOH(aq) \rightarrow Na_2C_2O_4(aq) + 2H_2O(l)$
1. n(acid) = 0.04984 × 25.0 ÷ 1000 = 1.246 × 10^{-3} mol
2. n(NaOH) = 2 × 1.246 × 10^{-3} = 2.492 × 10^{-3} mol
3. c(NaOH) = 2.492 × 10^{-3} ÷ (23.80 ÷ 1000) = **0.105 mol dm^{-3}**

Overall uncertainty about 1.3% (worked out in P4), so c(NaOH) = 0.105 ± 0.001 mol dm^{-3}.

CP4 Sources of error

- Solid left in the beaker or funnel (not rinsed in) → concentration too low.
- Overshooting the mark on the flask: the solution is too dilute. Start again; you can't remove water.
- Not inverting → the solution isn't uniform, so the titres aren't concordant.
- A hydrated acid that has lost some water of crystallisation → more moles per gram than calculated.
- Old NaOH: water evaporates (more concentrated) or it absorbs CO_2 (becomes partly carbonate), so the end-point is less sharp.

MARK-SCHEME LANGUAGE

- Standard solution, 4 marks (WCH13 Oct 2019):
  - M1 "dissolve in distilled / deionised water in a beaker";
  - M2 "use a volumetric flask";
  - M3 "add the washings and make up to the mark with distilled water";
  - M4 "invert / shake to mix" (only if M3 was given).
- In the 6-mark version, the first two marks are the M and mass calculation (M = 126, mass = 0.630 g for 100 cm^3 of 0.0500 mol dm^{-3}) (WCH13 Jan 2025).
- Washings: "so all the solid is transferred". Inverting: "so the concentration is the same throughout".
- Phenolphthalein: "colourless to (pale) pink" (NaOH from the burette) or "pink to colourless" (acid from the burette). Red or purple is not accepted.

Often asked as: describe how to make the standard solution; calculate the mass needed; calculate the concentration of the solution made; calculate the concentration of NaOH; compare the % uncertainty of the flask and the pipette; explain what one student did wrong.

COMMON MISTAKE

Making the solution up to 250 cm^3 in a beaker (not accurate) or dissolving it in a test tube.
Ignoring the water of crystallisation in the molar mass (use 126.0 for $H_2C_2O_4{\cdot}2H_2O$, not 90.0).
Saying "weigh accurately" without "by difference".

BEYOND THE SPEC, BUT EXAMINED

- **Mr of an unknown solid acid** H_2X from a titration: n(H_2X) = ½ n(NaOH); Mr = mass ÷ moles, e.g. 1.13 g ÷ 0.01257 mol ≈ 90 (WCH13 Oct 2019).
- **Citric acid** (three acidic H) reacts with NaOH in a 1 : 3 ratio. Lemon juice also contains small amounts of other acids, which have little effect on the result (WCH13 Jan 2024).

SPEC **CP5**

Rates of hydrolysis of halogenoalkanes

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (10.11): compare how fast different halogenoalkanes are hydrolysed, using aqueous silver nitrate in ethanol and the time taken for a silver halide precipitate to appear. Link the results to the C–X bond enthalpy and to whether the halogenoalkane is primary, secondary or tertiary.

CP5 The idea

- Water (from the aqueous silver nitrate) is a **nucleophile**. It attacks the **δ+ carbon** of the C–X bond, the C–X bond breaks, and a **halide ion** is released:
  $R–X + H_2O \rightarrow R–OH + H^+ + X^-$
- The halide ion immediately precipitates with silver ions:
  $Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$ (AgCl white, AgBr cream, AgI pale yellow)
- The **faster** the precipitate appears, the faster the hydrolysis. Rate ∝ **1 ÷ time**.
- **Ethanol** is a solvent that both the halogenoalkane and the aqueous silver nitrate mix with, so they can react.

CORE PRACTICAL 5: METHOD IN BRIEF

1. Put equal volumes of ethanol into three test tubes in a **water bath at about 50 °C**, alongside a tube of aqueous silver nitrate.
2. Add the **same number of drops** (same volume) of each halogenoalkane to its tube.
3. Add the **same volume** of the warmed silver nitrate solution to each, start the timer, and shake.
4. Record the time for a precipitate to first appear.
   Comparing halogens: 1-chlorobutane, 1-bromobutane, 1-iodobutane (all primary).
   Comparing structure: 1-bromobutane (primary), 2-bromobutane (secondary), 2-bromo-2-methylpropane (tertiary).

CP5 Variables and safety

- **IV**: the halogen (or the structure). **DV**: time for the precipitate to appear.
- **Control**: temperature (water bath), volume of halogenoalkane, volume and concentration of silver nitrate, volume of ethanol, same isomer type when comparing halogens (and the same halogen when comparing structures).
- **Safety**: ethanol and halogenoalkanes are **flammable**, so heat in a water bath with no flames. Halogenoalkanes are harmful. Silver nitrate is an irritant, stains skin and is harmful to the environment: gloves, don't pour down the sink.

CP5 Results and explanation

Comparison	Order of rate	Why
C–I, C–Br, C–Cl (primary)	**iodo > bromo > chloro**	bond enthalpy C–I (about 228 kJ mol^{-1}) < C–Br (about 290) < C–Cl (about 346): the weakest bond breaks most easily
tertiary, secondary, primary	**tertiary > secondary > primary**	tertiary halogenoalkanes react by a different route (via a carbocation), which is much faster

- Bond **polarity** predicts the opposite order (C–Cl is the most polar). Bond **enthalpy** is the deciding factor. This is a favourite trap.
- Chloroalkanes may need a long time or a higher temperature to show a precipitate. Fluoroalkanes don't react noticeably (C–F is very strong).

WORKED EXAMPLE: RATE FROM TIME

1-bromobutane gives a cream precipitate after 125 s.
Rate = 1 ÷ 125 = **8.0×10^{-3} s^{-1}** (relative rate; give 2 s.f., with the unit s^{-1})

CP5 Sources of error and improvements

- Judging when a precipitate "first appears" is subjective. Use a black background, or one observer for all tubes.
- Temperature changes after mixing. Pre-warm all solutions in the bath and keep the tubes in it.
- Drops vary in size. Use a graduated pipette or burette for the halogenoalkane.

MARK-SCHEME LANGUAGE

- Supporting a conclusion from times: compare **one variable at a time**. A tertiary chloride (fastest) against a primary chloride (320 s) shows that **structure** matters. A primary chloride (320 s) against a primary bromide (140 s) shows that the **halogen** matters, because the C–Br bond is weaker (WCH13 Jan 2021).
- Test for the halogen in an unknown halogenoalkane: "warm with aqueous NaOH", "acidify with dilute nitric acid", "add silver nitrate", then the colour and the ammonia test.

Often asked as: explain how the results show that the halogen and the structure affect the rate; name the precipitate and its colour; explain the order using bond enthalpy; suggest control variables; calculate 1 ÷ t.

COMMON MISTAKE

Explaining the halogen order by **polarity** or electronegativity. The answer is **bond enthalpy**.
Forgetting to acidify with nitric acid when NaOH(aq) has been used for the hydrolysis: OH^- would precipitate as brown silver oxide.
Comparing a primary chloride with a tertiary bromide: two variables change at once.

BEYOND THE SPEC, BUT EXAMINED

- The general **rate = 1 ÷ time** skill is used in other unfamiliar experiments, e.g. a "clock" reaction at different temperatures, where the rate rises more and more steeply with temperature (WCH13 Oct 2021).

SPEC CP6

Chlorination of 2-methylpropan-2-ol with concentrated hydrochloric acid

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (10.14): make 2-chloro-2-methylpropane from a tertiary alcohol and concentrated HCl in a separating funnel, then purify it by washing, drying and distillation.

CP6 The idea

$$(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$$

- Tertiary alcohols react quickly with concentrated HCl at room temperature. Primary and secondary ones don't.
- The product doesn't mix with water (it is non-polar enough), so it forms a **separate layer**. Its density (about 0.84–0.87 g cm^{-3}) is less than water's, so it is the **upper** layer.
- Boiling temperature of the product: about **51 °C**.

CORE PRACTICAL 6: METHOD IN BRIEF

1. **In a fume cupboard**, put about 10 cm^3 of 2-methylpropan-2-ol into a separating funnel. Add about 30 cm^3 of concentrated HCl **in small portions** (the reaction is exothermic).
2. Stopper, invert and shake. **Open the tap (funnel upside down) after each shake to release the pressure.** Repeat for several minutes, then stand the funnel upright and let the layers separate.
3. Run off and discard the **lower aqueous layer**.
4. Add **sodium hydrogencarbonate solution** to the organic layer, to remove the remaining acid. Shake, **venting often** (CO_2 forms), then run off the lower aqueous layer. (A wash with water can follow.)

5. Run the organic layer into a dry flask. Add **anhydrous sodium sulfate** (or calcium chloride or magnesium sulfate) and swirl until the liquid is **clear**.
6. **Decant** into a distillation flask with anti-bumping granules. Distil, collecting the fraction at about **50–52 °C** in a receiver cooled in ice.

(Separating funnel and distillation diagrams: see P5, organic techniques.)

CP6 Safety

Hazard	Precaution
concentrated HCl: corrosive; its fumes are toxic / irritant	fume cupboard; gloves; add in portions
pressure build-up (exothermic reaction, HCl vapour, CO_2 from the $NaHCO_3$)	vent through the tap often, with the funnel upside down and pointing away from people
2-methylpropan-2-ol and the product are **highly flammable**	no naked flames; distil with an electric heater or water bath

CP6 Percentage yield

WORKED EXAMPLE: % YIELD WITH A DENSITY STEP

12.0 cm^3 of 2-methylpropan-2-ol (density 0.78 g cm^{-3}) gives 7.25 g of pure 2-chloro-2-methylpropane.
1. mass of alcohol = 12.0 × 0.78 = 9.36 g
2. n(alcohol) = 9.36 ÷ 74.0 = 0.1265 mol (HCl is in excess)
3. 1 : 1 ratio, so the theoretical mass of product = 0.1265 × 92.5 = 11.70 g
4. % yield = 7.25 ÷ 11.70 × 100 = **62.0%**

Why the yield is low: product lost on transfers and in the separations, some product evaporates (volatile, b.p. 51 °C), incomplete reaction, product left in the drying agent and the distillation flask.

MARK-SCHEME LANGUAGE

- Venting: "pressure builds up" (from the exothermic reaction / gas) so "open the tap to release it" (WCH13 Oct 2025).
- Layers: the organic layer is on top because it is **less dense** than water. The layers don't mix because water molecules hydrogen-bond strongly to each other, while the halogenoalkane can only form London forces / dipole forces with water (WCH13 Jan 2021).
- $NaHCO_3$: "to remove / neutralise the acid", giving CO_2, hence venting. NaOH would **hydrolyse** the halogenoalkane (WCH13 Jan 2025).
- Drying agent: named **anhydrous** salt; "until the liquid goes clear".
- Distillation errors to spot (e.g. 3 marks): thermometer bulb not at the side arm; condenser water going the wrong way; sealed / stoppered receiver.
- Fume cupboard: "HCl vapour is toxic / corrosive". Comments about chlorine gas are ignored.

Often asked as: give two safety precautions (other than goggles and a lab coat); why is the tap opened; which layer is the product; why use $NaHCO_3$ and not NaOH; how do you know the product is dry; find the errors in the distillation diagram; calculate the % yield (often from a volume and density).

COMMON MISTAKE

Saying the reaction makes **chlorine gas**. The hazard is HCl.
Assuming the organic layer is always the lower one, or the upper one. Use the density data.
Forgetting the density step: 12.0 cm^3 is a volume, not a mass.

BEYOND THE SPEC, BUT EXAMINED

- **Mechanism** (named, not drawn at AS): nucleophilic substitution. The acid protonates the –OH, water leaves to form a tertiary carbocation, and Cl^- attacks it. Show the C(δ+)–O(δ−) dipole (WCH13 Oct 2025).
- Monitoring the reaction by **IR**: the broad O–H absorption (3750–3200 cm^{-1}) disappears and a C–Cl absorption (about 600–800 cm^{-1}) appears (WCH13 Oct 2025).
- **Bromoalkanes** are made from the alcohol with KBr and 50% sulfuric acid. Concentrated H_2SO_4 oxidises some bromide to orange **bromine**. Grinding the KBr increases the surface area (WCH13 Jun 2024).

SPEC CP7

Oxidation of propan-1-ol to propanal and propanoic acid

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (10.20): oxidise propan-1-ol with acidified potassium dichromate(VI), getting propanal by distilling it off as it forms, and propanoic acid by heating under reflux with excess oxidising agent. Test the products.

CP7 The chemistry

- Oxidising agent: **potassium dichromate(VI) acidified with dilute sulfuric acid**. Colour change **orange ($Cr_2O_7^{2-}$) → green (Cr^{3+})**.
- [O] stands for oxygen from the oxidising agent:
  $CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2CHO + H_2O$ (propanal)
  $CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$ (propanoic acid)

Substance	Boiling temperature	Why
propanal	49 °C	no O–H, so **no hydrogen bonds** between molecules
propan-1-ol	97 °C	hydrogen bonds
propanoic acid	141 °C	hydrogen bonds (stronger, and a larger molecule)

- The **aldehyde** has the lowest boiling temperature, so it can be **distilled off as it forms**, before it is oxidised further.
- To get the **acid**, heat under **reflux with excess oxidising agent**. The aldehyde vapour condenses and goes back into the flask, so it is oxidised completely.

CORE PRACTICAL 7: METHOD IN BRIEF

(a) Propanal: distil it off as it forms

1. Put the acidified dichromate(VI) solution in a pear-shaped flask set up for **distillation**, with anti-bumping granules.
2. Heat gently and add propan-1-ol **dropwise** (from a tap funnel), so the propanal distils off as it forms.

The alcohol is in excess.
3. Collect the distillate in a receiver **cooled in ice** (propanal is very volatile).

(b) Propanoic acid: reflux, then distil
1. Cool the acidified dichromate(VI) (an **excess**) in an **ice bath**. Add propan-1-ol **dropwise**, with cooling and swirling, because the reaction is **exothermic**.
2. Fit a vertical condenser and **heat under reflux** for about 20 minutes.
3. Rearrange for **distillation** and collect the fraction boiling near **141 °C**. Above 100 °C, heat with an electric heating mantle or an oil bath.

(Reflux and distillation diagrams: see P5, organic techniques.)

CP7 Testing the products

Test	Propanal	Propanoic acid	Propan-1-ol
warm with Fehling's or Benedict's	**blue → red precipitate**	no change	no change
add Na_2CO_3 or $NaHCO_3$	no change	**effervescence**; CO_2 turns limewater milky	no change
warm with acidified $K_2Cr_2O_7$	orange → green	no change	orange → green

CP7 Safety

- Potassium dichromate(VI): **toxic, carcinogenic, oxidising**. Wear gloves and avoid skin contact.
- Concentrated H_2SO_4 (used to make up the mixture) is corrosive.
- Propan-1-ol and propanal are **flammable**. Heat with an electric heater or a water bath, not a Bunsen.

CP7 Other alcohols

- **Secondary** (e.g. propan-2-ol) → **ketone** (propanone). Heat under reflux to make sure the reaction is complete; ketones are not oxidised further. Fehling's: no change.
- **Tertiary** (e.g. 2-methylpropan-2-ol): **no reaction**, so the dichromate stays orange. There's no hydrogen on the carbon bonded to the –OH.

MARK-SCHEME LANGUAGE

- Aldehyde: "distil off as it forms" to "prevent further oxidation (to the carboxylic acid)".
- Acid: "heat under reflux" so "the aldehyde is condensed and returned to the flask / oxidising agent" (WCH13 Jan 2026).
- Ice bath while adding the alcohol: "the reaction is exothermic".
- Electric heater or water bath, not Bunsen: "the alcohol and products are flammable".
- Oil bath for distilling the acid: "propanoic acid boils at 141 °C, above the temperature of a water bath".
- Choosing apparatus: a distillation set-up is wrong for making the acid (the aldehyde escapes); an open flask with no condenser loses reactants and products (WCH13 Jun 2023).
- Colour change: "orange to green" (blue or blue-green allowed).

Often asked as: write the equation using [O]; choose between three drawn set-ups; explain why reflux / distillation is used; give the colour change; describe a test to show the product is an aldehyde or an acid; explain why the aldehyde distils first; deduce which isomer of $C_4H_{10}O$ or C_3H_8O was oxidised.

COMMON MISTAKE

Using HCl, nitric acid, potassium chromate(VI) or manganate(VII) as the oxidant.
"Red solution" for Fehling's. It is a red **precipitate** from a **blue** solution.
Saying a tertiary alcohol "can't be oxidised because it is tertiary" without the reason (no H on the C–OH carbon).
Heating a flammable alcohol with a Bunsen burner.

BEYOND THE SPEC, BUT EXAMINED

- The **dichromate half-equation**: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. Combine it with the alcohol's half-equation for a full ionic equation.
- A positive aldehyde test can't pin down which **$C_4H_{10}O$** alcohol you started with: butan-1-ol and 2-methylpropan-1-ol are both primary (WCH13 Jun 2019).
- **Ethanal** (b.p. 21 °C) made by distillation escapes before it can be oxidised further (WCH13 Jan 2024).

SPEC CP8

Analysis of some inorganic and organic unknowns

WHAT THIS SPEC POINT ASKS

Unit 2 core practical (10.23): identify unknown inorganic salts and organic compounds by carrying out tests and interpreting the observations, using spectra where given.

CP8 A systematic approach

Inorganic solid

1. **Flame test** → Group 1 or 2 cation (or none: Mg^{2+}, or a non-metal cation such as NH_4^+).
2. **NaOH(aq) and warm**, damp red litmus → NH_4^+. NaOH also gives white precipitates with some Group 2 ions (Mg^{2+} strongly; Ba^{2+} none).
3. Dissolve in water. Add **dilute acid** → effervescence and limewater milky → CO_3^{2-} / HCO_3^-.
4. **Dilute HCl + $BaCl_2$(aq)** → white precipitate → SO_4^{2-}.
5. **Dilute HNO_3 + $AgNO_3$(aq)**, then ammonia → Cl^- / Br^- / I^-.
6. **Heat the solid** → O_2 only (Group 1 nitrate), brown $NO_2 + O_2$ (Group 2 nitrate), CO_2 (carbonate), $H_2O + CO_2$ (hydrogencarbonate).

Organic liquid

1. **Bromine water** → decolourised: C=C.
2. **PCl_5** → steamy fumes: –OH (alcohol or acid).
3. **$NaHCO_3$ / Na_2CO_3** → effervescence: –COOH.
4. **Fehling's / Benedict's** (warm) → red precipitate: –CHO.
5. **Acidified dichromate(VI)** (warm) → orange to green: primary or secondary alcohol (or aldehyde).
6. **IR** for the bonds present; **mass spectrum** for Mr (molecular ion) and fragments.

(All the tests and observations: P5.)

WORKED EXAMPLE: AN INORGANIC UNKNOWN

White solid X gives a **lilac** flame. Its solution, with dilute nitric acid then silver nitrate, gives a **cream** precipitate. The precipitate doesn't dissolve in dilute ammonia but does dissolve in concentrated ammonia.
Lilac → K^+. Cream precipitate, soluble only in concentrated NH_3 → Br^-. **X = KBr.**
Confirmation: with concentrated sulfuric acid, solid X gives steamy fumes **and** brown fumes (Br_2).

WORKED EXAMPLE: A NITRATE

White solid Y gives a **green** flame. On heating it gives a **brown gas** and a gas that **relights a glowing splint**, leaving a white solid. Y's solution gives a white precipitate with dilute sulfuric acid but no precipitate with NaOH(aq).

Green → Ba^{2+}. Brown gas + O_2 → a Group 2 **nitrate** (NO_2 + O_2). **Y = $Ba(NO_3)_2$**; the residue is BaO.

$2Ba(NO_3)_2(s) \rightarrow 2BaO(s) + 4NO_2(g) + O_2(g)$

$BaSO_4$ is insoluble (precipitate) but $Ba(OH)_2$ is soluble (no precipitate), which fits the Group 2 solubility trends.

WORKED EXAMPLE: AN ORGANIC UNKNOWN, C_3H_8O

Z gives **steamy fumes with PCl_5** but **no effervescence** with $NaHCO_3$ → an alcohol, not an acid. Warmed with acidified dichromate(VI) and distilled: **orange → green**. The distillate gives a **red precipitate** with Fehling's → an aldehyde formed, so Z is a **primary** alcohol.

Z = propan-1-ol, $CH_3CH_2CH_2OH$. Its mass spectrum has M^+ at m/z = 60 and a fragment at m/z = 31 (CH_2OH^+).

Its isomer propan-2-ol would give propanone (Fehling's stays blue) and a fragment at m/z = 45 (CH_3CHOH^+).

MARK-SCHEME LANGUAGE

- Give the identity **with charges** (K^+, Br^-), or the full formula of the compound if asked.
- Observations need the **start and end** colours, or the word "precipitate" with its colour.
- IR: name the bond and quote the wavenumber range from the data booklet.
- Mass spectrum fragments are **positive ions** (e.g. CH_3CO^+ at m/z = 43).
- Extra observations that are wrong lose the mark (e.g. adding "effervescence" where there is none).

Often asked as: identify A and B from a table of observations; give the formula of the precipitate; write the ionic equation with state symbols; suggest a test to confirm…; use the IR spectrum and mass spectrum to deduce the structure; explain why the compound is not the isomer.

COMMON MISTAKE

Swapping cations and anions, or writing an element instead of its ion ("bromine" for bromide).

Ignoring "functional group at the end of the chain" clues, e.g. giving propanone when an aldehyde test was positive.

Leaving out the starting colour (blue) or the precipitate in the Fehling's result.

BEYOND THE SPEC, BUT EXAMINED

- **Mixtures of halides**: chlorine water and hexane reveal iodide (purple upper layer). After dilute ammonia only AgCl dissolves, so a weighed residue can be used to find the composition (WCH13 Jan 2026).
- **Group 2 identification grids**: Ba^{2+} gives no precipitate with NaOH but does with carbonate and sulfate; Mg^{2+} gives a precipitate with NaOH and carbonate but not sulfate; Ca^{2+} gives slight precipitates with both NaOH and sulfate (WCH13 Oct 2024).
- **Mixing pairs of solutions** to identify five unknowns: HCl + carbonate fizzes; $AgNO_3$ + chloride gives a white precipitate that stays in dilute nitric acid; $AgNO_3$ + carbonate gives a precipitate that dissolves in nitric acid with fizzing (WCH13 Jan 2021).

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Unit 6: Practical Skills in Chemistry II

Revision notes: practical skills P1–P5 and Core Practicals 9–16

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

Contents

Practical skills

P1 Planning an investigation

P2 Carrying out practical work: apparatus, techniques and measurement

P3 Analysing data: tables, graphs and calculations

P4 Evaluating: errors, uncertainty and improvements

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Core practicals

CP9 Rates: iodine–propanone by titration (9a) and a clock reaction (9b)

CP10 Finding the activation energy of a reaction

CP11 Finding the Ka value for a weak acid

CP12 Investigating some electrochemical cells

CP13 Redox titrations: iron(II) with manganate(VII) (13a) and thiosulfate with iodine (13b)

CP14 The preparation of a transition-metal complex

CP15 Analysis of some inorganic and organic unknowns

CP16 The preparation of aspirin

Practical skills

SPEC P1 Planning an investigation

WHAT THIS SPEC POINT ASKS

Plan practical work: choose a method and apparatus, decide what to vary, measure and control, work out sensible quantities, and say how you will keep it safe and valid.

P1 What the WCH16 paper actually asks

Unit 6 never sets one long "plan an investigation" question. Planning comes in short **describe / outline / devise** parts worth 2–5 marks, marked as a list of separate points. Count the marks and give that many clear, ordered steps.

Typical stem	What gets the marks
"Describe how to make 250.0 cm^3 of a standard solution"	weigh, dissolve in a beaker of distilled water, transfer **with washings** to a **volumetric flask**, make up to the mark, stopper and invert
"Describe how to carry out the titration"	pipette + filler for the aliquot, indicator, burette, white tile, swirl, drop by drop near the end-point, read the bottom of the meniscus at eye level, repeat to concordant titres
"Outline how to find the mass of water in a hydrated salt"	weigh the crucible, add the salt, reweigh, heat, cool, reweigh, **heat to constant mass**
"State the changes needed to find the order for each reactant"	repeat, changing the volume of **one** reactant at a time, add water so the **total volume stays the same**, keep everything else (temperature) the same
"Suggest how to show the reaction is complete"	a test on what is given off or left, e.g. damp red litmus held **at the mouth** of the flask stays red (no more NH_3)

P1 Variables

- **Independent variable (IV):** the one you change on purpose, e.g. volume of KI(aq), temperature.
- **Dependent variable (DV):** what you measure, e.g. time for the blue-black colour, titre.
- **Control variables:** everything else that could affect the DV. Say **how** each is controlled:
  - concentration → keep the **total volume constant** by adding water, so the volume of a reactant is proportional to its concentration;
  - temperature → thermostatic water bath; record the temperature of each mixture;
  - amount of indicator, starch or catalyst → same volume each time, measured the same way.

P1 Choosing a method to follow a reaction

What changes during the reaction	Method	Watch out
colour of one species	colorimeter (filter of the complementary colour, calibration curve)	not useful if a coloured precipitate forms
a gas is given off	gas syringe (volume) or mass loss on a balance	mass loss is too small for a light gas such as H_2; use it for CO_2
acid, alkali or iodine is used up	remove samples, **quench**, titrate	quench must really stop the reaction
a fixed amount of product forms	clock reaction: time t for a sudden colour change, rate $\propto 1/t$	only valid near the start
H^+ concentration changes	pH meter	calibrate with buffers first

P1 Getting quantities right

- Use the equation and the data to choose amounts. For a titration aim for a titre of about **20–30 cm^3** (a small titre gives a large percentage uncertainty; a titre over 50 cm^3 needs a refill).
- If a reagent is "in excess", its exact amount doesn't matter, so a **measuring cylinder** is fine. The limiting (measured) reagent needs a **pipette** or **burette**.
- Heat to **constant mass**: heat, cool (in a desiccator), weigh, then repeat until two masses agree. A colour change alone does not show the reaction is complete.

MARK-SCHEME LANGUAGE

- "Heat ... **until mass is constant**" (reheat and reweigh until two readings are the same). "Until all the water is removed" scores nothing on its own.
- "Transfer the solution **and washings** to a **volumetric flask**; make up to the mark with distilled water; invert to mix." Beaker, conical flask or measuring cylinder for 250.0 cm^3 is wrong.
- "Add **drop by drop** using a **dropping (teat) pipette** (or burette)". "Add in small amounts" is not accepted.
- "Keep the **total volume** the same (add water) so that volume is **proportional to concentration**." "Fair test" is ignored.
- Method marks are independent: one mark per correct step, in a sensible order.

Often asked as: describe how to make a standard solution; describe a titration; outline heating to constant mass; state the changes needed to find the order with respect to another reactant; name two control variables.

COMMON MISTAKE

Describing how to make a standard solution when the question asks how to *carry out the titration* (or the other way round). Read the stem for which job is wanted.
Taking samples out when the question says the **whole mixture** is used, or describing a solution being heated when the sample is a solid.
"Hot filtration to collect the crystals." Hot filtration removes **insoluble impurities**; the crystals are collected later, after cooling.

BEYOND THE SPEC, BUT EXAMINED

- Checking that all the ammonia has been boiled off: hold **damp red litmus at the mouth** of the flask; repeat until it stays red. Never dip the paper into the liquid or "smell" the gas (WCH16 Oct 2022).
- Better temperature change for an enthalpy of solution: record T every 30 s for a few minutes, add the solid at a fixed time, keep recording, then extrapolate both lines to the time of mixing (WCH16 Oct 2021).
- Making sure a hydrated nitrate has fully decomposed: heat to constant mass, or check no more gas is given off (WCH16 Jan 2026).

SPEC P2

Carrying out practical work: apparatus, techniques and measurement

WHAT THIS SPEC POINT ASKS

Choose and use the right apparatus for each job, know its resolution, carry out the standard techniques (titration, weighing, heating, gas tests) correctly, and give a hazard with a matching control measure.

P2 Which piece of apparatus?

Job	Use	Not credited
a fixed, accurate volume (aliquot)	**volumetric (bulb) pipette** with a pipette filler	measuring cylinder, teat pipette
different volumes in each run, e.g. 7.5 cm^3	**burette** or graduated pipette	volumetric pipette, measuring cylinder
a small volume (e.g. 2.5 cm^3 of starch) so total volume stays constant	burette	measuring cylinder
a reagent **in excess** (e.g. 10 cm^3 of KI, 50 cm^3 of HCl)	**measuring cylinder** (accuracy isn't needed)	burette, pipette, volumetric flask
250.0 cm^3 of standard solution	**volumetric flask**	beaker, conical flask
a few drops until no further change	dropping / teat pipette (or burette)	"dropper", "small amounts"
removing a sample from a reacting mixture	pipette or syringe	burette, forceps
following a coloured species	colorimeter	titration
optical activity	polarimeter	spectrometer
crushing a solid or plant material	pestle and mortar	"grinder"
melting or boiling point above 100 °C	oil bath / Thiele tube / heating block	water bath (can't go above 100 °C)

Why: a pipette is more accurate than a cylinder, so it measures the reagent you calculate from. Mixing two solutions from separate tubes at once gives a clear start time; running one in from a burette is too slow.

P2 Resolution and uncertainty of common apparatus

Apparatus	Typical uncertainty	Notes
burette	±0.05 cm^3 per reading	a titre uses **two** readings, so ±0.10 cm^3
25.0 cm^3 pipette	±0.06 cm^3	one measurement
250.0 cm^3 volumetric flask	±0.2–0.3 cm^3	one measurement
2 d.p. balance	±0.005 g per reading	mass by difference uses two readings
thermometer (0.5 °C divisions)	±0.25 °C per reading	a temperature **change** uses two readings
measuring cylinder (100 cm^3)	±0.5 cm^3	much worse than a pipette

Always use the uncertainty the question gives you. Uncertainty is usually **half the smallest division**.

P2 Titration technique

1. Rinse the burette with distilled water, then with the **titrant** (so it isn't diluted). Rinse the pipette with the solution it will measure. Rinse the conical flask with **distilled water only**.
2. Fill the burette (funnel removed, tip filled, no air bubble). Read the **bottom of the meniscus at eye level**, to ±0.05 cm^3 (two decimal places, last digit 0 or 5).
3. Pipette the aliquot into a conical flask on a **white tile**; add indicator (not for MnO_4^-).
4. Do a rough titration, then accurate ones: **swirl**, add **drop by drop** near the end-point, wash the sides of the flask and the burette tip into the flask with distilled water.
5. Repeat until you have **concordant** titres (within 0.20 cm^3; many schools use 0.10 cm^3). Mean the concordant ones only.

P2 Gas tests: technique

- Hold **damp** indicator paper **at the mouth** of the tube, never in the liquid. State the starting colour ("damp **red** litmus turns blue").
- Ammonia with conc HCl on a glass rod gives white **smoke** (not fumes).
- Never "smell" a gas as a test.

P2 Hazards and control measures

The mark is for a **hazard paired with the precaution that matches it**. If the question says goggles and lab coat are worn, don't give them.

Hazard	Example	Precaution credited	Not credited
corrosive	conc HNO_3, conc HCl, ethanoic acid, phenol, benzoyl chloride	gloves	fume cupboard for a corrosive liquid
toxic gas or vapour	SO_2, NO_2, NH_3, HCl	fume cupboard (or well-ventilated lab)	mask / gas mask
toxic by skin absorption	phenylamine, nitrobenzene	gloves; small quantities; sealed waste	"toxic by inhalation" for a liquid in that step
oxidising	conc HNO_3, metal nitrates	keep away from flammables; no glowing splint into it	"flammable" (oxidising is not flammable)
flammable	ethanol, ether, methanol	water bath or electric heater, **no naked flame**	Bunsen burner
health hazard / environmental	carcinogens, heavy-metal waste	fume cupboard; dispose of waste safely, not down the sink	"causes pollution" with no action
exothermic addition	conc H_2SO_4 into water, nitrating mixture	add slowly, with cooling in an ice-water bath	"to stop an explosion"

Know the hazard pictograms: **corrosive** (liquid from two test tubes eating into a hand and a surface), **acute toxicity** (skull and crossbones), **oxidising** (flame over a circle), **flammable** (flame), **health hazard** (figure with a star on the chest: carcinogen, long-term harm), **environmental hazard** (dead tree and fish).

MARK-SCHEME LANGUAGE

- Titration: "add **dropwise** near the end-point", "**swirl**", "**white tile**", "read the **bottom of the meniscus** at **eye level**", "rinse the burette with the solution". "Do a rough titration" is ignored if it is your only point.
- Excess reagent: "measuring cylinder because the reagent is **in excess**, so an accurate volume is not needed".
- Hazards: "corrosive, so wear gloves"; "toxic gas, so use a fume cupboard"; "flammable, so use a water bath / no naked flames".
- Ammonia test: "**damp red** litmus turns **blue**" or "white **smoke** with conc HCl".

Often asked as: name the apparatus for a given volume and justify it; describe titration technique (3–5 marks); give a hazard and precaution for a named chemical; describe a hazard symbol; say why a burette, not a measuring cylinder, is used.

COMMON MISTAKE

"Mask" for a toxic gas: it must be a **fume cupboard**.
Calling an oxidising substance "flammable", or giving "fume cupboard" for a corrosive liquid.
Using a beaker for an accurately measured volume, or a pipette for something in excess. Choose the apparatus to match how accurate the volume needs to be.
Reading a burette to one decimal place. Burette readings are recorded to **±0.05 cm^3** (e.g. 23.45).

BEYOND THE SPEC, BUT EXAMINED

- Use **ice and water**, not ice alone, for a cooling bath: better contact with the flask (WCH16 Jun 2023).
- Boiling point of a small sample in a **Thiele tube** of oil with an inverted capillary: bubbles stream out, then on cooling the liquid is drawn back into the capillary at the boiling point (WCH16 Jun 2021).
- Generate a toxic gas such as SO_2 **in situ** (Na_2SO_3 + HCl) rather than from a cylinder, so less escapes (WCH16 Jan 2024).

SPEC P3

Analysing data: tables, graphs and calculations

WHAT THIS SPEC POINT ASKS

Record data in clear tables, plot and read graphs (straight lines, curves, tangents, half-lives, gradients), and carry out multi-step calculations to sensible significant figures.

P3 Tables

- Column headings carry the **quantity and unit**: "Volume / cm^3", "Time / s", "1/t / s^{-1}".
- Same number of decimal places down a column, matching the apparatus (burette 2 d.p.).
- Processed columns (1/t, ln k, 1/T) to the s.f. the question asks for, or the same s.f. as the raw data. Temperature in **kelvin**: T/K = T/°C + 273.

P3 Plotting a graph: the marks

1. **Axes the right way round**: the dependent variable (titre, 1/t, pH, ln k) on **y**, the independent variable (time, volume, 1/T) on **x**. Swapping them is the most common lost mark.
2. **Labels with units** on both axes.
3. A **sensible scale**: plotted points cover **at least half the grid in both directions**; even steps. The origin need not be included.
4. **Points accurate** to within half a small square, drawn as neat crosses.
5. **Line of best fit**: a ruled straight line if the trend is linear; a smooth **curve** if not (never dot-to-dot). Ignore an anomalous point, and say which it is.

P3 Reading a graph

Task	How
order from rate (or 1/t) against concentration (or volume)	straight line through the origin → **first order**; horizontal → zero order; upward curve → second order
order from concentration (or titre) against time	**straight line, negative gradient** → zero order; curve with **constant half-life** → first order
half-life	read the time for the value to halve; the next half-life starts **at the half value** and goes to a quarter, not from t = 0
rate at a time	draw a **tangent** at that time, make it long, rate = Δy/Δx with units
initial rate	tangent at t = 0
Ea	gradient of ln k (or ln 1/t) against 1/T = −Ea/R

P3 Significant figures and working

- Give the answer to the **same s.f. as the least precise data** (usually 2 or 3). Never 1 s.f. unless asked.

- Keep full values in the calculator; **round only at the end**.
- **Show working every time.** Some mark schemes give full marks for a bare correct answer; others give **zero**.
- Units: concentration mol dm^{-3}; rate mol dm^{-3} s^{-1} (a graph-based rate can be in cm^3 s^{-1} or cm^3 min^{-1}); enthalpy kJ mol^{-1} with sign.

P3 Calculations that come up again and again

Calculation	The steps
titration	n = c × V/1000 → ratio from equation → scale up (×10 for 25.0 cm^3 of 250.0 cm^3) → mass or %
back titration	amount added − amount left (found by titration) = amount that reacted
excess reagent	mass = volume × density → moles → compare using the equation ratio, and **state** which is bigger
percentage yield	(actual mass ÷ theoretical mass) × 100, to 2–3 s.f.
empirical formula by combustion	mass C = m(CO_2) × 12/44; mass H = m(H_2O) × 2/18; O by difference; mole ratio
water of crystallisation	moles from titration → Mr of hydrate → (Mr − Mr anhydrous) ÷ 18.0
enthalpy change	q = m c ΔT (m = mass of water/solution only); ΔH = −q ÷ n, with sign and kJ mol^{-1}
gas volume	n = V ÷ 24 000 (cm^3 at r.t.p.)
Ka from half-neutralisation	pKa = pH at half the equivalence volume; Ka = 10^{-pH}
E⊖cell	E⊖(reduction, more positive) − E⊖(oxidation, less positive)

WORKED EXAMPLE: ENTHALPY OF SOLUTION

4.24 g of LiCl (Mr 42.4) is dissolved in 50.0 g of water. The extrapolated temperature rise is 17.5 °C. c = 4.18 J g^{-1} K^{-1}.

q = m c ΔT = 50.0 × 4.18 × 17.5 = 3658 J (use the mass of **water**, not water + solid; ΔT is the same in K and °C, so don't add 273)

n(LiCl) = 4.24 ÷ 42.4 = 0.100 mol

ΔH = −3658 ÷ 0.100 = −36 580 J mol^{-1} = **−36.6 kJ mol^{-1}** (temperature rose, so exothermic: negative sign)

WORKED EXAMPLE: EMPIRICAL FORMULA BY COMBUSTION

0.600 g of an organic compound containing C, H and O burns completely to give 1.760 g of CO_2 and 0.360 g of H_2O.

mass C = 1.760 × 12.0/44.0 = 0.480 g → 0.480 ÷ 12.0 = 0.0400 mol

mass H = 0.360 × 2.0/18.0 = 0.0400 g → 0.0400 ÷ 1.0 = 0.0400 mol

mass O = 0.600 − 0.480 − 0.0400 = 0.0800 g → 0.0800 ÷ 16.0 = 0.00500 mol

ratio C : H : O = 0.0400 : 0.0400 : 0.00500 = 8 : 8 : 1 → **C_8H_8O**

Working with masses like this is safer than starting from moles of CO_2 and H_2O.

MARK-SCHEME LANGUAGE

- Graph: "axes the right way round, labelled with units"; "scale uses at least half the grid in both directions"; "all points plotted correctly"; "straight line of best fit" or "smooth curve".
- "Correct answer with some working scores full marks"; incorrect rounding is penalised once; TE (transferred error) applies at each step.
- Rate units from a graph: $cm^3\ h^{-1}$ or $cm^3\ s^{-1}$ are accepted; they need not be $mol\ dm^{-3}\ s^{-1}$.
- In combustion analysis: anhydrous **$CaCl_2$** absorbs the water, **soda lime** absorbs the CO_2 (water absorber first).

Often asked as: complete a table (1/t, ln k, 1/T); plot a graph and draw the best-fit line; use the graph to find order, half-life, tangent rate or gradient; a 4–6 mark multi-step titration calculation; percentage yield; empirical formula from combustion data.

COMMON MISTAKE

Plotting the dependent variable on x. Joining points dot-to-dot with a ruler instead of drawing a smooth curve.
Measuring the second half-life from zero, so it looks twice as long. Each half-life starts where the last one ended.
Rounding mid-calculation, then losing the final mark. Giving 4 or 5 s.f. when the data have 3.
Adding the mass of the solid to the mass of water in q = mcΔT, or converting ΔT to kelvin by adding 273.

BEYOND THE SPEC, BUT EXAMINED

- **Cooling correction**: when the temperature keeps changing after mixing, plot temperature against time and extrapolate the lines back to the time of mixing to find ΔT (WCH16 Oct 2021).
- **Kc with units** from titration data, e.g. $Ca(OH)_2(s) \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$: $Kc = [Ca^{2+}][OH^-]^2$, units **$mol^3\ dm^{-9}$** (WCH16 Jun 2025).
- Moles of gas from volume at r.t.p. (n = V/24 000) to find an Ar and identify a metal (WCH16 Jan 2026).

SPEC P4

Evaluating: errors, uncertainty and improvements

WHAT THIS SPEC POINT ASKS

Judge the quality of data: accuracy, precision and uncertainty; explain how an error changes the result; suggest and justify improvements; and comment on product purity and yield.

P4 Key words

- **Accurate**: close to the true value.
- **Precise**: repeat readings close to each other (concordant titres). Precise results can still be inaccurate.
- **Random error**: unpredictable scatter (e.g. judging an end-point). Reduce by repeating and averaging.
- **Systematic error**: shifts every reading the same way (e.g. heat loss, a burette not rinsed with titrant). Repeating doesn't help; change the method.
- **Uncertainty**: the range within which the true value is likely to lie, from the apparatus (± half a division per reading).

P4 Percentage uncertainty

$$\%\ \text{uncertainty} = (\text{uncertainty} \div \text{quantity measured}) \times 100$$

- Anything found from **two readings** (a titre, a temperature change, a mass by difference) has **twice** the single-reading uncertainty.
- The **bigger** the quantity measured, the **smaller** the percentage uncertainty. So: use a larger titre, a bigger temperature change, a larger mass.
- To combine several measurements, **add** their percentage uncertainties.

WORKED EXAMPLE: PERCENTAGE UNCERTAINTY

(a) A thermometer reads to ±0.25 °C. The temperature rises from 20.0 °C to 32.5 °C.
ΔT = 12.5 °C from two readings, so uncertainty = 2 × 0.25 = ±0.50 °C
% uncertainty = 0.50 ÷ 12.5 × 100 = **4.0 %**
(b) A titre of 22.40 cm^3 is read from a burette to ±0.05 cm^3 per reading.
% uncertainty = (2 × 0.05) ÷ 22.40 × 100 = **0.45 %**
(c) The 25.0 cm^3 pipette (±0.06 cm^3) adds 0.06 ÷ 25.0 × 100 = 0.24 %. Total for the titration ≈ 0.45 + 0.24 = **0.69 %**.
(d) If the solution were diluted so the titre was only 11.20 cm^3, the burette uncertainty would double to 0.89 %.

P4 "What effect would this error have?"

Answer as one chain: **what happens chemically → titre (or reading) goes up or down → calculated value goes up or down.**

Error	Chain	Effect on result
NH_3 not all boiled off before back-titrating excess NaOH with HCl	NH_3 also neutralises HCl → bigger titre → less NaOH seems to have reacted	**lower** % ammonium salt
HNO_3 not neutralised before adding KI (copper in brass)	HNO_3 oxidises I^- to extra I_2 → bigger thiosulfate titre	**higher** % Cu
HCl used to acidify MnO_4^-	MnO_4^- oxidises Cl^- → more MnO_4^- used → bigger titre	**higher** apparent amount of reductant
tin (reducing agent) left in the flask	tin also reduces MnO_4^- → bigger titre	too high
delay before titrating a quenched sample (quench incomplete)	reaction continues → less reactant left → smaller titre	rate wrong
precipitate not washed / not dried	extra mass of water or soluble salts	mass too high
crystals still damp when weighed	mass too high	yield too high (even over 100 %)
burette not rinsed with titrant (water left inside)	titrant diluted → bigger titre	too high

P4 Improvements that score

- **Reduce heat loss**: polystyrene cup with a lid ("reduces", never "prevents"); cooling correction graph.
- **Reduce percentage uncertainty**: larger volumes/masses; a burette or pipette in place of a cylinder; a thermometer with finer divisions; a larger temperature change (more concentrated solutions).
- **Clock reactions**: a water bath; a light sensor / colorimeter instead of judging the colour change by eye; mix in the bath once both solutions have reached the bath temperature.
- **Titrations**: concordant repeats; a titre near 25 cm^3.

- Judge a suggested improvement against the actual method (e.g. quenching in ice "is not needed if the reaction is already slow at room temperature").

P4 Purity and yield

- **Pure solid**: melts **sharply** (range of about 1–2 °C) at a temperature that **matches the data-book value**. Both ideas are needed.
- **Impure solid**: melts **lower** and over a **wider range**. Give both.
- **Yield over 100 %**: product still **damp** (water or solvent).
- **Yield below 100 %**: incomplete (equilibrium) reaction, side reactions (e.g. dinitration), product left in solution during recrystallisation, losses on transfer and filtration.

MARK-SCHEME LANGUAGE

- "% uncertainty = 2 × 0.25 ÷ 12.5 × 100 = 4 %". The ×2 is the commonly lost mark.
- "Larger titre / volume, so **smaller** percentage uncertainty."
- Effect-of-error answers need the **direction** of both the titre and the final value.
- Purity: "**sharp** melting temperature" **and** "**close to the data-book value**". Impure: "**lower** and over a **range**".
- Polystyrene: "good insulator", "**reduces** heat loss".

Often asked as: calculate a % uncertainty; explain the effect of a stated error on the titre and on the result; suggest an improvement and explain why it helps; give the advantage and disadvantage of diluting a sample; explain an apparent yield over 100 %; predict the melting range of a crude sample.

COMMON MISTAKE

Forgetting the factor of 2 for a difference between two readings.
"Repeat to make it more accurate." Repeating improves **reliability**/precision and spots anomalies; it doesn't remove systematic error.
"Human error" or "measured wrongly" as an explanation. Name the actual error and its direction.
"Impure starting materials" or "transfer losses" on their own as the reason for a low yield: give the chemistry (side reaction, product left in solution).

BEYOND THE SPEC, BUT EXAMINED

- **Percentage error** compares your value with an expected one: (experimental − expected) ÷ expected × 100. E.g. a measured $Ca(OH)_2$ titre of 9.50 cm^3 against an expected 10.55 cm^3; explained by the lab being above 20 °C, where $Ca(OH)_2$ is less soluble (WCH16 Jun 2025).
- **Advantage of diluting** a sample before titration: enough solution for a rough and several accurate titres; **disadvantage**: smaller titre, so bigger % uncertainty (WCH16 Jun 2024).
- Predicted melting range of a crude solid: at least ~4 °C wide and topping out **below** the pure value (WCH16 Jun 2023).

SPEC P5 Chemistry-specific skills: tests, titrations and organic techniques

WHAT THIS SPEC POINT ASKS

Recall and interpret tests for cations (especially transition-metal ions), anions, gases and organic functional groups; use spectra to identify compounds; carry out and calculate redox titrations; and use the techniques for preparing and purifying organic compounds.

P5 Transition-metal ions with NaOH(aq) and NH_3(aq)

Always say **colour and state**: "blue **precipitate**", "dark blue **solution**". Add the reagent **drop by drop**, then **in excess**.

Ion and colour in water	Few drops NaOH(aq)	Excess NaOH(aq)	Few drops NH_3(aq)	Excess NH_3(aq)	Also
Cu^{2+}, blue $[Cu(H_2O)_6]^{2+}$	blue ppt $Cu(OH)_2(H_2O)_4$	no change	pale blue ppt	dissolves → **deep (dark) blue solution** $[Cu(NH_3)_4(H_2O)_2]^{2+}$	conc HCl → green then **yellow** $[CuCl_4]^{2-}$
Fe^{2+}, pale green	**green** ppt $Fe(OH)_2$	no change	green ppt	no change	ppt turns **brown on standing** (air oxidises it to $Fe(OH)_3$)
Fe^{3+}, yellow / orange-brown	**red-brown (orange-brown)** ppt $Fe(OH)_3$	no change	red-brown ppt	no change	oxidises I^- to I_2
Cr^{3+}, green (violet $[Cr(H_2O)_6]^{3+}$)	**grey-green** ppt $Cr(OH)_3(H_2O)_3$	dissolves → **dark green solution** $[Cr(OH)_6]^{3-}$	grey-green ppt	dissolves (slowly) → **purple/violet** $[Cr(NH_3)_6]^{3+}$	+ H_2O_2 in alkali, warm → **yellow** CrO_4^{2-}
Mn^{2+}, very pale (treat as colourless)	**off-white / pale brown** ppt $Mn(OH)_2$	no change	off-white ppt	no change	ppt **darkens on standing** (air oxidation)
Co^{2+}, pink $[Co(H_2O)_6]^{2+}$	blue ppt	no change	blue ppt	dissolves → **straw / yellow-brown** $[Co(NH_3)_6]^{2+}$, darkens in air	conc HCl → **blue** $[CoCl_4]^{2-}$
Ni^{2+}, green	green ppt	no change	green ppt	dissolves → blue-violet $[Ni(NH_3)_6]^{2+}$	
Zn^{2+}, colourless	**white** ppt $Zn(OH)_2$	dissolves → **colourless** $[Zn(OH)_4]^{2-}$	white ppt	dissolves → **colourless** $[Zn(NH_3)_4]^{2+}$	Zn is d-block but **not** a transition element

Green solutions to keep apart: Fe^{2+}, Cr^{3+}, Ni^{2+}, V^{3+}. Fe^{2+} and Cr^{3+} are told apart by excess NaOH or excess NH_3: $Cr(OH)_3$ dissolves, $Fe(OH)_2$ doesn't (and goes brown).

P5 Three types of reaction

What happens	Type	Example equation
OH^- or a little NH_3 removes H^+ from water ligands → neutral hydroxide ppt	**deprotonation** (acid–base)	$[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(OH)_2(H_2O)_4] + 2H_2O$
		$[Cu(H_2O)_6]^{2+} + 2NH_3 \rightarrow [Cu(OH)_2(H_2O)_4] + 2NH_4^+$
ppt dissolves in excess OH^- (amphoteric hydroxide)	**deprotonation**	$[Cr(OH)_3(H_2O)_3] + 3OH^- \rightarrow [Cr(OH)_6]^{3-} + 3H_2O$
excess NH_3 (or Cl^-) replaces the ligands	**ligand exchange**	$[Cu(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$
		$[Cr(H_2O)_6]^{3+} + 6NH_3 \rightarrow [Cr(NH_3)_6]^{3+} + 6H_2O$
		$[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$
oxidation state of the metal changes	**redox**	$Fe(OH)_2 \rightarrow Fe(OH)_3$ in air; $2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 2OH^- + 8H_2O$

The number of OH^- needed to make the neutral hydroxide equals the charge on the ion. Amphoteric = reacts with both acids and bases.

P5 Chromium and vanadium colours

- Cr(VI): CrO_4^{2-} **yellow** ⇌ $Cr_2O_7^{2-}$ **orange**: $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$ (acid–base, **not** redox; Cr stays +6). $Ba^{2+} + CrO_4^{2-} \rightarrow$ yellow ppt $BaCrO_4$.
- $Cr_2O_7^{2-}/H^+$ warmed with a primary/secondary alcohol or an aldehyde: **orange** → **green** (Cr^{3+}). Zinc and acid take Cr(III) on to **blue** Cr^{2+}, which turns green again in air.
- Vanadium with zinc and acid, step by step:

Oxidation state	Ion	Colour
+5	VO_2^+	yellow
+4	VO^{2+}	blue
+3	V^{3+}	green
+2	V^{2+}	violet (purple)

The colours seen are yellow → green (yellow + blue mixture) → blue → green → violet. V^{2+} left in air goes back to green V^{3+}.

P5 Other cations

- **Flame tests** (nichrome wire dipped in conc HCl, then the solid, in a blue flame): Li^+ red, Na^+ yellow-orange, K^+ lilac, Ca^{2+} orange-red (brick-red), Sr^{2+} red (crimson), Ba^{2+} (pale) green.
- **Ammonium**: warm with NaOH(aq) → NH_3, which turns **damp red litmus blue**.

P5 Anion tests

Anion	Test	Positive result
Cl^-, Br^-, I^-	add **dilute HNO_3**, then $AgNO_3$(aq)	Cl^- **white** ppt; Br^- **cream** ppt; I^- **yellow** ppt
confirm halide	add NH_3(aq) to the ppt	AgCl dissolves in **dilute** NH_3; AgBr dissolves only in **concentrated** NH_3; AgI insoluble in both
halides with conc H_2SO_4 (solid salt)	add conc H_2SO_4	Cl^-: **misty (steamy) fumes** of HCl; Br^-: misty fumes + **orange-brown** Br_2 (+ SO_2); I^-: **purple vapour / black solid** I_2, yellow S, rotten-egg H_2S
SO_4^{2-}	add **dilute HCl** (or HNO_3), then $BaCl_2$(aq)	**white** ppt $BaSO_4$
CO_3^{2-} / HCO_3^-	add dilute acid; bubble gas into limewater	**effervescence**; limewater turns **cloudy**
NO_3^- (beyond spec)	warm with NaOH(aq) and aluminium	NH_3 turns damp red litmus blue

Why the acid first? HNO_3 or HCl removes **carbonate** (and sulfite or hydroxide) ions that would also give a precipitate with Ag^+ or Ba^{2+}. Don't use HCl before $AgNO_3$ (it adds Cl^-), or H_2SO_4 before $BaCl_2$ (it adds SO_4^{2-}).

P5 Gas tests

Gas	Test	Result
H_2	lighted splint	squeaky pop
O_2	glowing splint	relights
CO_2	limewater	turns cloudy (milky)
NH_3	damp red litmus / conc HCl on a rod	turns blue / white smoke (NH_4Cl)
HCl	damp blue litmus / NH_3 on a rod	turns red / white smoke; misty fumes in moist air
Cl_2	damp blue litmus	turns red, then **bleached** white
SO_2	acidified $K_2Cr_2O_7$ paper	orange → green
H_2O	anhydrous $CoCl_2$ paper / anhydrous $CuSO_4$	blue → **pink** / white → blue
NO_2	look	brown gas (toxic)

P5 Organic functional-group tests

Group	Test	Positive result	Negative / notes
C=C (alkene)	bromine water, shake	**orange → colourless**	alkane: no change
C=O (aldehyde or ketone)	Brady's reagent (2,4-DNPH)	**orange/yellow precipitate**; melting point of the purified derivative identifies the compound	must say "precipitate"
aldehyde (–CHO)	Tollens' reagent, warm	**silver mirror** (grey/black ppt)	ketone: **no change**
aldehyde	Fehling's or Benedict's, warm	blue solution → **red (brick-red) ppt** of Cu_2O	ketone: no change
1° or 2° alcohol, aldehyde	acidified $K_2Cr_2O_7$, warm	**orange → green**	3° alcohol and ketone: no change
$CH_3CO–$ or $CH_3CH(OH)–$	iodine + NaOH(aq), warm	**pale yellow ppt** of CHI_3, antiseptic smell	ethanal and ethanol also positive
–COOH	$NaHCO_3$(aq) or Na_2CO_3(aq)	**effervescence**; gas turns limewater cloudy	alcohols, phenol: no fizz with carbonate
–OH (alcohol or acid)	PCl_5	**misty (steamy) fumes** of HCl	can't tell alcohol from acid
alcohol	carboxylic acid + conc H_2SO_4, warm	sweet, **fruity smell** (ester)	
acyl chloride	add water	**misty fumes** (HCl); white smoke with NH_3	
halogenoalkane	warm with NaOH(aq), add dilute HNO_3, then $AgNO_3$(aq)	white / cream / yellow ppt (Cl / Br / I)	must acidify first
phenol	bromine water	decolourised; **white ppt** (2,4,6-tribromophenol)	
amine	universal indicator; Cu^{2+}(aq)	alkaline (blue); pale blue ppt then **deep blue** solution	
aromatic or alkene (high C:H)	burn on a spatula	**smoky (sooty) flame**	

Distinguishing a pair ("one chemical test"): name the reagent, then give the result for **both** compounds. Propan-1-ol vs propanoic acid: $NaHCO_3$(aq) (acid fizzes, alcohol no change). Propanal vs propanone: Tollens' (silver mirror vs no change). Pentan-2-one vs pentan-3-one: iodine/NaOH (yellow ppt vs no change).

P5 Spectra and identification

- **Mass spectrum**: the molecular ion (highest m/z, M^+) gives Mr. Fragments must be written as **positive ions** with no loose bond: CH_3^+ 15, $C_2H_5^+$ 29, CHO^+ 29, CH_3CO^+ 43, $C_3H_7^+$ 43, $COOH^+$ 45. If 43 could be CH_3CO^+ or $C_3H_7^+$, use the other data to decide.
- **Infrared**: quote the **bond and the wavenumber range** from the data booklet. Typical ranges: O–H (alcohol) 3750–3200 broad; O–H (carboxylic acid) 3300–2500 very broad; N–H 3500–3300; C=O 1750–

1630 (ester 1750–1735, aldehyde/ketone and acid about 1740–1700, amide 1700–1630); C–H about 2950–2850.

- **^{13}C NMR**: number of peaks = number of carbon environments. Propan-2-ol has 2 (the two CH_3 carbons are equivalent).
- **^{1}H NMR**: number of peaks = proton environments; **peak areas** (integration) = ratio of protons; splitting by the **n + 1 rule**, where n is the number of protons on **adjacent** carbon atoms. A triplet + quartet in a 3 : 2 ratio = CH_3CH_2–. A 9H singlet = $(CH_3)_3C$–. H on the carbon next to C=O appears at about 2.1–2.6 ppm.
- **"Fully justify" questions**: link each test result to a group, each spectrum feature to a part of the structure, then give the structure. One clear deduction per mark.

P5 Redox titrations

Titration	Key equation	Indicator and end-point	Ratio
MnO_4^- with Fe^{2+}	$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$	none (**self-indicating**): first **permanent pale pink**	MnO_4^- : Fe^{2+} = 1 : 5
MnO_4^- with ethanedioate	$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$	permanent pale pink; warm to ~60 °C at the start	MnO_4^- : $C_2O_4^{2-}$ = 2 : 5
iodine with thiosulfate	$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$	starch added when **pale yellow (straw)**: **blue-black → colourless**	I_2 : $S_2O_3^{2-}$ = 1 : 2
Cu^{2+} (e.g. brass) via iodine	$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$	as above	Cu^{2+} : $S_2O_3^{2-}$ = 1 : 1

Manganate(VII) rules:

- $KMnO_4$ goes in the **burette** (read the top of the meniscus if it is too dark to see the bottom).
- Acidify with **excess dilute H_2SO_4**. Not HCl: MnO_4^- oxidises Cl^- to Cl_2, so the titre is too big. Not HNO_3: it is an oxidising agent and would oxidise Fe^{2+} itself. Too little acid gives a **brown precipitate of MnO_2**.
- With ethanedioate the reaction starts slowly (two negative ions repel), so warm to about 60 °C. Once started it speeds up because the **Mn^{2+} formed catalyses it (autocatalysis)**.

Iodine–thiosulfate rules:

- The oxidising agent is added to **excess KI**, releasing I_2; the I_2 is titrated with thiosulfate.
- Add **starch near the end-point** (pale yellow). If added at the start, a lot of iodine binds to the starch and the end-point is hard to see.
- For brass/copper, neutralise the nitric acid first (Na_2CO_3 until a faint ppt, then a little ethanoic acid) so it can't oxidise I^-.

Back titration: add a known excess of reagent, let it react, then titrate what is left. Amount reacted = amount added − amount left. Used when the substance is insoluble, volatile (NH_3) or reacts slowly.

WORKED EXAMPLE: MnO_4^- TITRATION (PURITY OF IRON(II) SULFATE)

3.20 g of impure $FeSO_4 \cdot 7H_2O$ (Mr 277.9) is dissolved and made up to 250.0 cm^3. 25.0 cm^3 portions are acidified with dilute H_2SO_4 and titrated with 0.0100 mol dm^{-3} $KMnO_4$. Titres: rough 23.10, then 22.45, 22.35, 22.85 cm^3.

Concordant titres: 22.45 and 22.35 (22.85 is not within 0.20 cm^3). Mean = **22.40 cm^3**

$n(MnO_4^-) = 0.0100 \times 22.40/1000 = 2.240 \times 10^{-4}$ mol

$n(Fe^{2+})$ in 25.0 $cm^3 = 5 \times 2.240 \times 10^{-4} = 1.120 \times 10^{-3}$ mol

$n(Fe^{2+})$ in 250.0 $cm^3 = 1.120 \times 10^{-2}$ mol (×10)

mass $FeSO_4 \cdot 7H_2O = 1.120 \times 10^{-2} \times 277.9 = 3.112$ g

% purity = 3.112 ÷ 3.20 × 100 = **97.3 %**

WORKED EXAMPLE: WATER OF CRYSTALLISATION BY MnO_4^- TITRATION

1.26 g of $H_2C_2O_4 \cdot xH_2O$ is made up to 250.0 cm^3. 25.0 cm^3 portions, acidified and warmed to 60 °C, need a mean titre of 20.00 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$.

$n(MnO_4^-) = 0.0200 \times 20.00/1000 = 4.00 \times 10^{-4}$ mol

$n(C_2O_4^{2-}) = 4.00 \times 10^{-4} \times 5/2 = 1.00 \times 10^{-3}$ mol in 25.0 $cm^3 \rightarrow 1.00 \times 10^{-2}$ mol in 250.0 cm^3

Mr of hydrate = $1.26 \div 1.00 \times 10^{-2} = 126$

$H_2C_2O_4 = 90.0$, so water = 126 − 90.0 = 36.0 → x = 36.0 ÷ 18.0 = **2**

WORKED EXAMPLE: BACK TITRATION (AMMONIUM SULFATE IN A FERTILISER)

2.50 g of fertiliser is boiled with 50.0 cm^3 of 1.00 mol dm^{-3} NaOH (excess) until no more NH_3 is given off. The cooled solution is made up to 250.0 cm^3. 25.0 cm^3 portions need a mean titre of 20.40 cm^3 of 0.100 mol dm^{-3} HCl.

$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$, so $(NH_4)_2SO_4$: NaOH = 1 : 2

NaOH added = 1.00 × 50.0/1000 = 0.0500 mol

NaOH left in 25.0 cm^3 = n(HCl) = $0.100 \times 20.40/1000 = 2.040 \times 10^{-3}$ mol → in 250.0 $cm^3 = 2.040 \times 10^{-2}$ mol

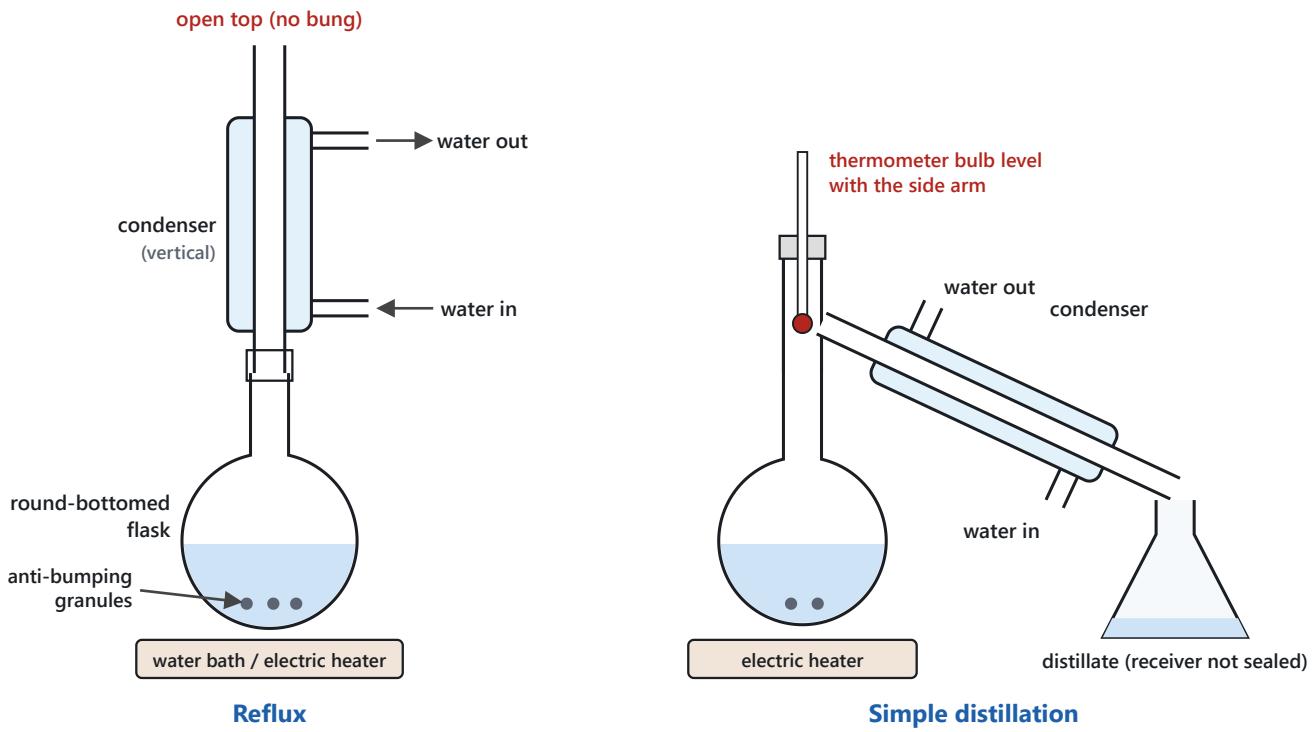
NaOH that reacted = 0.0500 − 0.0204 = 0.0296 mol = $n(NH_4^+)$

$n((NH_4)_2SO_4) = 0.0296 \div 2 = 0.0148$ mol; mass = 0.0148 × 132.1 = 1.955 g

% = 1.955 ÷ 2.50 × 100 = **78.2 %**

Known slips the mark scheme checks for: forgetting ÷2, forgetting ×10, or using the NaOH left instead of the NaOH that reacted.

P5 Organic preparation and purification



Reflux (left): vertical condenser, open top, water in at the bottom. Distillation (right): thermometer bulb level with the side arm, receiver not sealed.

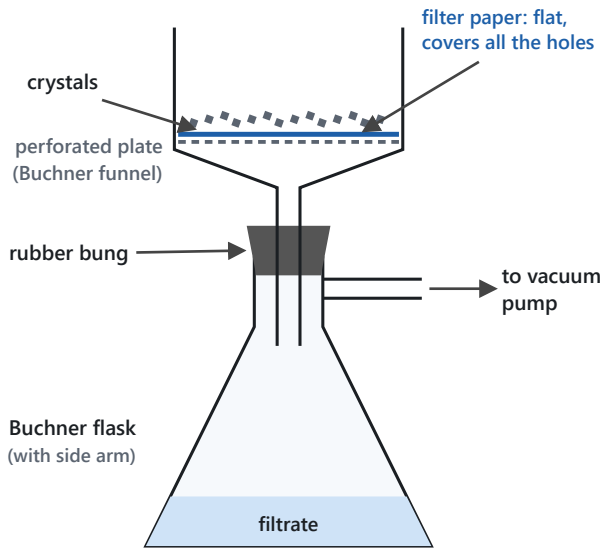
Reflux: heating a reaction mixture for a long time with a **vertical condenser**, so vapour condenses and runs back into the flask.

- Why: heating **increases the rate** so the reaction goes to completion, **and** the condenser stops volatile (flammable, toxic) reactants or products escaping. With acidified dichromate it keeps the intermediate aldehyde in the flask so it is oxidised all the way to the carboxylic acid.
- Diagram marks: round-bottomed (or pear-shaped) flask with liquid in it; heat source; vertical condenser; water **in at the bottom, out at the top**; **open top**; flask and condenser drawn as two pieces joined with no gap; anti-bumping granules.
- Errors you may be asked to spot: a **sealed top** (pressure builds up, it could explode); water flowing the wrong way (condenser doesn't fill, so less effective); a conical flask; no anti-bumping granules (violent, uneven boiling); a Bunsen with a flammable mixture. A thermometer is not an error unless it seals the top.

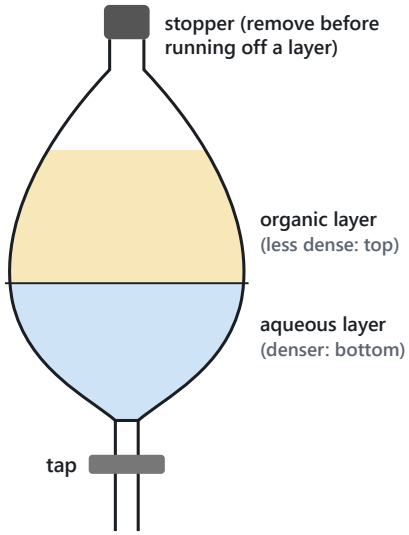
Distillation: separates a liquid by its boiling temperature.

- Thermometer **bulb level with the entrance to the side arm** (it measures the temperature of the vapour that is distilling). Collect the fraction that distils over a narrow range around the boiling point of the product (e.g. within 2 °C).
- Condenser sloping down, water in at the lower end. Receiver open to the air (never a sealed system).
- Distilling **as it forms** (no reflux) collects the aldehyde from the oxidation of a primary alcohol before it can be oxidised further.

Steam distillation: steam is passed through a mixture containing an **immiscible**, heat-sensitive organic liquid (e.g. limonene from orange peel, phenylamine from a reaction mixture). The mixture distils **below 100 °C**, so the product distils without **decomposing** at its own high boiling point. The distillate is **cloudy** (an emulsion of oil in water).



Suction (reduced-pressure) filtration



Separating funnel

Suction filtration: flat filter paper covering all the holes, Buchner flask with side arm to a pump. Separating funnel: layers decided by density.

Separating funnel (washing and solvent extraction)

- Add the wash (water, Na_2CO_3(aq) or $NaHCO_3$(aq)) or the extracting solvent; **stopper**, invert and shake, **opening the tap regularly to release pressure**; clamp and let the layers separate; **remove the stopper**; run off the lower layer.
- Which layer is which: the **less dense** liquid is on top. Most organic liquids (density below 1 g cm^{-3}) are on top; halogenated ones are below. Test by adding a few drops of water and seeing which layer gets bigger.
- **Na_2CO_3 / $NaHCO_3$ wash**: removes acid (unreacted carboxylic acid, the acid catalyst, HCl). It gives off CO_2 ($H^+ + HCO_3^- \rightarrow H_2O + CO_2$), which is why the pressure must be released.
- **Water wash**: removes soluble impurities such as salts, acids, alcohols.
- **Solvent extraction**: the product is more soluble in an immiscible organic solvent (e.g. ether) than in water, so it moves into the organic layer. Several small extractions remove more than one big one. Remove the ether by distillation using a **hot water bath** (very flammable, no flame).
- **Salting out**: adding NaCl to the aqueous layer makes the organic compound less soluble in it.

Drying a liquid: add an anhydrous drying agent (**anhydrous $MgSO_4$, Na_2SO_4, $CaCl_2$**), swirl, and keep adding until the liquid is **clear** (no longer cloudy) and the solid stays **free-flowing** rather than clumping. Then **filter or decant**. Drying agents that are not accepted: anhydrous $CuSO_4$, $CoCl_2$, conc H_2SO_4.

Recrystallisation (asked almost every year)

1. Dissolve the impure solid in the **minimum volume of hot solvent**: the solution will be **saturated** on cooling, so most of the product crystallises out. (Don't heat above the solid's melting point.)
2. **Filter while hot** (pre-warmed funnel) to remove **insoluble impurities**. Hot, so crystals don't form in the funnel.
3. **Cool** the filtrate (then in an ice bath) so the product crystallises. **Soluble impurities** stay in solution because they are present in small amounts.
4. **Filter under reduced pressure** (Buchner) to collect the crystals.
5. **Wash** with a small volume of **ice-cold** solvent, to remove the solution of soluble impurities from the crystals without dissolving them.
6. **Dry**: on filter paper, in a **desiccator**, or in a warm (not hot) oven.

A good recrystallisation solvent dissolves the compound **well when hot** and **poorly when cold**, and doesn't react with it.

Suction (reduced-pressure) filtration: Buchner funnel with a perforated plate, **flat filter paper covering all the holes**, sealed with a bung into a **Buchner flask with a side arm** connected to a vacuum pump. It is **faster** than gravity filtration and gives a **drier** product.

Melting temperature: a small amount of **dry, powdered** sample in a **capillary tube sealed at one end**, held next to the thermometer bulb in an oil bath (Thiele tube) or a melting-point apparatus; heat **slowly** near the melting point; record the range. Pure: sharp and matching the data-book value. Impure: lower and wider range.

Boiling temperature: distil and read the thermometer (bulb at the side arm) while the liquid distils steadily; or use a Thiele tube with an inverted capillary. Atmospheric pressure affects it.

Temperature control in preparations

Preparation	Condition	Why
nitration of an arene (e.g. methyl benzoate)	keep below about 10–15 °C; add the nitrating mixture slowly with an ice bath	exothermic; higher temperature gives **dinitration** and other isomers
diazotisation (phenylamine + HNO_2)	keep **cold**, above 0 °C but no higher than about 5–10 °C	above it the diazonium ion **decomposes** (to a phenol + N_2); below 0 °C too slow / water freezes. HNO_2 is made **in situ** ($NaNO_2$ + HCl) because it is unstable
aspirin, acylation	cool when mixing	mixing is exothermic
flammable solvent	hot water bath or electric heater	no naked flame

Reagent roles: conc H_2SO_4 in esterification and aspirin = **catalyst**. Tin with conc HCl in reducing nitrobenzene = **reducing agent**. Excess of one reagent pushes an equilibrium (or ensures the limiting reagent fully reacts); prove which is in excess by comparing moles.

MARK-SCHEME LANGUAGE

- Ion tests: colour **and** "precipitate" or "solution"; "**dark/deep blue** solution" (not just blue) for copper with excess NH_3; "**off-white**" $Mn(OH)_2$ that "darkens"; "green ppt turns **brown** on standing" for Fe^{2+}.
- Type of reaction: "deprotonation (acid–base)" for NaOH; "ligand exchange (substitution)" for excess NH_3; "redox" when the oxidation number changes.
- Halides: "add **dilute nitric** acid, then silver nitrate"; "white ppt soluble in **dilute** ammonia". Sulfate: "acidify with dilute **HCl**, then $BaCl_2$: **white** ppt".
- Organic: "**orange precipitate**" with Brady's; "**silver mirror**"; "**red precipitate**" with Fehling's; "**no change**" (not "no observation") for a negative; aldehyde group written **–CHO**.
- Titrations: "permanent pale pink"; starch "near the end-point (**straw/pale yellow**)"; "**blue-black to colourless**"; "dilute **sulfuric** acid in excess".
- Reflux: "heat to **increase the rate**" + "**condenser prevents loss of volatile** reactants/products". Recrystallisation: "**minimum** volume of **hot** solvent", "filter hot to remove **insoluble** impurities", "wash with **cold** solvent". Suction filtration: "**faster**" and "**drier**". Drying: "add until the liquid is **clear**" / "solid no longer clumps".

Often asked as: identify ions A–D from a table of observations; complete observation/inference cells; state the type of reaction; write the formula of a complex; choose one test to distinguish a pair; deduce a structure from tests plus IR, MS and NMR; why starch is added late; why H_2SO_4 and not HCl; spot three errors in a reflux or filtration diagram; why a sodium carbonate wash; describe recrystallisation and say what each step removes.

COMMON MISTAKE

Writing a colour without "precipitate" or "solution". "Blue" instead of "dark blue" for $[Cu(NH_3)_4(H_2O)_2]^{2+}$.
Calling the reaction of NaOH with an aqua ion "ligand exchange". It is **deprotonation**.
Hexaammine copper $[Cu(NH_3)_6]^{2+}$, or $[Zn(NH_3)_6]^{2+}$ and $[Co(NH_3)_4]^{2+}$: copper and zinc take **four** NH_3, cobalt takes **six**.
Giving "pale pink" for $Mn^{2+}(aq)$: treat it as colourless. Confusing off-white $Mn(OH)_2$ with red-brown $Fe(OH)_3$.
Mixing the tests: HCl before $AgNO_3$, or H_2SO_4 before $BaCl_2$.
PCl_5 or sodium to tell an alcohol from a carboxylic acid: both react. Use $NaHCO_3$.
Fragment ions with no + charge, a negative charge, or a dangling bond.
"Copper(II) oxide" for Fehling's: the red ppt is **copper(I) oxide**.
Reflux diagrams with a bung in the top, or water entering at the top.

BEYOND THE SPEC, BUT EXAMINED

- **Silver halide ppt filtered off before testing with ammonia**, because a coloured metal hydroxide (e.g. green $Fe(OH)_2$) would also form and hide the result (WCH16 Jan 2023).
- **Feasibility vs E⊖**: Cu^{2+} oxidising I^- looks unfeasible from E⊖ values (E⊖cell = −0.39 V), but it happens because CuI precipitates, removing Cu^+ and raising E for Cu^{2+}/Cu^+ (WCH16 Jun 2022).
- **Salting out** with NaCl, and the cloudy steam distillate, in making phenylamine (WCH16 Jan 2022).
- **Thermometer bulb level with the side arm** and a pear-shaped (not conical) flask for distillation (WCH16 Oct 2021).
- A solid product can't be dried with a solid drying agent: you couldn't separate them. Dry it in a desiccator instead (WCH16 Jun 2023).
- Lone pair on **oxygen** delocalised **into the ring** explains why phenol brominates without a halogen carrier (WCH16 Jan 2025).

Core practicals

SPEC CP9 Rates: iodine–propanone by titration (9a) and a clock reaction (9b)

WHAT THIS SPEC POINT ASKS

9a: follow the acid-catalysed reaction of iodine with propanone by taking samples, quenching them and titrating the iodine left. 9b: use a clock reaction to measure initial rates and find orders.

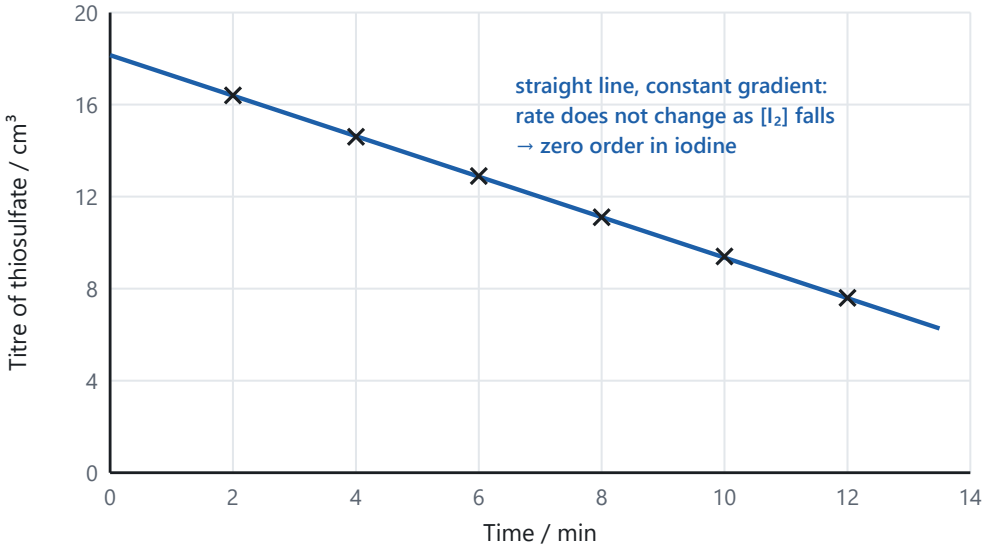
CP9 9a Aim and principle

$$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)\ (H^+ \text{ catalyst})$$

- Propanone and acid are in **large excess**, so their concentrations hardly change. Only $[I_2]$ changes, so the experiment shows how the rate depends on **iodine alone**.
- Samples are removed at timed intervals and **quenched** in sodium hydrogencarbonate solution. $NaHCO_3$ **neutralises the acid catalyst**, so the reaction stops and the iodine left is fixed.
- The iodine left is titrated with sodium thiosulfate: $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$. The **titre is directly proportional to $[I_2]$**, so titre can be plotted instead of concentration.

CORE PRACTICAL 9A: METHOD IN BRIEF

1. Mix 25 cm^3 of 1.0 mol dm^{-3} propanone and 25 cm^3 of 1.0 mol dm^{-3} H_2SO_4 in a flask.
2. Add 50 cm^3 of 0.020 mol dm^{-3} iodine solution, swirl and **start the clock**.
3. At about 5 minutes, withdraw a **10.0 cm^3 sample** with a pipette and run it into a conical flask containing about 10 cm^3 of $NaHCO_3(aq)$. Record the time when half the sample has run in.
4. Titrate the iodine with 0.010 mol dm^{-3} $Na_2S_2O_3$ until **pale yellow**, add starch, continue until **blue-black → colourless**.
5. Repeat sampling every few minutes (6–8 samples).
6. Plot **titre against time**.
7. To find the order for propanone and for H^+, repeat with a different starting concentration of one of them (others the same) and compare the gradients.



Iodine–propanone: titre falls in a straight line, so the rate stays constant as $[I_2]$ falls. The reaction is zero order with respect to iodine.

CP9 9a Variables and safety

- IV: time. DV: titre (∝ $[I_2]$). Control: temperature (water bath), sample volume (pipette), concentrations of thiosulfate and $NaHCO_3$.
- Propanone: **flammable**, no naked flames. Iodine solution: harmful, stains; gloves. Sulfuric acid: irritant at this concentration; goggles.

CP9 9a Processing the results

- **Straight line with a negative gradient** → rate doesn't change as $[I_2]$ falls → **zero order with respect to I_2.**
- Doubling [propanone] doubles the gradient, and doubling $[H^+]$ doubles it too, so:

$$\text{rate} = k[CH_3COCH_3][H^+]$$

- Iodine is not in the rate equation, so it reacts **after** the rate-determining step. The slow step involves propanone and H^+ (acid-catalysed formation of the enol); iodine then reacts quickly with the enol.

WORKED EXAMPLE: RATE FROM THE TITRE–TIME GRAPH

Samples of 10.0 cm^3 were titrated with 0.0100 mol dm^{-3} $Na_2S_2O_3$. Titres: 16.4 cm^3 at 2 min and 7.6 cm^3 at 12 min (on the straight line).
gradient = (7.6 − 16.4) ÷ (12 − 2) = −0.88 cm^3 min^{-1}
thiosulfate per minute = 0.88/1000 × 0.0100 = 8.8×10^{-6} mol min^{-1} → I_2 = 4.4×10^{-6} mol min^{-1} (ratio 1 : 2)
in 10.0 cm^3: rate = $4.4 \times 10^{-6} \div 0.0100$ dm^3 = 4.4×10^{-4} mol dm^{-3} min^{-1} = **7.3×10^{-6} mol dm^{-3} s^{-1}**

CP9 9b Aim and principle: the clock reaction

The Harcourt–Esson reaction between peroxodisulfate and iodide:

$$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq) \text{ (slow)}$$

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq) \text{ (fast)}$$

- A **small, fixed amount** of thiosulfate and some starch are in the mixture. The thiosulfate reacts with the iodine as fast as it forms, so the **colour change is delayed**.
- When all the thiosulfate is used up, free I_2 appears and turns the starch **blue-black** suddenly. The time t is recorded.
- The same amount of iodine has formed by time t in every run, so **rate ∝ 1/t**.
- This is only an **initial rate** if very little reactant has been used by time t, i.e. the thiosulfate is small compared with the other reactants, so the rate is almost constant during the timed period.

Other clocks you may meet use the same idea: $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$ with thiosulfate and starch; or bromate(V) + bromide + acid making Br_2, with **phenol** removing the Br_2 until it is used up, after which the Br_2 **bleaches methyl red** (see CP10).

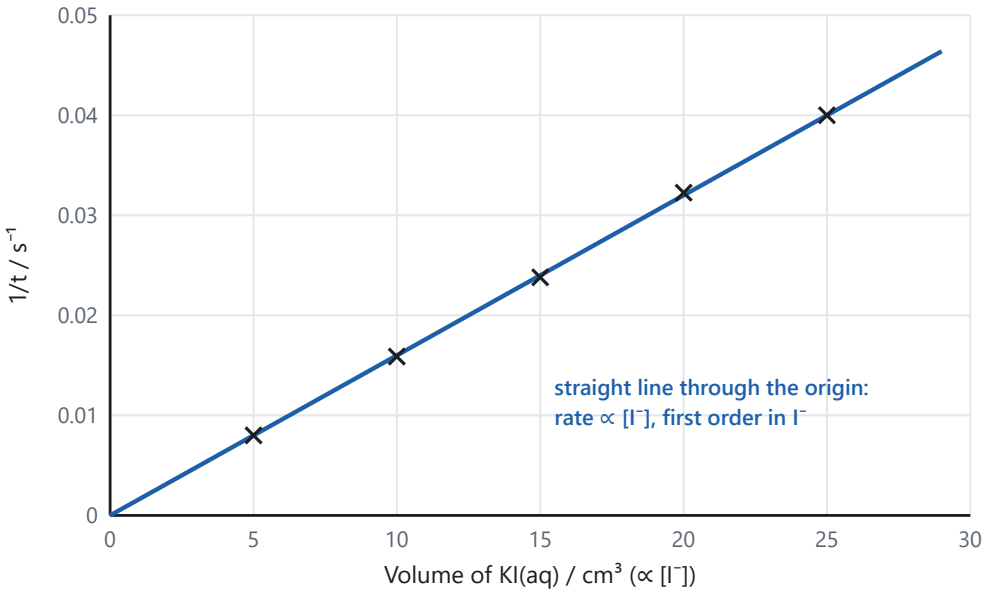
CORE PRACTICAL 9B: METHOD IN BRIEF

1. Using burettes, put into a beaker: 10.0 cm^3 KI(aq), 5.0 cm^3 $Na_2S_2O_3$(aq), 2.0 cm^3 starch, and water so that the **total volume is always the same**.
2. Put 10.0 cm^3 of $K_2S_2O_8$(aq) in a separate boiling tube (both at the same temperature).
3. Pour the peroxodisulfate in, mix, and **start the clock**. Stop it when the blue-black colour appears. Record the temperature.
4. Repeat with 5, 15, 20, 25 cm^3 of KI(aq), changing the water to keep the total volume constant.
5. Then keep $[I^-]$ fixed and vary the volume of $K_2S_2O_8$(aq) the same way.

6. Plot **1/t against volume** of the reactant varied.

CP9 9b Variables and safety

- IV: volume (∝ concentration) of one reactant. DV: time to colour change (1/t ∝ rate). Control: **total volume** (add water), volume of thiosulfate and starch, temperature (water bath), concentration of the other reactant.
- A **burette** (or graduated pipette) is used for the small, different volumes so the total volume is exact; a measuring cylinder is too inaccurate.
- Potassium peroxodisulfate is an **oxidising** solid and can cause skin and breathing sensitisation; avoid contact and dust. Low concentrations of the others are low hazard.



Clock reaction: 1/t against volume of KI(aq) (total volume constant) is a straight line through the origin, so the reaction is first order in I^-.

WORKED EXAMPLE: ORDER FROM A CLOCK REACTION

Total volume 50.0 cm^3 in every run. KI(aq) is 0.200 mol dm^{-3}.

Volume KI / cm^3	**5.0**	**10.0**	**15.0**	**20.0**	**25.0**
t / s	125	63	42	31	25
1/t / s^{-1}	0.0080	0.0159	0.0238	0.0323	0.0400

Doubling the volume from 5.0 to 10.0 cm^3 halves t, so 1/t (the rate) doubles. The graph of 1/t against volume is a **straight line through the origin**, so rate ∝ $[I^-]$: **first order in I^-**.
$[I^-]$ in the 10.0 cm^3 run = 0.200 × 10.0 ÷ 50.0 = 0.0400 mol dm^{-3} (calculate it; don't read it off the table).
If the peroxodisulfate runs give the same result: rate = $k[S_2O_8^{2-}][I^-]$, **second order overall**.

CP9 Errors and improvements (9a and 9b)

- Judging the colour change by eye is subjective: use a **light sensor or colorimeter**, or a white tile behind the beaker.
- Temperature drifts: use a **water bath**; bring both solutions to temperature before mixing; record the temperature of each mixture.
- Clock: the start time is uncertain if mixing is slow; pour quickly from a boiling tube.

- 9a: the time of quenching is uncertain (half the pipette); $NaHCO_3$ quench must be **immediate**; if titrating is delayed and quenching is incomplete, iodine keeps reacting and titres come out too small.
- A very short t (under ~10 s) has a large percentage uncertainty: use more dilute solutions.

MARK-SCHEME LANGUAGE

- 9a: "acid and propanone in (large) excess so **only [I_2] changes**" (not "iodine is limiting"); $NaHCO_3$ "**neutralises/removes the H^+ catalyst**, so the reaction stops"; "titre is **proportional to** the concentration of iodine"; "**straight line**, so **zero order**"; rate equation **with k**: rate = k[CH_3COCH_3][H^+].
- 9b: thiosulfate "**reacts with the iodine** as it forms" + "so the colour change is **delayed**". Not "slows the reaction", not "a catalyst", not "quenches".
- "Total volume constant, so **volume is proportional to concentration**."
- "1/t is proportional to rate because the **same amount** of iodine forms each time"; valid only when the amount of thiosulfate is **small** so the rate is nearly constant.
- Starch colour in a forward clock: "colourless to **blue-black**".

Often asked as: why the quench works; why the reactants are in excess; plot titre or 1/t and deduce the order; calculate [I^-] in a mixture; write the rate equation; find half-lives on a given graph; name two control variables; why a burette, not a cylinder; why the rate is taken as 1/t.

COMMON MISTAKE

Leaving **k** out of the rate equation, or putting [I_2] in it.
Saying the overall order from data where one reactant (water, OH^-, H^+) was in large excess: you can only find the order for the reactant that changed.
Reading [I^-] off the table instead of calculating it from volume × concentration ÷ total volume.
Measuring a second half-life from t = 0, so the two look different.

BEYOND THE SPEC, BUT EXAMINED

- **Other quenches**: propanone added to slow (not stop) a haloalkane hydrolysis, which is why titrations must be done straight away (WCH16 Jan 2023); KI added to use up MnO_4^- in a sample, then the I_2 titrated (WCH16 Jun 2024); ice to slow an ester hydrolysis (WCH16 Oct 2022).
- **Autocatalysis** by Mn^{2+} in the MnO_4^-/ethanedioate reaction: the concentration–time curve is slow, then fast, then slow, so it must be drawn as a smooth S-shaped curve, not a straight line (WCH16 Jun 2024).
- **Ester hydrolysis followed by titrating the acid formed**: plot ($V_\infty - V_t$), which is proportional to the concentration of the **ester**; constant half-life → first order (WCH16 Oct 2022).
- Continuous monitoring with a **colorimeter** of a fading dye; OH^- in large excess so only the order in the dye can be found (WCH16 Jan 2025).

SPEC CP10 Finding the activation energy of a reaction

WHAT THIS SPEC POINT ASKS

Measure the rate of a reaction at several temperatures and use an Arrhenius plot (ln k or ln 1/t against 1/T) to find the activation energy.

CP10 Aim and principle

$$k = A\,e^{-Ea/RT} \text{ so } \ln k = -Ea/R \times (1/T) + \ln A$$

- A graph of **ln k against 1/T** is a straight line with **gradient = −Ea/R**. R = 8.31 J mol^{-1} K^{-1}; T in **kelvin**.
- In a clock reaction the same amount of reaction happens by time t at every temperature, so **k ∝ 1/t**. Plot **ln(1/t)** against 1/T: the gradient is the same.
- The usual reaction is bromate(V) with bromide in acid:

$$BrO_3^-(aq) + 5Br^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

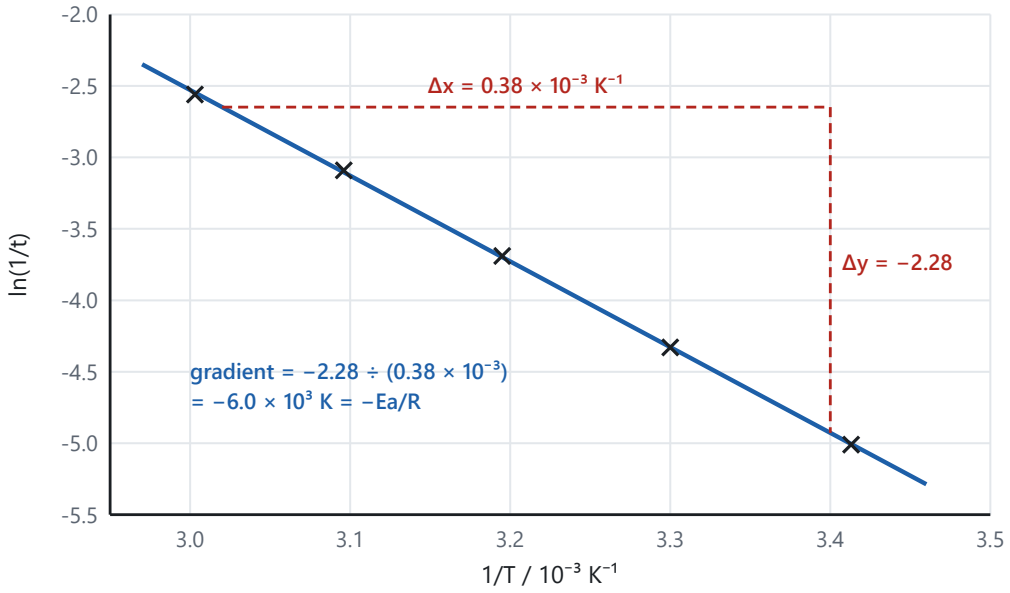
A fixed amount of **phenol** reacts with the bromine as it forms. Once the phenol is used up, free bromine **bleaches the methyl red** indicator. Time t is from mixing until the red colour disappears. (Your school may use another clock, such as peroxodisulfate–iodide, or $CaCO_3$ + HCl timed until the solid disappears; the analysis is the same.)

CORE PRACTICAL 10: METHOD IN BRIEF

1. In one boiling tube put fixed volumes of KBr/$KBrO_3$ solution, phenol solution and methyl red (from burettes). In a second tube put a fixed volume of dilute H_2SO_4.
2. Stand both tubes in a water bath at about 20 °C until both reach the bath temperature (check with a thermometer).
3. Pour the acid into the first tube, start the clock, mix, keep in the bath. Stop the clock when the red colour disappears. Record the temperature of the mixture.
4. Repeat at about 30, 40, 50 and 60 °C, using the **same volumes** each time.
5. Calculate 1/T (T in K) and ln(1/t). Plot **ln(1/t) against 1/T**, draw the best-fit line and find the gradient.
6. Ea = −gradient × R.

CP10 Variables and safety

- IV: temperature. DV: time t (rate ∝ 1/t). Control: volumes and concentrations of every solution (same total volume), amount of phenol and indicator.
- **Phenol** is **corrosive** and toxic by skin absorption: wear **gloves**. Bromine formed is toxic: use small amounts, ventilate. Hot water bath: take care.



Arrhenius plot from the worked example. Use a large gradient triangle on the line, not the data points.

WORKED EXAMPLE: ARRHENIUS GRAPH GRADIENT → EA

T / °C	20	30	40	50	60
T / K	293	303	313	323	333
t / s	150	76	40	22	13
$1/T$ / 10^{-3} K^{-1}	3.41	3.30	3.19	3.10	3.00
ln(1/t)	−5.01	−4.33	−3.69	−3.09	−2.56

Two points **on the best-fit line**: (3.02 × 10^{-3}, −2.65) and (3.40 × 10^{-3}, −4.93)
gradient = (−4.93 − (−2.65)) ÷ ((3.40 − 3.02) × 10^{-3}) = −2.28 ÷ (0.38 × 10^{-3}) = −6.0 × 10^{3} K
Ea = −gradient × R = 6.0 × 10^{3} × 8.31 = 49 900 J mol^{-1} = **+49.9 kJ mol^{-1}** (≈ +50 kJ mol^{-1})
Check: Ea must be **positive**. A negative answer means a sign slip.

WORKED EXAMPLE: TWO-POINT METHOD (NO GRAPH)

t = 150 s at 293 K and t = 40 s at 313 K. Take k ∝ 1/t.
Write ln k = −Ea/RT + ln A for each and subtract, so ln A cancels:
$\ln(k_2/k_1) = (Ea/R)(1/T_1 - 1/T_2)$
ln(150/40) = 1.322; 1/293 − 1/313 = 3.4130 × 10^{-3} − 3.1949 × 10^{-3} = 2.181 × 10^{-4} K^{-1}
Ea = 8.31 × 1.322 ÷ (2.181 × 10^{-4}) = 50 400 J mol^{-1} = **+50.4 kJ mol^{-1}**
Keep 1/T unrounded: rounding 1/T to 3 s.f. here changes Ea a lot.

CP10 Errors and improvements

- Temperature falls while the reaction runs (especially at 60 °C): keep the tube in the bath; record the temperature at the start and end and use the mean.
- Judging when the colour disappears: same white background; a colorimeter or light sensor.
- Both solutions must be at the bath temperature **before** mixing.
- At high temperature t is short, so its percentage uncertainty is large; at low temperature the reaction is slow but fine.
- More temperatures give a more reliable line.

MARK-SCHEME LANGUAGE

- "Plot **ln(1/t)** (or ln k, ln rate) against **1/T**"; "find the **gradient**"; "Ea = −gradient × R" (multiply by 8.31). "ln 1/T" or "1/t against T" is not credited.
- Answer: **positive**, with units **kJ mol^{-1}** (or J mol^{-1}), 2–3 s.f.
- "1/t is a measure of the rate because the **same amount** of reaction has taken place (same amount of phenol / $CaCO_3$ used up)".
- "Burette (not measuring cylinder) for small volumes so the **total volume is constant**." "Water-bath temperature checked in both tubes **before** mixing."

Often asked as: complete a table of 1/T and ln(1/t); plot the Arrhenius graph and find Ea from the gradient; Ea from two temperatures; explain how the graph gives Ea; why phenol is needed; suggest an improvement to temperature control; hazard of phenol.

COMMON MISTAKE

Using °C instead of K. Using $\log_{10}$ instead of ln.
Ea = gradient × R (gives a negative Ea). Leaving out "per mole" in the unit.
Using the two separate equations and getting two different Ea values instead of subtracting them.
Treating the constant term in a given equation as if it were k.

BEYOND THE SPEC, BUT EXAMINED

- The **intercept** on an ln k against 1/T graph is ln A at 1/T = 0, far off the left of any real graph, so a suggested value of A can't be checked from the plotted data (WCH16 Jan 2025).
- $CaCO_3$ + HCl timed until the solid disappears: HCl in large **excess** (show by moles) so its concentration is effectively constant, and the same mass of $CaCO_3$ reacts each time (WCH16 Oct 2025).

SPEC **CP11**

Finding the Ka value for a weak acid

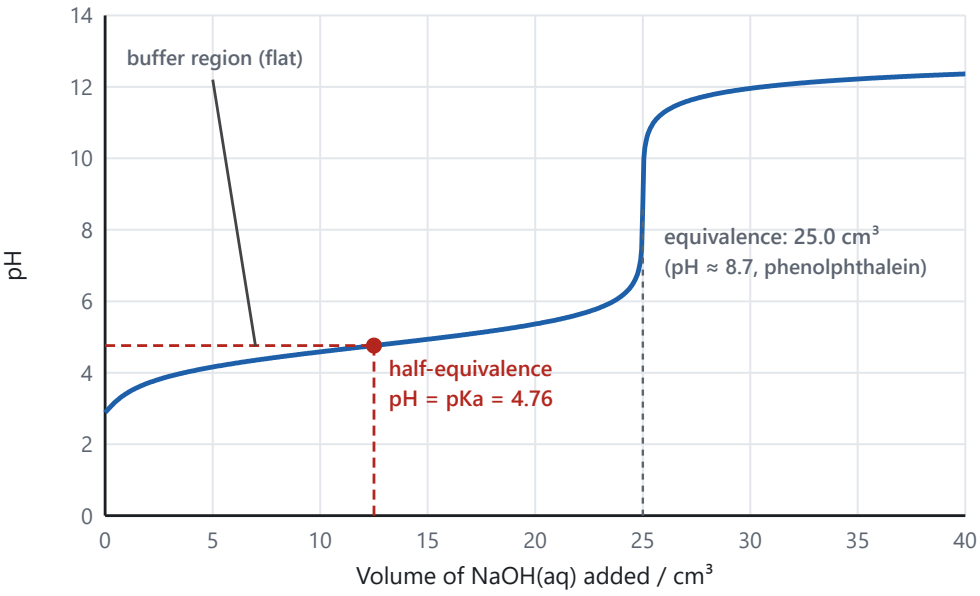
WHAT THIS SPEC POINT ASKS

Find Ka of a weak acid (e.g. ethanoic acid) by measuring the pH of a half-neutralised solution, using a titration to find the equivalence volume.

CP11 Aim and principle

$$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq) \quad Ka = [H^+][CH_3COO^-] \div [CH_3COOH]$$

- When the acid is **half-neutralised**, $[CH_3COOH] = [CH_3COO^-]$, so they cancel: **Ka = $[H^+]$**, i.e. **pKa = pH**.
- So: find the equivalence volume by titration, make a half-neutralised solution, measure its pH with a calibrated pH meter, then **Ka = 10^{-pH}**.



25.0 cm^3 of 0.100 mol dm^{-3} ethanoic acid with 0.100 mol dm^{-3} NaOH. At half the equivalence volume, pH = pKa.

CORE PRACTICAL 11: METHOD IN BRIEF

1. **Calibrate the pH meter** with buffer solutions of known pH (e.g. pH 4 and pH 7). Rinse the probe with distilled water between solutions.

2. Pipette 25.0 cm^3 of the weak acid into a conical flask, add phenolphthalein and titrate with NaOH(aq) from a burette until the **first permanent pink**. Repeat to concordant titres. Mean titre = V.
3. Pipette a fresh 25.0 cm^3 of acid and add exactly **V/2** of NaOH from the burette. (Or: neutralise 25.0 cm^3 exactly, then add a further 25.0 cm^3 of the acid.) Either way, [HA] = $[A^-]$.
4. Measure the pH of this half-neutralised solution. pKa = pH; Ka = 10^{-pH}.
5. Alternative: add NaOH in 1 cm^3 portions (smaller near the end-point), recording pH each time; plot pH against volume; read the pH at half the equivalence volume.

CP11 Variables and safety

- Measured: titre V, pH. Control: temperature (Ka depends on temperature), same acid sample, calibrated meter.
- NaOH(aq) is **corrosive/irritant** to eyes: goggles. Ethanoic acid at this concentration is an irritant.

CP11 Processing the results

WORKED EXAMPLE: KA FROM THE HALF-NEUTRALISATION PH

25.0 cm^3 of ethanoic acid needs 25.0 cm^3 of NaOH(aq). A fresh 25.0 cm^3 with 12.5 cm^3 of NaOH has pH 4.76.

pKa = pH = 4.76

Ka = $10^{-4.76}$ = **1.74×10^{-5} mol dm^{-3}**

Cross-check from the pH of the acid alone: 0.100 mol dm^{-3} acid has pH 2.88, so $[H^+] = 10^{-2.88} = 1.32 \times 10^{-3}$ mol dm^{-3}.

Assuming $[H^+] = [A^-]$ and [HA] ≈ 0.100: Ka = $(1.32 \times 10^{-3})^2 \div 0.100$ = **1.74×10^{-5} mol dm^{-3}**

CP11 Errors and improvements

- pH meter not calibrated or not rinsed: systematic error. Calibrate with two buffers either side of the expected pH.
- End-point judged imprecisely, so V/2 is not exactly half: repeat to concordant titres; add NaOH dropwise near the end.
- Temperature different from 25 °C: Ka changes; record it.
- In the pH-curve method, use **smaller additions near the end-point** so the equivalence volume isn't missed.

MARK-SCHEME LANGUAGE

- "pKa = pH when the acid is **half-neutralised**" because "[HA] = $[A^-]$"; "Ka = 10^{-pH}".
- Method points: titrate to find the end-point volume with an indicator; add **half that volume** to a **fresh** sample of acid (same volume); measure the pH with a **pH meter**.
- "Pipette for the 20.0/25.0 cm^3 of acid"; "rinse the burette with NaOH"; "smaller portions near the end-point".
- Explanations of how to do the calculation are not needed in a method answer.

Often asked as: outline a method to find Ka; why pH = pKa at half-neutralisation; calculate Ka from a pH; read pKa from a pH curve; choose an indicator; suggest why the value differs from the data book.

COMMON MISTAKE

Measuring the pH **at the equivalence point** instead of at half-neutralisation.
Saying "pH = pKa" without describing how you make the half-neutralised solution.
Using an uncalibrated meter, or universal indicator paper, to measure a pH to 2 d.p.

BEYOND THE SPEC, BUT EXAMINED

- For the **first** Ka of a diprotic acid (ethanedioic acid), the first end-point is at low pH, so phenolphthalein would show the second end-point instead; use a pH probe and the first half-equivalence point (WCH16 Oct 2024).
- Saturated **$Ca(OH)_2$** left for a day to reach equilibrium, filtered and titrated with HCl to find [OH^-] and Kc (WCH16 Jun 2025).

SPEC **CP12**

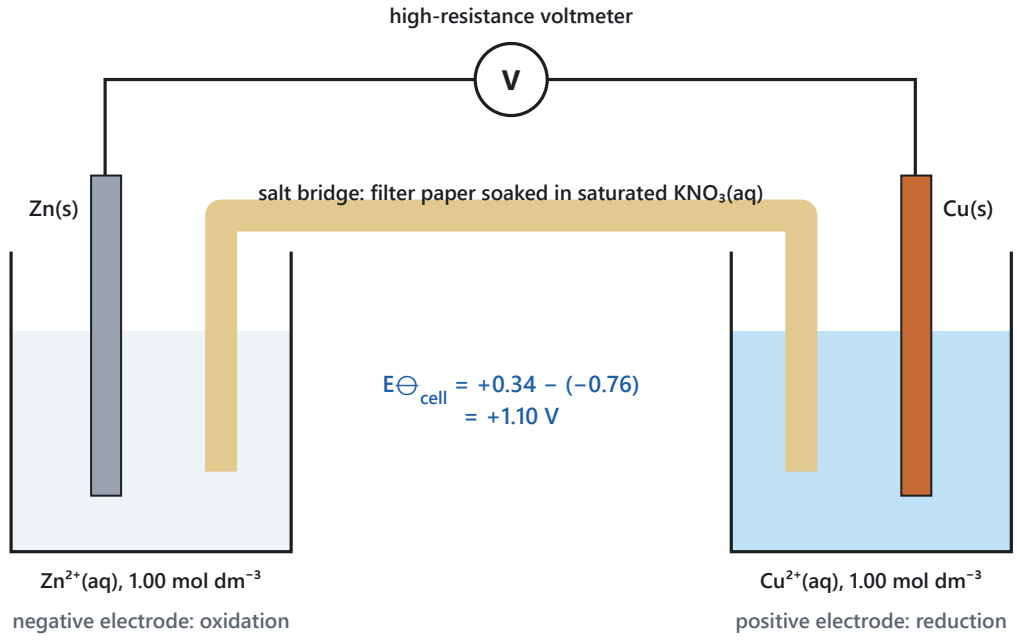
Investigating some electrochemical cells

WHAT THIS SPEC POINT ASKS

Set up cells from two half-cells joined by a salt bridge, measure the cell emf with a high-resistance voltmeter, and compare it with the value from standard electrode potentials.

CP12 Aim and principle

- A **half-cell** is a metal in a solution of its ions (Zn | Zn^{2+}), or **platinum** in a solution containing both oxidation states of an ion (Pt | Fe^{3+}, Fe^{2+}).
- Two half-cells are joined by a **wire through a high-resistance voltmeter** and a **salt bridge**. The salt bridge lets ions move to complete the circuit without mixing the solutions.
- $E^\ominus_{cell}$ = $E^\ominus$(more positive, reduction) − $E^\ominus$(less positive, oxidation). It must come out **positive** for the feasible direction.
- Standard conditions: **1.00 mol dm^{-3}** solutions, **298 K**, gases at **100 kPa (1 atm)**.



A Zn | Zn^{2+} || Cu^{2+} | Cu cell. Zinc is the negative electrode, where oxidation happens.

CORE PRACTICAL 12: METHOD IN BRIEF

1. Clean the metal strips with emery paper (remove the oxide layer), then rinse.
2. Half-fill two beakers with 1.00 mol dm^{-3} solutions (e.g. $ZnSO_4$ and $CuSO_4$) and put each metal in its own solution.
3. Make a **salt bridge**: a strip of filter paper soaked in **saturated KNO_3(aq)**, dipping into both solutions.
4. Connect the metals to a **high-resistance voltmeter** with leads and crocodile clips. Record the reading (it should be steady).
5. Repeat with other pairs (e.g. Cu/Ag, Zn/Fe, Cu with Fe^{3+}/Fe^{2+} using a **platinum** electrode). Use a fresh salt bridge each time.
6. Compare each measured emf with $E^{\ominus}$cell from the data booklet.

CP12 Why each choice

- **High-resistance voltmeter**: almost no current flows, so you measure the maximum potential difference (the emf).
- **KNO_3**: its ions are soluble, don't react with the electrodes or solutions and don't form precipitates. **KCl** is unsuitable with Ag^+ (white AgCl forms); KOH is wrong (forms hydroxide precipitates).
- **Platinum**: inert, conducts, and gives a surface for the electron transfer between Fe^{3+} and Fe^{2+}.

WORKED EXAMPLE: CELL EMF AND CELL DIAGRAM

$E^{\ominus}$ data: Zn^{2+}/Zn −0.76 V; Cu^{2+}/Cu +0.34 V; Fe^{3+}/Fe^{2+} +0.77 V.
(a) Zinc–copper: $E^{\ominus}$cell = +0.34 − (−0.76) = **+1.10 V**
Cell diagram: Zn(s) | Zn^{2+}(aq) || Cu^{2+}(aq) | Cu(s)
Overall: Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s)
(b) Copper with Fe^{3+}/Fe^{2+}: $E^{\ominus}$cell = +0.77 − (+0.34) = **+0.43 V**
Cell diagram: Cu(s) | Cu^{2+}(aq) || Fe^{3+}(aq), Fe^{2+}(aq) | Pt(s)
Overall: Cu(s) + 2Fe^{3+}(aq) → Cu^{2+}(aq) + 2Fe^{2+}(aq). $E^{\ominus}$ is **not** doubled when the half-equation is doubled.

CP12 Variables, safety and errors

- Control: concentration (1.00 mol dm^{-3}), temperature, clean electrodes.
- Hazards: $CuSO_4$ and $AgNO_3$ are harmful/irritant and **toxic to aquatic life**: dispose of as directed, not down the sink. $AgNO_3$ stains skin: gloves.
- Why measured values differ from the data booklet: concentrations not exactly 1.00 mol dm^{-3}; temperature not 298 K; oxide on the metal; resistance of the salt bridge; a small current flowing.

MARK-SCHEME LANGUAGE

- "(High-resistance) **voltmeter**" (not "voltameter", not a power supply).
- "Salt bridge: filter paper soaked in (saturated) **potassium nitrate** solution" (or KCl, $NaNO_3$ where no precipitate forms). KOH is wrong.
- "**Platinum** electrode" for a half-cell with two aqueous ions.
- Overall equation with a single arrow; $E^{\ominus}$cell with **sign** and **V**.

Often asked as: label a cell diagram; spot three mistakes (power supply instead of voltmeter, wire instead of salt bridge, iron instead of platinum); calculate $E^{\ominus}$cell; write the cell diagram and overall equation; say why KNO_3 and why a high-resistance voltmeter; explain why the measured emf differs from $E^{\ominus}$cell.

COMMON MISTAKE

Doubling E⊖ for a half-equation multiplied by two.
Using a wire as a salt bridge, or replacing **both** metals with platinum.
A negative E⊖cell for the cell as set up: subtract the **less** positive from the **more** positive.

BEYOND THE SPEC, BUT EXAMINED

- Predicting feasibility from E⊖, then explaining why a reaction "predicted not to happen" does: non-standard conditions, e.g. **CuI precipitating** removes Cu^+ and shifts the equilibrium (WCH16 Jun 2022).

SPEC CP13 Redox titrations: iron(II) with manganate(VII) (13a) and thiosulfate with iodine (13b)

WHAT THIS SPEC POINT ASKS

13a: titrate iron(II) ions with acidified potassium manganate(VII). 13b: titrate iodine with sodium thiosulfate using starch. Use the results in multi-step calculations.

CP13 13a Aim and principle

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$$

- **MnO_4^- : Fe^{2+} = 1 : 5** (Mn goes +7 → +2, gaining 5 electrons; each Fe^{2+} loses 1).
- **Self-indicating**: purple MnO_4^- is decolourised while Fe^{2+} remains; the end-point is the **first permanent pale pink** (the first drop of excess MnO_4^-).
- Acidify with **excess dilute sulfuric acid**. HCl is oxidised by MnO_4^- (titre too large); HNO_3 is itself an oxidising agent. Too little acid → brown MnO_2.

CORE PRACTICAL 13A: METHOD IN BRIEF

1. Weigh the iron(II) sample (e.g. iron tablets or ammonium iron(II) sulfate) and dissolve it in **dilute sulfuric acid**; make up to 250.0 cm^3 (or 100.0 cm^3) in a volumetric flask with washings; invert to mix.
2. Pipette 25.0 cm^3 into a conical flask and add about 10 cm^3 of dilute H_2SO_4 (measuring cylinder; it is in excess).
3. Titrate with $KMnO_4$(aq) from the burette, swirling, until a **pale pink colour stays** (permanent for ~30 s).
4. Rough titration, then accurate ones to concordant titres.

WORKED EXAMPLE: IRON IN AN IRON TABLET

One tablet (0.650 g) is dissolved in dilute H_2SO_4 and made up to 100.0 cm^3. 25.0 cm^3 portions need a mean of 29.50 cm^3 of 0.00200 mol dm^{-3} $KMnO_4$.
$n(MnO_4^-) = 0.00200 \times 29.50/1000 = 5.90 \times 10^{-5}$ mol
$n(Fe^{2+})$ in 25.0 $cm^3 = 5 \times 5.90 \times 10^{-5} = 2.95 \times 10^{-4}$ mol
$n(Fe^{2+})$ in 100.0 $cm^3 = 2.95 \times 10^{-4} \times 4 = 1.18 \times 10^{-3}$ mol (scale factor is ×4 here, not ×10)
mass of iron = $1.18 \times 10^{-3} \times 55.8 = 0.0658$ g = **65.8 mg per tablet** (10.1 % by mass)

CP13 13b Aim and principle

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

- Used to find **oxidising agents**: the oxidant is added to **excess KI**, which it oxidises to I_2; the iodine is then titrated with standard thiosulfate.
- Examples: $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$ (IO_3^- : $S_2O_3^{2-}$ = 1 : 6); $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$ (**Cu^{2+} : $S_2O_3^{2-}$ = 1 : 1**).
- Indicator: **starch**, added when the iodine colour has faded to **pale yellow (straw)**; end-point **blue-black → colourless**. With copper, the off-white CuI precipitate stays, so the end-point mixture looks white/cream.

CORE PRACTICAL 13B: METHOD IN BRIEF (COPPER IN BRASS)

1. Weigh the brass. In a **fume cupboard**, dissolve it in concentrated nitric acid (brown NO_2 forms; the solution goes blue-green).
2. Dilute, **cool**, then add Na_2CO_3(aq) slowly until a faint permanent precipitate forms (the acid is neutralised); add ethanoic acid dropwise until it just clears.
3. Transfer with washings into a **250.0 cm^3 volumetric flask**, make up to the mark, invert.
4. Pipette 25.0 cm^3 into a conical flask; add about 10 cm^3 of KI(aq) (**excess**, so a measuring cylinder is fine). A white precipitate (CuI) forms in a brown iodine solution.
5. Titrate with 0.100 mol dm^{-3} $Na_2S_2O_3$ until **pale yellow**, add starch (blue-black), continue dropwise until the **blue-black colour just disappears**.

WORKED EXAMPLE: PERCENTAGE OF COPPER IN BRASS

1.75 g of brass is dissolved and made up to 250.0 cm^3. 25.0 cm^3 portions with excess KI need a mean titre of 17.60 cm^3 of 0.100 mol dm^{-3} $Na_2S_2O_3$.
$n(S_2O_3^{2-}) = 0.100 \times 17.60/1000 = 1.760 \times 10^{-3}$ mol
$2Cu^{2+} \rightarrow I_2 \rightarrow 2S_2O_3^{2-}$, so $n(Cu^{2+})$ in 25.0 cm^3 = 1.760×10^{-3} mol
n(Cu) in 250.0 cm^3 = 1.760×10^{-2} mol
mass Cu = $1.760 \times 10^{-2} \times 63.5 = 1.118$ g
% Cu = 1.118 ÷ 1.75 × 100 = **63.9 %**

CP13 Variables, safety and errors (13a and 13b)

- Hazards: **conc HNO_3 corrosive → gloves**; **NO_2 toxic → fume cupboard**; $KMnO_4$ oxidising and harmful, stains; dilute H_2SO_4 irritant.
- Cool before adding Na_2CO_3: the neutralisation is vigorous and the solution could froth over.
- Errors: unneutralised HNO_3 oxidises I^- → titre too high → % Cu too high. Starch added too early → iodine–starch complex, end-point unclear. Not enough acid in 13a → MnO_2 forms, titre wrong. Air oxidising Fe^{2+} before titrating → titre too small.

MARK-SCHEME LANGUAGE

- 13a: "dilute **sulfuric** acid"; "**permanent pale pink**" (not "pink to colourless", not purple); "no indicator needed because MnO_4^- is **self-indicating**"; HCl "is **oxidised** by MnO_4^- (to Cl_2)".
- 13b: starch "**near the end-point** / when the solution is **pale yellow**"; "**blue-black to colourless**"; "KI in **excess**, so a measuring cylinder is accurate enough".
- Calculations: n = cV/1000; correct ratio (5 : 1, 1 : 1, 5 : 2); ×10 (or the right scale factor); Ar Cu 63.5, Fe 55.8; answer to 2–3 s.f.

Often asked as: 4–6 mark titration calculations (iron, copper in brass, hydrated ethanedioic acid); why H_2SO_4 and not HCl or HNO_3; end-point colours; why starch is added late; why the nitric acid must be neutralised; three techniques for accurate burette readings; effect of a named error on the result.

COMMON MISTAKE

Forgetting the scale factor for the aliquot, or doubling 63.5 for copper.
Giving a black or brown end-point, or "pink to colourless" for MnO_4^- in the burette.
"Nitric acid reacts with copper" as the reason for neutralising: the point is that HNO_3 would **oxidise iodide** to extra iodine.

BEYOND THE SPEC, BUT EXAMINED

- **Back titration** of chlorate(V) with excess Fe^{2+}, the leftover Fe^{2+} titrated with MnO_4^-; Cl^- left in the flask also uses up MnO_4^-, making the calculated chlorate lower (WCH16 Jan 2021).
- Vanadium(V) reduced by zinc or tin, then titrated back up with MnO_4^-; tin left in the flask also reduces MnO_4^-, so the titre is too large (WCH16 Jan 2024).
- Equation for brass dissolving: $Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$ (WCH16 Oct 2021).

SPEC CP14

The preparation of a transition-metal complex

WHAT THIS SPEC POINT ASKS

Prepare a transition-metal complex by ligand exchange, isolate and purify the crystals, and calculate the percentage yield.

CP14 Aim and principle

The usual example is tetraamminecopper(II) sulfate-1-water. Excess ammonia replaces four water ligands around Cu^{2+} (ligand exchange):

$$[Cu(H_2O)_6]^{2+}(aq) + 4NH_3(aq) \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 4H_2O(l)$$

$$CuSO_4{\cdot}5H_2O + 4NH_3 \rightarrow [Cu(NH_3)_4]SO_4{\cdot}H_2O + 4H_2O$$

- First a **pale blue precipitate** of $Cu(OH)_2$ forms (deprotonation by the first drops of NH_3); in excess it dissolves to a **deep blue solution**.
- The complex is much **less soluble in aqueous ethanol** than in water, so adding ethanol makes it crystallise.

CORE PRACTICAL 14: METHOD IN BRIEF

1. Weigh about 2.5 g of $CuSO_4{\cdot}5H_2O$ into a beaker; dissolve in the **minimum** of distilled water (a few cm^3).
2. **In a fume cupboard**, add concentrated ammonia solution (about 5 cm^3, measuring cylinder) with stirring: pale blue precipitate, then a **deep blue solution**.
3. Slowly add about 8 cm^3 of **ethanol**; cool in an ice bath. Deep blue crystals form.
4. Collect by **suction filtration**. **Wash** with a little **ice-cold ethanol**.
5. Dry on filter paper or in a desiccator. Weigh and calculate the percentage yield.

CP14 Why each step

- Measuring cylinders are fine: the volumes are approximate and the ammonia is in **excess**.
- Fume cupboard: conc NH_3 gives off **toxic, corrosive** ammonia gas.
- Ethanol: lowers the solubility of the complex, so more crystallises.
- Cold ethanol wash: removes **soluble impurities** (unreacted $CuSO_4$, excess NH_3) without **dissolving the product**; ethanol also evaporates quickly so the crystals dry fast.

CP14 Safety

- Conc NH_3: **corrosive**, irritating vapour → fume cupboard, gloves.
- Ethanol: **flammable** → no naked flames.
- Copper(II) sulfate: harmful, **environmental hazard** → don't pour down the sink.

WORKED EXAMPLE: PERCENTAGE YIELD

2.50 g of $CuSO_4 \cdot 5H_2O$ (Mr 249.6) gave 1.98 g of $[Cu(NH_3)_4]SO_4 \cdot H_2O$ (Mr 245.6).
$n(CuSO_4 \cdot 5H_2O) = 2.50 \div 249.6 = 0.01002$ mol = n(complex), 1 : 1
theoretical mass = $0.01002 \times 245.6 = 2.460$ g
% yield = $1.98 \div 2.460 \times 100$ = **80.5 %**
A "yield" over 100 % means the crystals were still **damp**. Inverting the ratio, or skipping the Mr conversion and dividing the two masses, scores nothing.

CP14 Errors and improvements

- Low yield: product left dissolved in the filtrate (too much water at the start, or warm ethanol for washing); losses on transfer.
- Crystals contaminated with $CuSO_4$: wash properly with cold ethanol.
- Dry to constant mass before weighing.

MARK-SCHEME LANGUAGE

- "**Deep/dark blue** solution" (not just "blue").
- "Product is **less soluble in (aqueous) ethanol** than in water, so it crystallises."
- "Wash with cold ethanol to **remove soluble impurities**"; "cold so the product does **not dissolve**".
- "Measuring cylinder: approximate volumes / NH_3 in excess, so the accuracy of a pipette isn't needed."
- Yield > 100 %: "**damp** crystals" (not "hydrated salt").

Often asked as: observations on adding NH_3; why ethanol; why a fume cupboard; draw the suction-filtration apparatus; why the washing solvent is cold; calculate the percentage yield using both Mr values; explain a yield over 100 %.

COMMON MISTAKE

"Blue solution" for the ammine complex: it must be deep or dark blue.
"Wash the crystals" without saying what the wash removes.
Writing the complex as $[Cu(NH_3)_6]^{2+}$.

BEYOND THE SPEC, BUT EXAMINED

- **Vanadium oxidation states**: ammonium vanadate(V) reduced by zinc in acid, flask loosely plugged with **cotton wool** to keep air out but let hydrogen escape; colours yellow → green → blue → green → violet (WCH16 Jan 2024).
- **Cobalt complexes**: pink $[Co(H_2O)_6]^{2+}$ to blue $[CoCl_4]^{2-}$ with conc HCl; Co^{2+} with a little NH_3 gives a blue precipitate by **deprotonation**, not ligand exchange (WCH16 Oct 2022).

SPEC CP15 Analysis of some inorganic and organic unknowns

WHAT THIS SPEC POINT ASKS

Carry out tests on unknown substances, record the observations, and use them (with any other data) to identify the ions or functional groups present.

CP15 Aim and approach

This practical is the source of Q1 and often Q2 on every WCH16 paper. The tests and observations are all in **P5** above. In the lab, and in the exam, work in this order:

1. **Look**: colour of the solid or solution (blue → Cu^{2+}; green → Fe^{2+}, Cr^{3+}, Ni^{2+}; pink → Co^{2+}; yellow-brown → Fe^{3+}; orange → $Cr_2O_7^{2-}$; purple → MnO_4^-).
2. **Cation**: NaOH(aq) drop by drop, then in excess; then NH_3(aq) the same way on a fresh sample. Flame test for s-block metals. Warm with NaOH for NH_4^+.
3. **Anion**: dilute acid (effervescence → carbonate); HNO_3 + $AgNO_3$ (halide), then NH_3 to confirm; HCl + $BaCl_2$ (sulfate).
4. **Organic**: test each functional group, then use spectra (IR → groups; MS → Mr and fragments; NMR → environments, ratios, neighbours).

CORE PRACTICAL 15: HOW TO RECORD TESTS

Use three columns: **Test** (reagent, how much, conditions) → **Observation** (what you saw: colour, ppt or solution, gas, smell) → **Inference** (what it shows, with formula).
Add reagents **dropwise** with a teat pipette, shake, then add **excess**.
Use a **fresh sample** for each test, and test gases at the mouth of the tube.

WORKED EXAMPLE: IDENTIFYING AN INORGANIC SALT

Test	Observation	Inference
look at the solution	pale green	Fe^{2+} (or Cr^{3+}, Ni^{2+})
NaOH(aq), then excess	green ppt, insoluble in excess; turns brown at the surface on standing	$Fe(OH)_2$ oxidised to $Fe(OH)_3$: **Fe^{2+}**
dilute HNO_3 then $AgNO_3$(aq)	no change	no Cl^-, Br^- or I^-
dilute HCl then $BaCl_2$(aq)	white ppt	**SO_4^{2-}** ($BaSO_4$)

Conclusion: **iron(II) sulfate**, $FeSO_4$. (If you give a name and a formula, both must be right.)

WORKED EXAMPLE: IDENTIFYING AN ORGANIC COMPOUND, C_4H_8O

Evidence	Deduction
Brady's reagent: orange ppt	C=O present (aldehyde or ketone)
Tollens' reagent, warm: no change	not an aldehyde, so a **ketone**
iodine/NaOH, warm: pale yellow ppt	**$CH_3CO–$** group (methyl ketone)
IR: strong peak at 1715 cm^{-1}	C=O (ketone, 1720–1700 cm^{-1})
MS: m/z 72 (M^+), 43	Mr 72; CH_3CO^+ fragment
^1H NMR: 3 peaks, ratio 3 : 3 : 2: triplet, singlet, quartet	CH_3 next to CH_2 (triplet), isolated CH_3 (singlet), CH_2 next to CH_3 (quartet)

Structure: $CH_3COCH_2CH_3$, **butanone**.

MARK-SCHEME LANGUAGE

- "If name and formula are both given, **both must be correct**"; the **oxidation state** must be in a name (iron(II)); charges must be on complex ions.
- "precipitate / ppt / solid" accepted; shades of colour usually ignored; "**no change**", not "no observation" (penalised once).
- **Distinguish** questions: one reagent, then the result for **each** compound. The result mark depends on the reagent mark.
- "Justify": each test linked to the group it shows; each spectrum feature linked to part of the structure.

Often asked as: name or formula of ions A–D from a results table; complete missing observations; deduce a structure from tests plus IR, MS and NMR; choose a test to tell two compounds apart; write ionic equations for precipitates; state the type of reaction.

COMMON MISTAKE

Reading cobalt (Co) as copper (Cu): check the symbol you write.
Giving several answers where one is asked: a wrong extra answer cancels the right one.
"Amide" or "ammine" when you mean **amine**; writing –COH for an aldehyde.
Answering about a different sample from the one the question tested.

BEYOND THE SPEC, BUT EXAMINED

- **Combustion analysis train**: anhydrous $CaCl_2$ (absorbs H_2O) before soda lime (absorbs CO_2); then the empirical formula from the mass gains (WCH16 Jun 2021).
- A **sooty flame** on burning suggests an aromatic ring or high C : H ratio (WCH16 Jan 2025).
- Brady's derivative recrystallised and its melting point matched against a data table to identify the carbonyl compound (WCH16 Jun 2022).

SPEC CP16

The preparation of aspirin

WHAT THIS SPEC POINT ASKS

Prepare aspirin from 2-hydroxybenzoic acid and ethanoic anhydride, purify it by recrystallisation and check its purity by melting temperature; apply the same techniques to other organic preparations.

CP16 Aim and principle

$$HOC_6H_4COOH + (CH_3CO)_2O \rightarrow CH_3COOC_6H_4COOH + CH_3COOH$$

2-hydroxybenzoic acid + ethanoic anhydride → aspirin (2-ethanoyloxybenzoic acid) + ethanoic acid

- The phenol –OH of 2-hydroxybenzoic acid is **esterified** by ethanoic anhydride. A few drops of **concentrated sulfuric (or phosphoric) acid** act as a **catalyst**.
- Ethanoic anhydride is used **in excess** so all the (more expensive) 2-hydroxybenzoic acid reacts.
- Adding cold water at the end **hydrolyses the excess anhydride** to ethanoic acid (which stays in solution) and makes the aspirin **crystallise**, as it is only slightly soluble in cold water.

CORE PRACTICAL 16: METHOD IN BRIEF

1. Weigh 2.00 g of 2-hydroxybenzoic acid into a pear-shaped flask.
2. In a **fume cupboard**, add 5.0 cm^3 of ethanoic anhydride and 5 drops of conc H_2SO_4, with the flask in **cold water** (mixing is exothermic). Swirl.
3. Warm in a **water bath** at about 60–80 °C for about 15–20 minutes (or reflux), swirling.
4. Cool, then carefully add cold water / crushed ice to destroy the excess anhydride and crystallise the aspirin. Cool in an ice bath.
5. Collect the crude crystals by **suction filtration**; wash with a little ice-cold water.
6. **Recrystallise**: dissolve in the minimum of hot ethanol (or ethanol–water), filter hot if needed, cool slowly then in ice, filter under reduced pressure, wash with ice-cold solvent, dry.
7. Weigh and find the **melting temperature** (pure aspirin: about 136 °C).

CP16 Variables and safety

- Ethanoic anhydride: **corrosive**, irritating vapour (lachrymator), reacts with water → fume cupboard, gloves.
- Conc H_2SO_4: **corrosive** → gloves, add dropwise.
- Ethanol: **flammable** → water bath, no flame.
- Control for a fair yield comparison: masses, heating time and temperature, volume of solvent in recrystallisation.

WORKED EXAMPLE: EXCESS REAGENT AND PERCENTAGE YIELD

2.00 g 2-hydroxybenzoic acid (Mr 138.0); 5.0 cm^3 ethanoic anhydride (density 1.08 g cm^{-3}, Mr 102.0); 1.85 g of dry aspirin (Mr 180.0) obtained.
n(acid) = 2.00 ÷ 138.0 = 0.0145 mol
mass of anhydride = 5.0 × 1.08 = 5.40 g → n = 5.40 ÷ 102.0 = 0.0529 mol
Ratio is 1 : 1 and **0.0529 > 0.0145**, so ethanoic anhydride is **in excess**.
theoretical mass aspirin = 0.0145 × 180.0 = 2.61 g
% yield = 1.85 ÷ 2.61 × 100 = **70.9 %**

CP16 Errors and improvements

- Yield below 100 %: reaction incomplete; some aspirin stays dissolved in the recrystallisation solvent and the washings; losses on transfer; some aspirin hydrolysed back if heated with water too long.
- Purity: a melting range that is **sharp** (≈1–2 °C) and close to 136 °C. A lower, wider range means impurities (often unreacted 2-hydroxybenzoic acid). Recrystallising again improves purity but lowers the yield.
- A useful extra check (not required): iron(III) chloride solution turns purple with a phenol, so it shows up any unreacted 2-hydroxybenzoic acid; pure aspirin gives little colour.

CP16 The organic preparation family on WCH16

Every paper's last question is a preparation like this one with a new compound. The same technique set comes up each time:

Preparation seen	Techniques asked about
esters (ethyl butanoate, methyl salicylate, phenyl benzoate)	reflux; excess by calculation; Na_2CO_3 wash in a separating funnel (release CO_2 pressure); drying agent; distil and collect the right fraction
nitration (methyl 3-nitrobenzoate)	ice bath, temperature below ~10–15 °C, add nitrating mixture slowly; recrystallisation; melting point
azo dyes (via phenylamine and a diazonium salt)	0–10 °C, HNO_2 made in situ; coupling with a phenol in alkali; suction filtration; recrystallisation
phenylamine from nitrobenzene	Sn + conc HCl (reducing agent); NaOH to free the amine; steam distillation; salting out; ether extraction (hot water bath)
amides (phenylethanamide)	acyl chloride added slowly (vigorous, HCl given off); IR to show C=O of an amide
steam distillation of natural oils (limonene, lavender)	why steam distillation; which layer is the oil; drying the oil

MARK-SCHEME LANGUAGE

- Conc H_2SO_4 is a "**catalyst**" ("speeds up the reaction"); "provides H^+" or "increases yield" is ignored.
- Cold water / crushed ice: "to **react with (hydrolyse) the excess ethanoic anhydride**" and "to **crystallise** the aspirin". "To quench / stop the reaction" scores 0.
- Excess: masses → moles for **both** reagents, then a **comparison** ("0.053 > 0.0145").
- Suction filtration: "**faster**" and "gives a **drier** product" (comparatives).
- Purity: "**sharp** melting temperature" **and** "**close to the data-book value**". Boiling temperature or yield is not a purity test.

Often asked as: why the flask is cooled when mixing; role of conc H_2SO_4; show by calculation which reagent is in excess; spot three errors in a reflux diagram; why water is added at the end; advantage of suction filtration; describe recrystallisation; test of purity; percentage yield.

COMMON MISTAKE

Giving four or five errors when three are asked: a wrong extra one costs a mark.
"Dry product" for suction filtration: it gives a **drier** product.
Saying a thermometer or direct Bunsen heating is "needed" in a reflux set-up.
Using yield, or a boiling point, as evidence of purity for a solid.

BEYOND THE SPEC, BUT EXAMINED

- **Steam distillation** of limonene from orange peel: the oil is the **upper** layer because its density (0.85 g cm^{-3}) is below water's; optical activity checked with a **polarimeter** (WCH16 Jun 2022).
- **Recrystallisation temperature** kept below the product's melting point (e.g. 60 °C for a solid melting at 70 °C) so it doesn't melt into an oil (WCH16 Jun 2025).
- **Sulfonic acid group** on an azo dye makes it more soluble in water, so it washes out of fabric more easily (WCH16 Oct 2024).